统计学习-讲义及案例

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1 课程简介

1.1 课程内容

生活中数据几乎无处不在。然而，数据给人的感觉可能千差万别：或冰冷枯燥，让人望而生畏、百思不解其意；或生动有趣，让人豁然开朗。数据需要通过一种科学且直观的方式来展示，从而让人们一目了然地了解到它背后的含义，这也是统计数据分析及可视化技术的价值所在。它能让数据讲出科学的语言，给人们带来优雅的视觉感受，并助力科学的预测和决策。本课程内容主要包括：

- Tidy 式数据处理框架
- ggplot2 可视化绘图
- 从数理统计中参数估计、假设检验、回归分析等经典问题入手，利用随机模拟的手段介绍这些经典统计方法的内涵与意义，并从应用与决策的角度介绍这些方法的价值
- 利用实际案例介绍利用统计进行数据处理、数据挖掘分析与可视化的价值
- This course will teach you how to do data science with R: You'll learn how to get your data into R, get it into the most useful structure, transform it and visualize. [from book: R for Data Science (2e)]

1.2 What's new

大模型（DeepSeek 等）给我们学习编程提供了新的利器。通过大模型给我们提供的程序框架，我们可以从框架中学习我们想要的内容。这给我们的数据分析与编程带来了新的学习思路。

现在的学习不必过多拘泥于细节。我们也会在课程教学中尝试与大模型的互动教学，体现更现代化的学习方式。

但是，在这个时代，还是要记住一句话：人类的一小步，计算机的一大步。思考，肯定是比大模型更可靠的巨人。站在大模型的肩膀上，肯定不如站在思考的肩膀上看得透、学得深！

2 数据操作

```
library(magrittr)
library(tidyverse)
```

```
Warning: 程辑包'ggplot2'是用R版本4.3.3 来建造的
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.3     v readr     2.1.4
v forcats   1.0.0     v stringr   1.5.0
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.0
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x tidyr::extract()   masks magrittr::extract()
x dplyr::filter()    masks stats::filter()
x dplyr::lag()       masks stats::lag()
x purrr::set_names() masks magrittr::set_names()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become e
```

2.1 管道函数

```
x = runif(100, 1, 5)
sqrt(mean(x))
```

```
[1] 1.750074
```

```
x  %>%  mean  %>% sqrt
```

```
[1] 1.750074
```

- mpg 数据集，以 drv 变量分组，统计 hwy 的均值

  非管道操作

- 一般流程

```
mean(mpg[mpg$drv=="f",]$hwy)
```

```
[1] 28.16038
```

```
mean(mpg[mpg$drv=="4",]$hwy)
```

```
[1] 19.17476
```

```
mean(mpg[mpg$drv=="r",]$hwy)
```

```
[1] 21
```

- 如果有很多组怎么办

```
levelValue = levels(factor(mpg$drv))
aveVec = rep(0, length(levelValue))
for(i in 1:length(levelValue)){
  aveVec[i] = mean(mpg[mpg$drv==levelValue[i],]$hwy)
}
```

- base 里面的分组统计操作

```
aveVec = tapply(mpg$hwy, INDEX = mpg$drv, FUN = mean)
aveVec
```

```
       4        f        r
19.17476 28.16038 21.00000
```

管道操作

```
d1 <- mpg %>%
group_by(drv)
d1
```

```
# A tibble: 234 x 11
# Groups:   drv [3]
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
 1 audi         a4           1.8  1999     4 auto~ f        18    29 p     comp~
 2 audi         a4           1.8  1999     4 manu~ f        21    29 p     comp~
 3 audi         a4           2    2008     4 manu~ f        20    31 p     comp~
 4 audi         a4           2    2008     4 auto~ f        21    30 p     comp~
 5 audi         a4           2.8  1999     6 auto~ f        16    26 p     comp~
 6 audi         a4           2.8  1999     6 manu~ f        18    26 p     comp~
 7 audi         a4           3.1  2008     6 auto~ f        18    27 p     comp~
 8 audi         a4 quattro   1.8  1999     4 manu~ 4        18    26 p     comp~
 9 audi         a4 quattro   1.8  1999     4 auto~ 4        16    25 p     comp~
10 audi         a4 quattro   2    2008     4 manu~ 4        20    28 p     comp~
# i 224 more rows
```

```
d1 %>%
summarise(hwy_ave = mean(hwy))
```

```
# A tibble: 3 x 2
  drv   hwy_ave
  <chr>   <dbl>
1 4        19.2
2 f        28.2
3 r        21
```

2.1.1 管道函数非第一参数

```
mpg %>%
group_split(drv)  %>%
map(~mean(.x$hwy))
```

```
[[1]]
[1] 19.17476

[[2]]
[1] 28.16038

[[3]]
[1] 21
```

- . 可以在当层管道中多次使用，但是要放在 {} 环境中

```
x = c(n = 10, m = 5, lambda = 2)
```

从 x 出发，生成 n 个参数为 λ 的指数分布随机数

错误用法

```
x  %>%
rexp(.[1],.[3])
```

正确用法

```
x  %>%
{rexp(.[1],.[3])}
```

```
 [1] 0.002169885 0.217899511 0.170162662 0.260192699 0.690796820 0.150923823
 [7] 0.525410922 0.385764673 0.019092259 0.370246957
```

```
x  %>%
 {list(rnum = rexp(.[1], .[3]), m = .[2])}  %>%
 {.$rnum[1:.$m]}
```

```
[1] 0.5709187 0.5002119 0.4865998 0.3482133 0.4384037
```

- map 循环机制

  计算 mpg 数据集 hwy 与 cty 两个变量的均值

```
mpg1 <- mpg[,c("hwy","cty")]
mpg1  %>%
map(mean)
```

```
$hwy
[1] 23.44017

$cty
[1] 16.85897
```

可以控制返回值的类型

```
mpg1  %>%
map_dbl(mean)
```

```
     hwy      cty
23.44017 16.85897
```

```
map2_dbl(iris[,1], iris[,2], ~.x + .y)
```

```
  [1]  8.6  7.9  7.9  7.7  8.6  9.3  8.0  8.4  7.3  8.0  9.1  8.2  7.8  7.3  9.8
 [16] 10.1  9.3  8.6  9.5  8.9  8.8  8.8  8.2  8.4  8.2  8.0  8.4  8.7  8.6  7.9
 [31]  7.9  8.8  9.3  9.7  8.0  8.2  9.0  8.5  7.4  8.5  8.5  6.8  7.6  8.5  8.9
 [46]  7.8  8.9  7.8  9.0  8.3 10.2  9.6 10.0  7.8  9.3  8.5  9.6  7.3  9.5  7.9
```

```
 [61]  7.0  8.9  8.2  9.0  8.5  9.8  8.6  8.5  8.4  8.1  9.1  8.9  8.8  8.9  9.3
 [76]  9.6  9.6  9.7  8.9  8.3  7.9  7.9  8.5  8.7  8.4  9.4  9.8  8.6  8.6  8.0
 [91]  8.1  9.1  8.4  7.3  8.3  8.7  8.6  9.1  7.6  8.5  9.6  8.5 10.1  9.2  9.5
[106] 10.6  7.4 10.2  9.2 10.8  9.7  9.1  9.8  8.2  8.6  9.6  9.5 11.5 10.3  8.2
[121] 10.1  8.4 10.5  9.0 10.0 10.4  9.0  9.1  9.2 10.2 10.2 11.7  9.2  9.1  8.7
[136] 10.7  9.7  9.5  9.0 10.0  9.8 10.0  8.5 10.0 10.0  9.7  8.8  9.5  9.6  8.9
```

将 mpg 数据集按 drv 变量进行分组，然后统计各组 hwy 的均值

```
mpg %>%
group_split(drv)  %>%
map(mean)
```

```
Warning in mean.default(.x[[i]], ...): 参数不是数值也不是逻辑值: 回覆NA

Warning in mean.default(.x[[i]], ...): 参数不是数值也不是逻辑值: 回覆NA

Warning in mean.default(.x[[i]], ...): 参数不是数值也不是逻辑值: 回覆NA

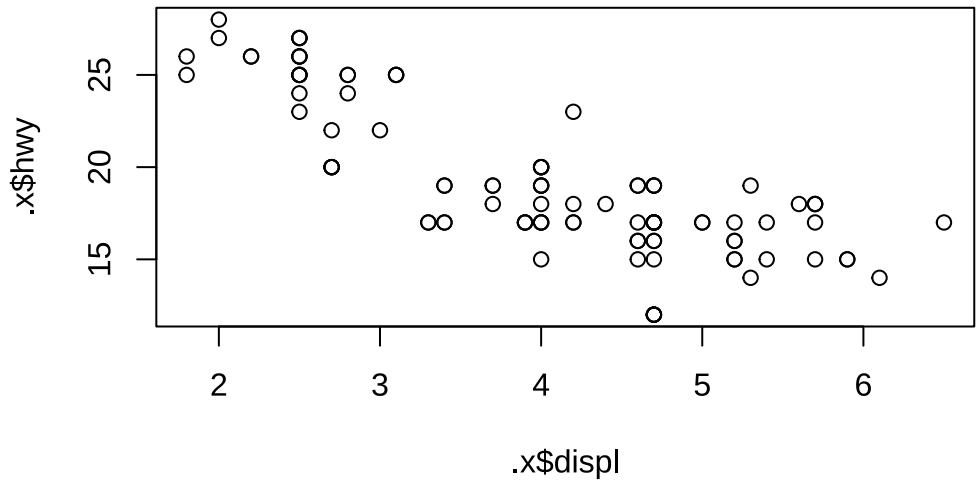
[[1]]
[1] NA

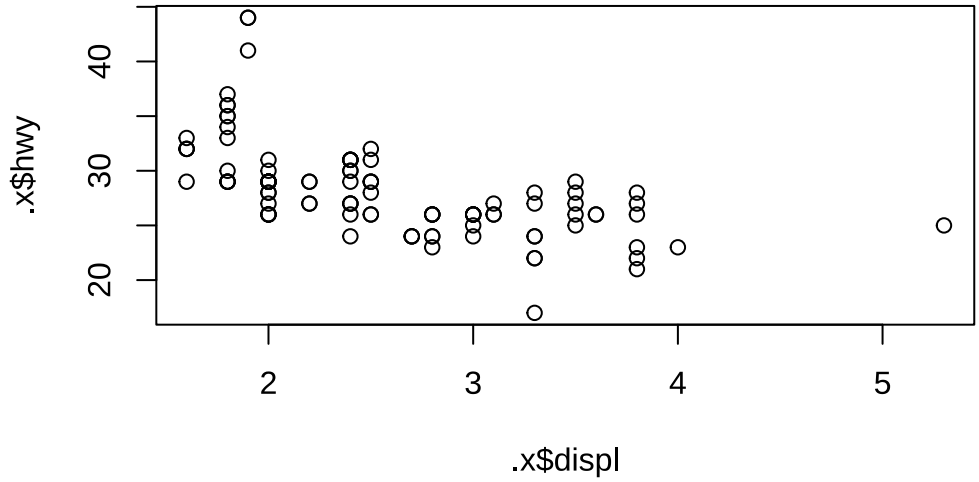
[[2]]
[1] NA

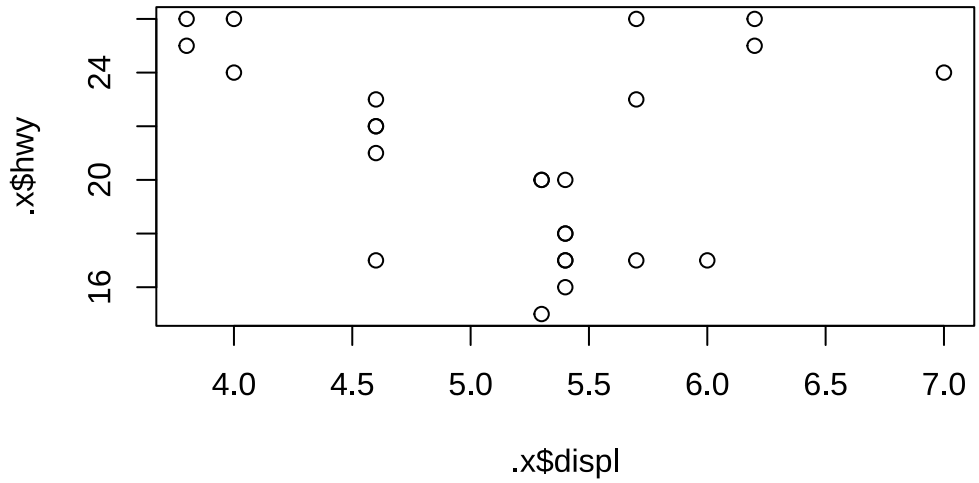
[[3]]
[1] NA
```

将 mpg 数据集按 drv 变量进行分组，然后画出各组 displ 与 hwy 的散点图

```
mpg %>%
group_split(drv)  %>%
map(~plot(.x$displ, .x$hwy))
```


.x$hwy
15
20
25
2
3
4
5
6
.x$displ


.x$hwy
20
30
40
2
3
4
5
.x$displ


.x$hwy
16
20
24
4.0
4.5
5.0
5.5
6.0
6.5
7.0
.x$displ

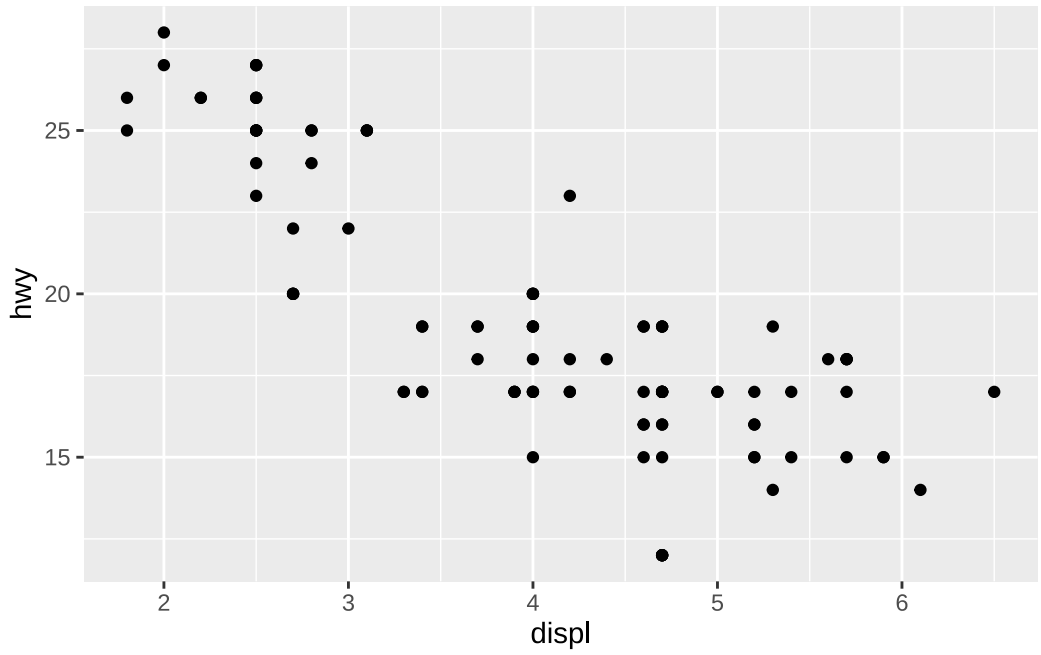
```
[[1]]
NULL

[[2]]
NULL

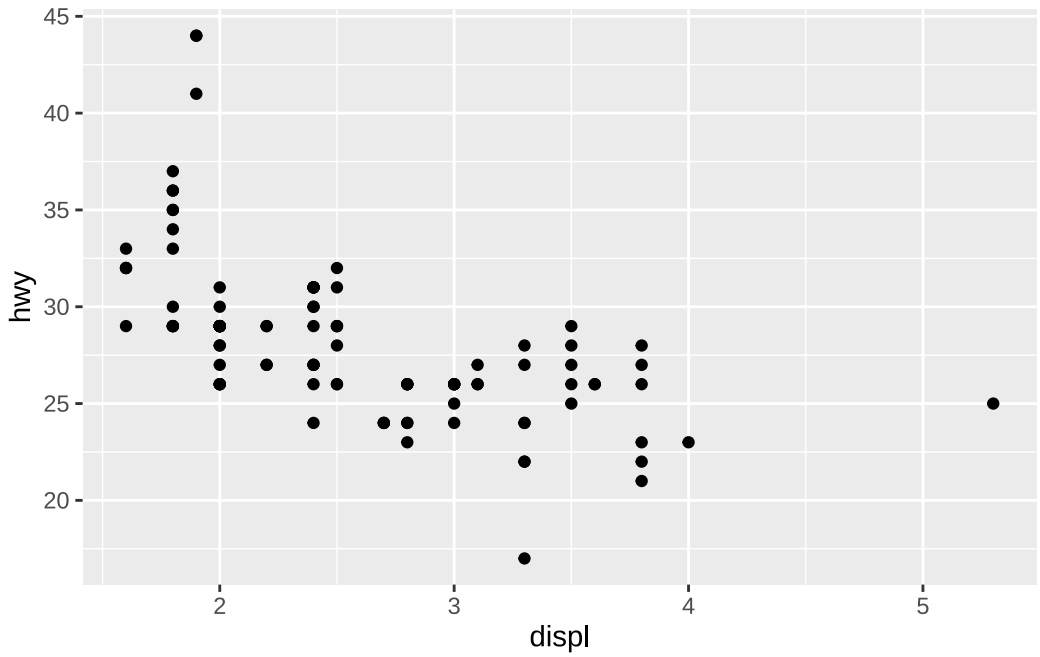
[[3]]
NULL
```

```
plotfun <- function(xx){
  p1 <- ggplot(data = xx, aes(x = displ, y  = hwy)) +
  geom_point()
  p1
}
mpg %>%
group_split(drv)  %>%
map(plotfun)
```

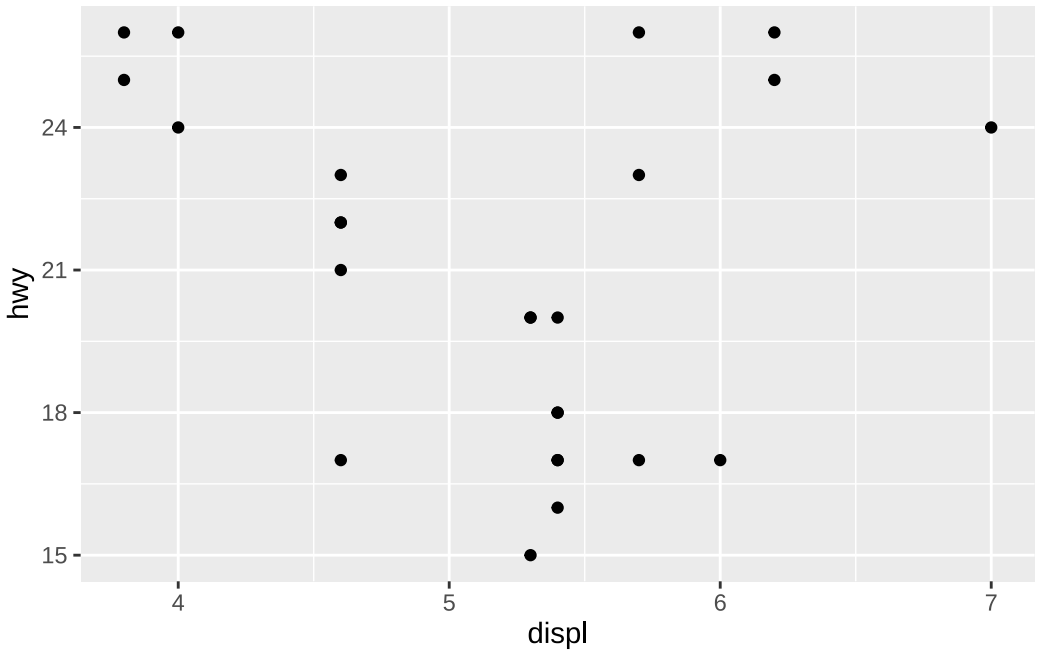
```
[[1]]
```



```
[[2]]
```



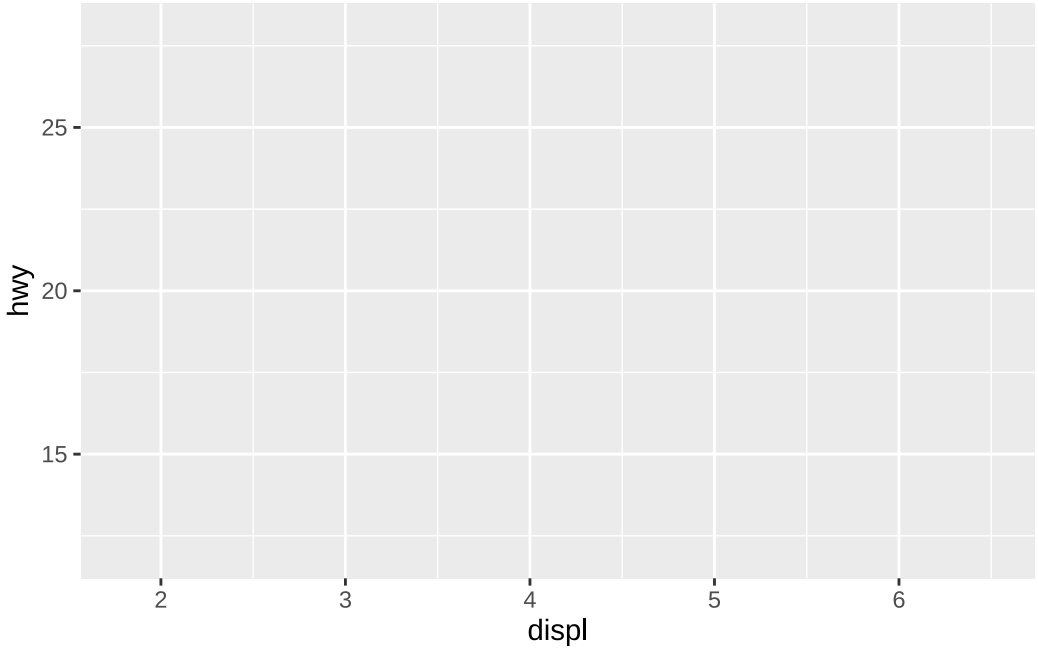

```
[[3]]
```



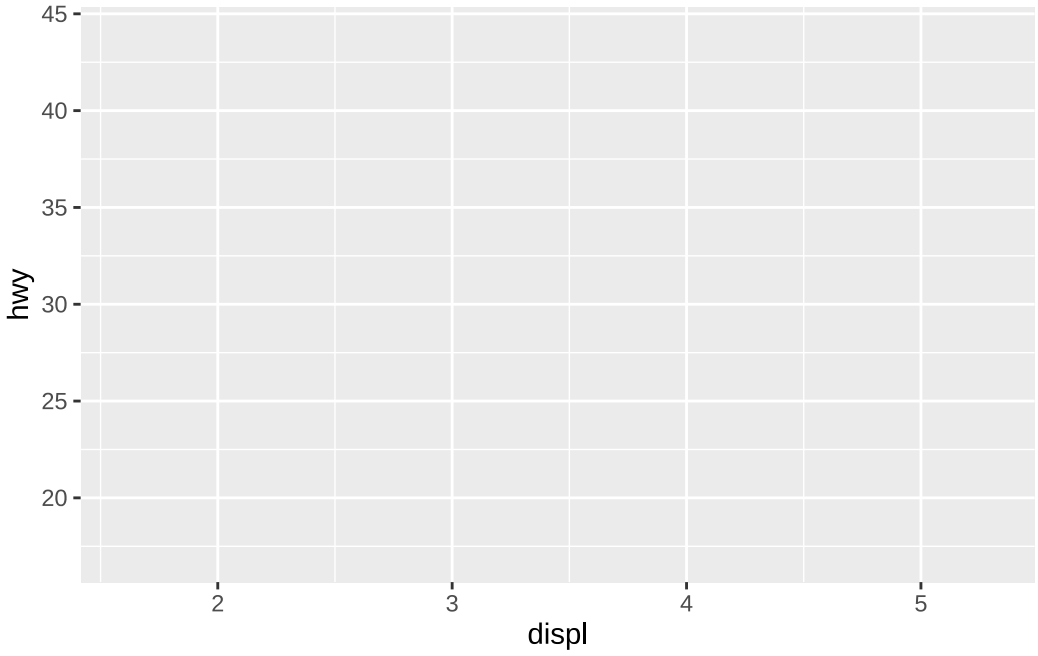
使用匿名函数

```
mpg %>%
group_split(drv)  %>%
map(~ggplot(.x, aes(x = displ, y = hwy)))
```

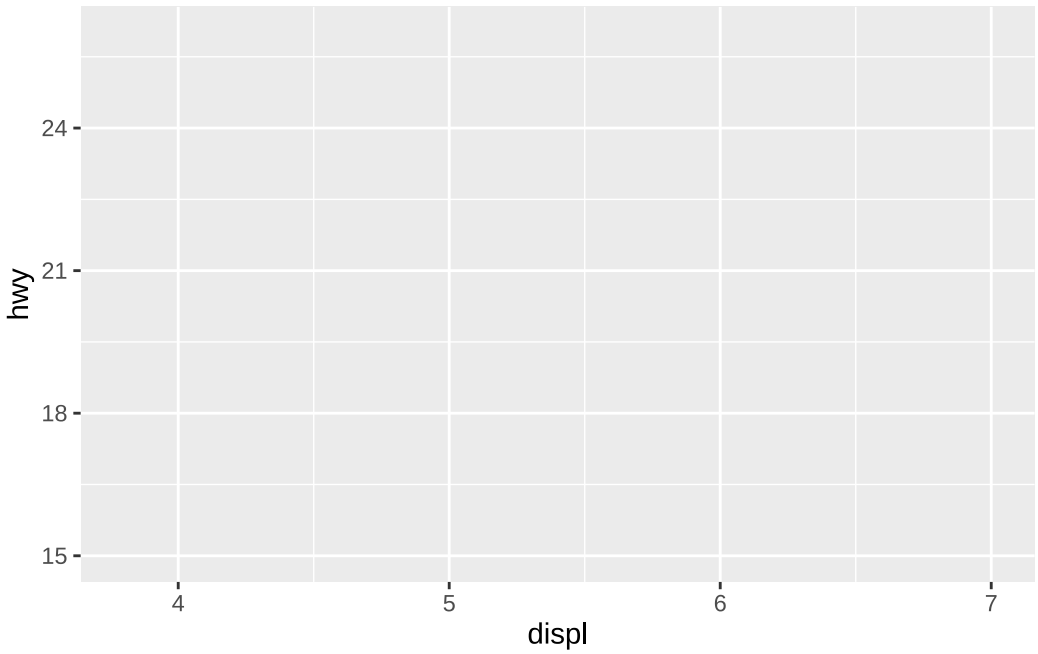
```
[[1]]
```




```
[[2]]
```



```
[[3]]
```



2.2 数据变换

mutate 增加列

```
iris %>%
  mutate(Sepal.LW = Sepal.Length+Sepal.Width) %>%
  head(5)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species Sepal.LW
1          5.1         3.5          1.4         0.2  setosa      8.6
2          4.9         3.0          1.4         0.2  setosa      7.9
3          4.7         3.2          1.3         0.2  setosa      7.9
4          4.6         3.1          1.5         0.2  setosa      7.7
5          5.0         3.6          1.4         0.2  setosa      8.6
```

2.3 分组统计

```
mpg  %>%
group_by(drv)  %>%
summarise(meanHwy = mean(hwy))
```

```
# A tibble: 3 x 2
  drv   meanHwy
  <chr>   <dbl>
1 4        19.2
2 f        28.2
3 r        21
```

```
mpg  %>%
group_by(drv)  %>%
summarise(NumbyDrv = n())
```

```
# A tibble: 3 x 2
  drv   NumbyDrv
  <chr>    <int>
1 4          103
2 f          106
3 r           25
```

```
mpg  %>%
group_split(drv) %>%
map(~mean(.x$hwy))
```

```
[[1]]
[1] 19.17476

[[2]]
[1] 28.16038
```

```
[[3]]
[1] 21
```

```
mpg  %>%
group_split(drv) %>%
map_dbl(~length(.x$hwy))
```

```
[1] 103 106  25
```

2.4 filter 函数

mpg 数据以 drv 分组，找到各组数据 hwy 值大于均值 + 标准差的样本

- 体会 group_by 与 group_split 的区别

```
mpg  %>%
group_by(drv)  %>%
filter(hwy > mean(hwy)+sd(hwy))
```

```
# A tibble: 40 x 11
# Groups:   drv [3]
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
 1 audi         a4 quattro   1.8  1999     4 manu~ 4        18    26 p     comp~
 2 audi         a4 quattro   1.8  1999     4 auto~ 4        16    25 p     comp~
 3 audi         a4 quattro   2    2008     4 manu~ 4        20    28 p     comp~
 4 audi         a4 quattro   2    2008     4 auto~ 4        19    27 p     comp~
 5 audi         a4 quattro   2.8  1999     6 auto~ 4        15    25 p     comp~
 6 audi         a4 quattro   2.8  1999     6 manu~ 4        17    25 p     comp~
 7 audi         a4 quattro   3.1  2008     6 auto~ 4        17    25 p     comp~
 8 audi         a4 quattro   3.1  2008     6 manu~ 4        15    25 p     comp~
 9 audi         a6 quattro   2.8  1999     6 auto~ 4        15    24 p     mids~
10 audi         a6 quattro   3.1  2008     6 auto~ 4        17    25 p     mids~
# i 30 more rows
```

```
mpg  %>%
group_split(drv) %>%
map(~filter(.x, hwy >mean(hwy)+sd(hwy)))
```

```
[[1]]
# A tibble: 23 x 11
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
 1 audi         a4 quattro   1.8  1999     4 manu~ 4        18    26 p     comp~
 2 audi         a4 quattro   1.8  1999     4 auto~ 4        16    25 p     comp~
 3 audi         a4 quattro   2    2008     4 manu~ 4        20    28 p     comp~
 4 audi         a4 quattro   2    2008     4 auto~ 4        19    27 p     comp~
 5 audi         a4 quattro   2.8  1999     6 auto~ 4        15    25 p     comp~
 6 audi         a4 quattro   2.8  1999     6 manu~ 4        17    25 p     comp~
 7 audi         a4 quattro   3.1  2008     6 auto~ 4        17    25 p     comp~
 8 audi         a4 quattro   3.1  2008     6 manu~ 4        15    25 p     comp~
 9 audi         a6 quattro   2.8  1999     6 auto~ 4        15    24 p     mids~
10 audi         a6 quattro   3.1  2008     6 auto~ 4        17    25 p     mids~
# i 13 more rows

[[2]]
# A tibble: 11 x 11
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
 1 honda        civic        1.6  1999     4 manu~ f        28    33 r     subc~
 2 honda        civic        1.8  2008     4 manu~ f        26    34 r     subc~
 3 honda        civic        1.8  2008     4 auto~ f        25    36 r     subc~
 4 honda        civic        1.8  2008     4 auto~ f        24    36 c     subc~
 5 toyota       corolla      1.8  1999     4 auto~ f        24    33 r     comp~
 6 toyota       corolla      1.8  1999     4 manu~ f        26    35 r     comp~
 7 toyota       corolla      1.8  2008     4 manu~ f        28    37 r     comp~
 8 toyota       corolla      1.8  2008     4 auto~ f        26    35 r     comp~
 9 volkswagen   jetta        1.9  1999     4 manu~ f        33    44 d     comp~
10 volkswagen   new beetle   1.9  1999     4 manu~ f        35    44 d     subc~
11 volkswagen   new beetle   1.9  1999     4 auto~ f        29    41 d     subc~
```

```
[[3]]
# A tibble: 6 x 11
  manufacturer model    displ  year   cyl trans    drv     cty   hwy fl    class
  <chr>        <chr>    <dbl> <int> <int> <chr>    <chr> <int> <int> <chr> <chr>
1 chevrolet    corvette   5.7  1999     8 manual(~ r        16    26 p     2sea~
2 chevrolet    corvette   6.2  2008     8 manual(~ r        16    26 p     2sea~
3 chevrolet    corvette   6.2  2008     8 auto(s6) r        15    25 p     2sea~
4 ford         mustang    3.8  1999     6 manual(~ r        18    26 r     subc~
5 ford         mustang    3.8  1999     6 auto(l4) r        18    25 r     subc~
6 ford         mustang    4    2008     6 manual(~ r        17    26 r     subc~
```

```
mpg  %>%
group_split(drv) %>%
map(~filter(.x, .x$hwy >mean(.x$hwy)+sd(.x$hwy)))
```

```
[[1]]
# A tibble: 23 x 11
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
 1 audi         a4 quattro   1.8  1999     4 manu~ 4        18    26 p     comp~
 2 audi         a4 quattro   1.8  1999     4 auto~ 4        16    25 p     comp~
 3 audi         a4 quattro   2    2008     4 manu~ 4        20    28 p     comp~
 4 audi         a4 quattro   2    2008     4 auto~ 4        19    27 p     comp~
 5 audi         a4 quattro   2.8  1999     6 auto~ 4        15    25 p     comp~
 6 audi         a4 quattro   2.8  1999     6 manu~ 4        17    25 p     comp~
 7 audi         a4 quattro   3.1  2008     6 auto~ 4        17    25 p     comp~
 8 audi         a4 quattro   3.1  2008     6 manu~ 4        15    25 p     comp~
 9 audi         a6 quattro   2.8  1999     6 auto~ 4        15    24 p     mids~
10 audi         a6 quattro   3.1  2008     6 auto~ 4        17    25 p     mids~
# i 13 more rows

[[2]]
# A tibble: 11 x 11
   manufacturer model      displ  year   cyl trans drv     cty   hwy fl    class
```

```
   <chr>        <chr>      <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>
 1 honda        civic        1.6  1999     4 manu~ f        28    33 r     subc~
 2 honda        civic        1.8  2008     4 manu~ f        26    34 r     subc~
 3 honda        civic        1.8  2008     4 auto~ f        25    36 r     subc~
 4 honda        civic        1.8  2008     4 auto~ f        24    36 c     subc~
 5 toyota       corolla      1.8  1999     4 auto~ f        24    33 r     comp~
 6 toyota       corolla      1.8  1999     4 manu~ f        26    35 r     comp~
 7 toyota       corolla      1.8  2008     4 manu~ f        28    37 r     comp~
 8 toyota       corolla      1.8  2008     4 auto~ f        26    35 r     comp~
 9 volkswagen   jetta        1.9  1999     4 manu~ f        33    44 d     comp~
10 volkswagen   new beetle   1.9  1999     4 manu~ f        35    44 d     subc~
11 volkswagen   new beetle   1.9  1999     4 auto~ f        29    41 d     subc~

[[3]]
# A tibble: 6 x 11
  manufacturer model    displ  year   cyl trans    drv     cty   hwy fl    class
  <chr>        <chr>    <dbl> <int> <int> <chr>    <chr> <int> <int> <chr> <chr>
1 chevrolet    corvette   5.7  1999     8 manual(~ r        16    26 p     2sea~
2 chevrolet    corvette   6.2  2008     8 manual(~ r        16    26 p     2sea~
3 chevrolet    corvette   6.2  2008     8 auto(s6) r        15    25 p     2sea~
4 ford         mustang    3.8  1999     6 manual(~ r        18    26 r     subc~
5 ford         mustang    3.8  1999     6 auto(l4) r        18    25 r     subc~
6 ford         mustang    4    2008     6 manual(~ r        17    26 r     subc~
```

3 ggplot2 画图系统

载入需要的程辑包：sysfonts

载入需要的程辑包：showtextdb

4 ggplot2 画图系统

4.1 数据集介绍

```
library(tidyverse)
```

```
Warning: 程辑包'ggplot2'是用R版本4.3.3 来建造的
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.3     v readr     2.1.4
v forcats   1.0.0     v stringr   1.5.0
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.0
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become e
```

```
dim(mpg)
```

```
[1] 234  11
```

```
head(mpg)
```

```
# A tibble: 6 x 11
  manufacturer model displ  year   cyl trans      drv     cty   hwy fl    class
  <chr>        <chr> <dbl> <int> <int> <chr>      <chr> <int> <int> <chr> <chr>
1 audi         a4      1.8  1999     4 auto(l5)   f        18    29 p     compa~
2 audi         a4      1.8  1999     4 manual(m5) f        21    29 p     compa~
3 audi         a4      2    2008     4 manual(m6) f        20    31 p     compa~
4 audi         a4      2    2008     4 auto(av)   f        21    30 p     compa~
5 audi         a4      2.8  1999     6 auto(l5)   f        16    26 p     compa~
6 audi         a4      2.8  1999     6 manual(m5) f        18    26 p     compa~
```

数据集大小为 234 行 *11 列，数据集主要变量介绍

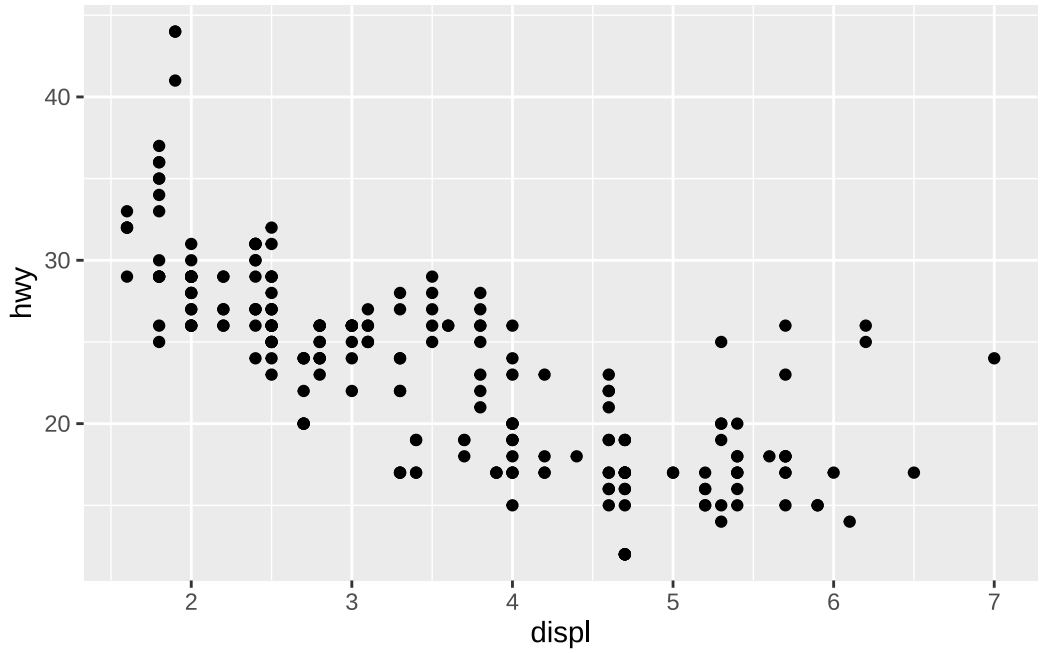
- manufacturer: 生产商
- model: 型号
- displ: 发动机排量（Displacement, litres）, 连续变量
- year: 生产年份
- cyl: 汽车的汽缸数，离散变量
- trans: type of transmission
- drv: 驱动方式（f, r, 4）
- cty: city miles per gallon
- hwy: highway miles per gallon
- fl: fuel type
- class: type of car

将数据集保存到本地

```
write.csv(file="./output/mpg.csv", mpg)
```

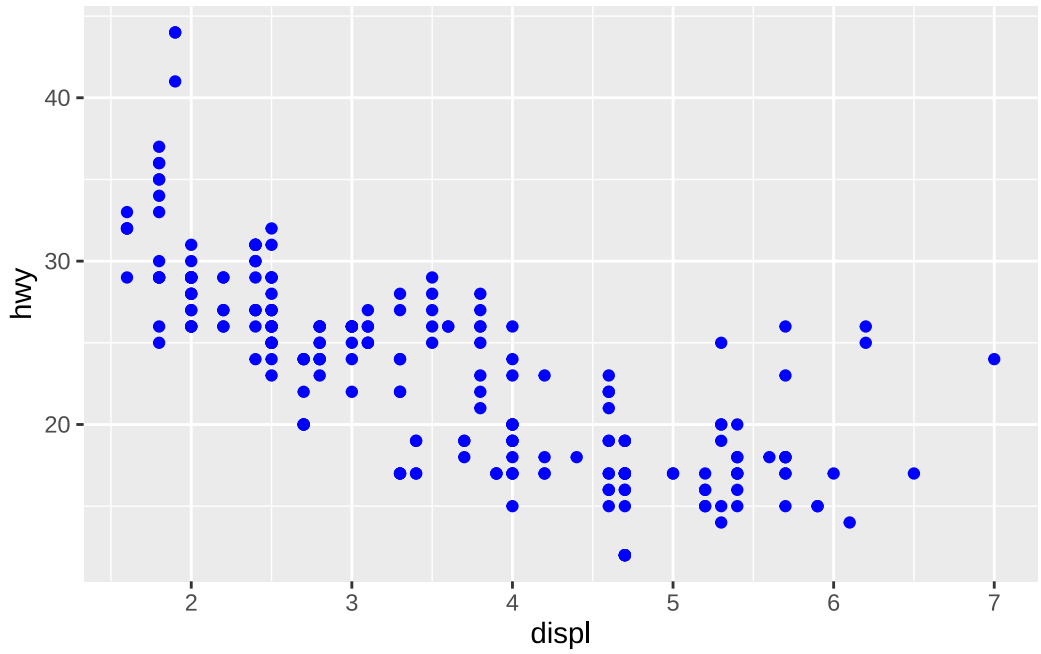
横轴为 displ, 纵轴为 hwy，画散点图

```
p0 <- ggplot(data = mpg, aes(x = displ, y = hwy))
p0 + geom_point()
```



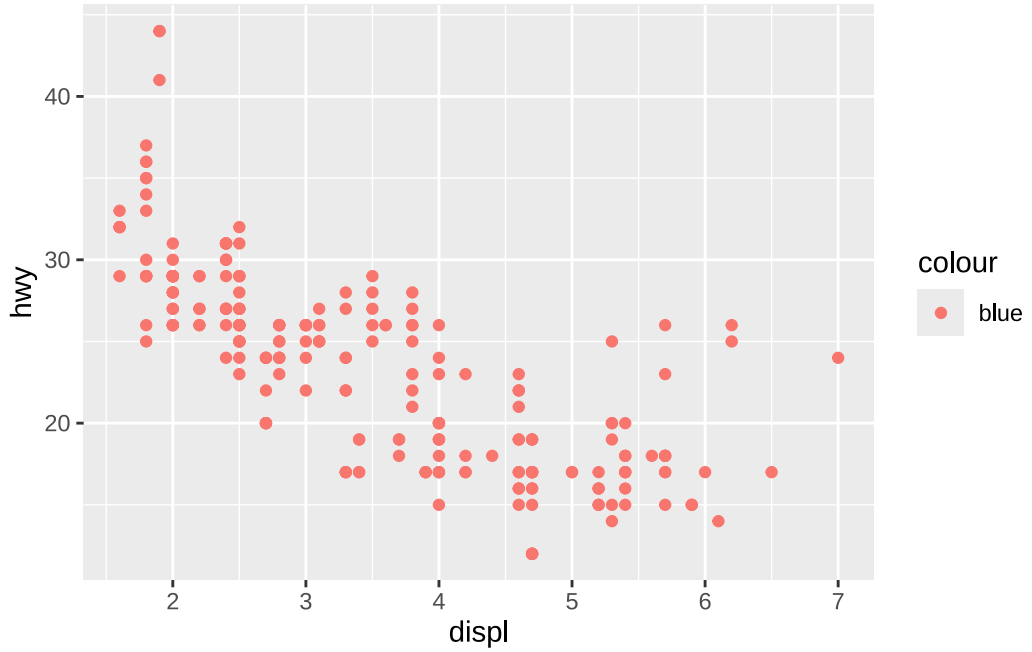
把点的颜色设置为蓝色

```
p0 + geom_point(color = "blue")
```

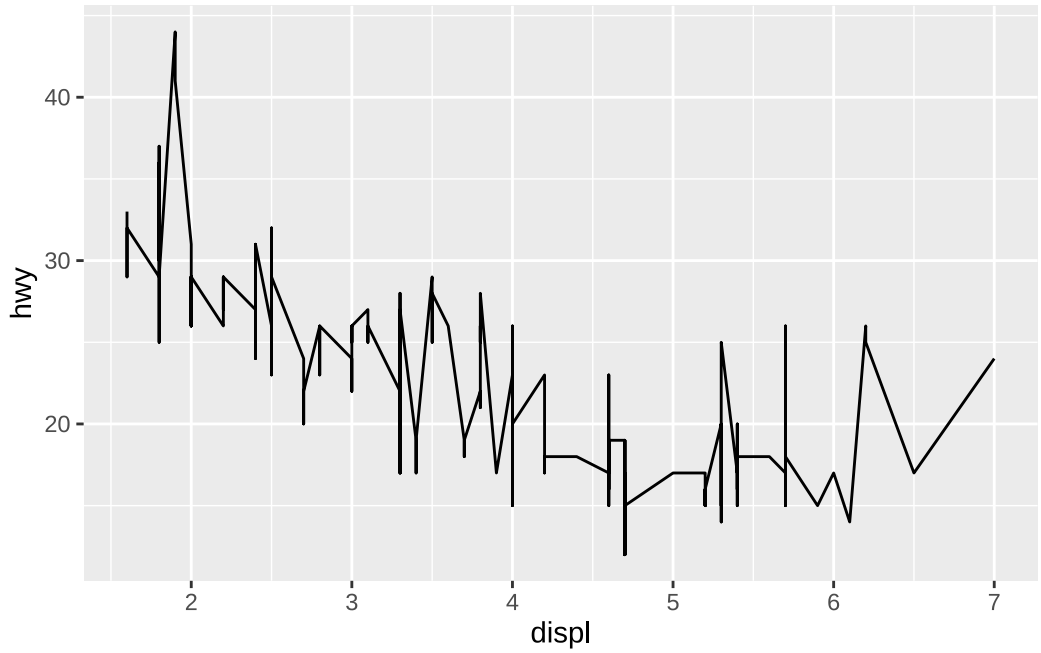


- 放到 aes 里面的是 mapping，不是 setting

```
p0 +  geom_point(aes(color = "blue"))
```

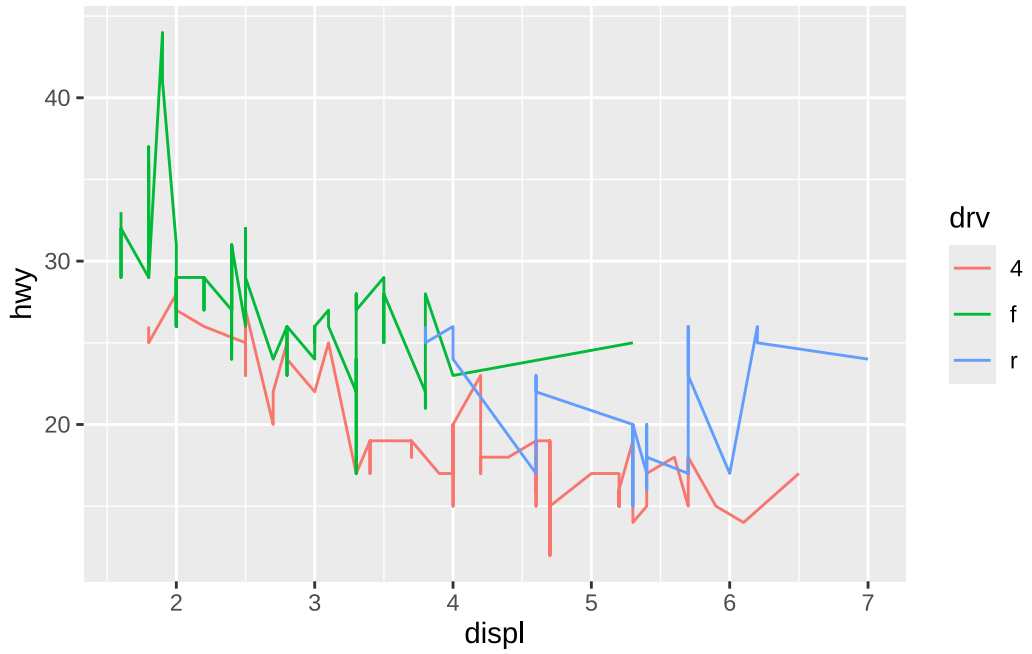


```
p0 + geom_line()
```

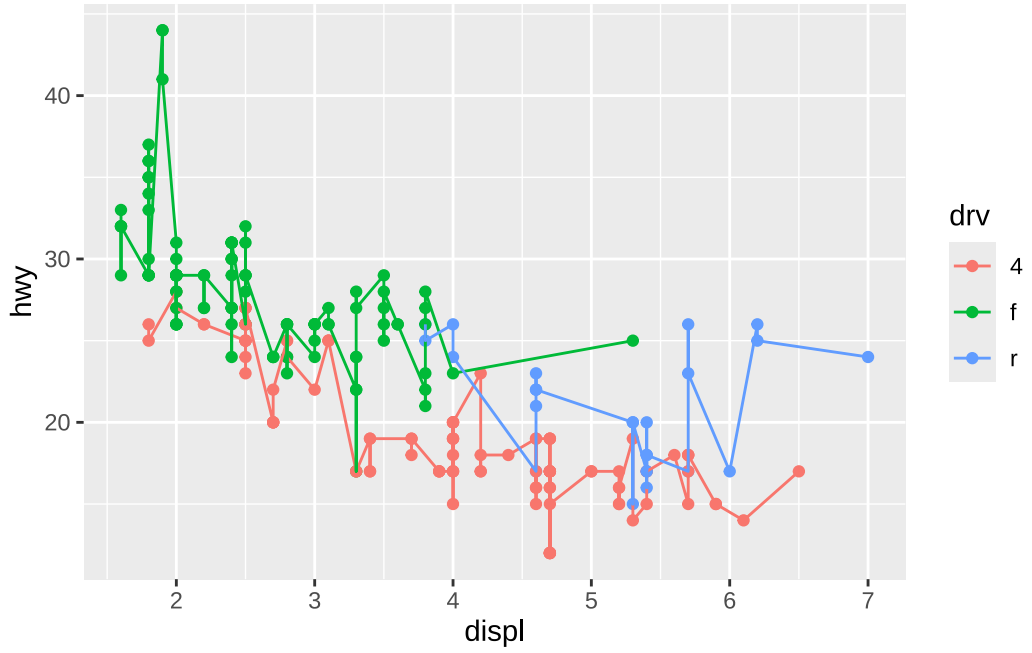


点图与线图 mapping 的区别

```
p0 + geom_line(aes(color = drv))
```




```
p0 + geom_point(aes(color = drv)) +
geom_line(aes(color = drv))
```

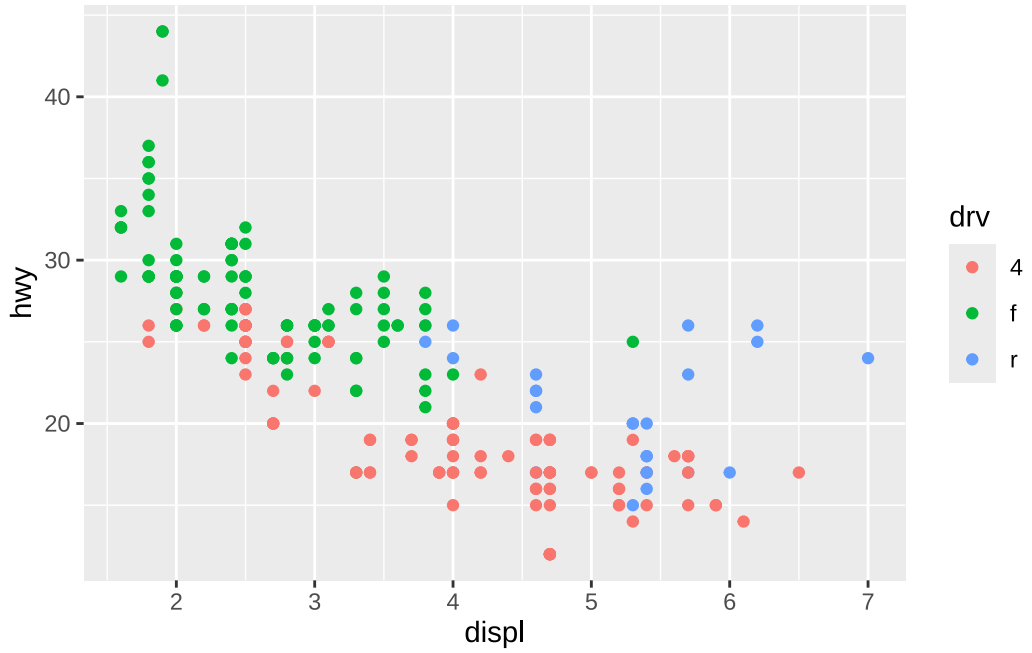


4.2 分组

分组操作：aes(color = drv)

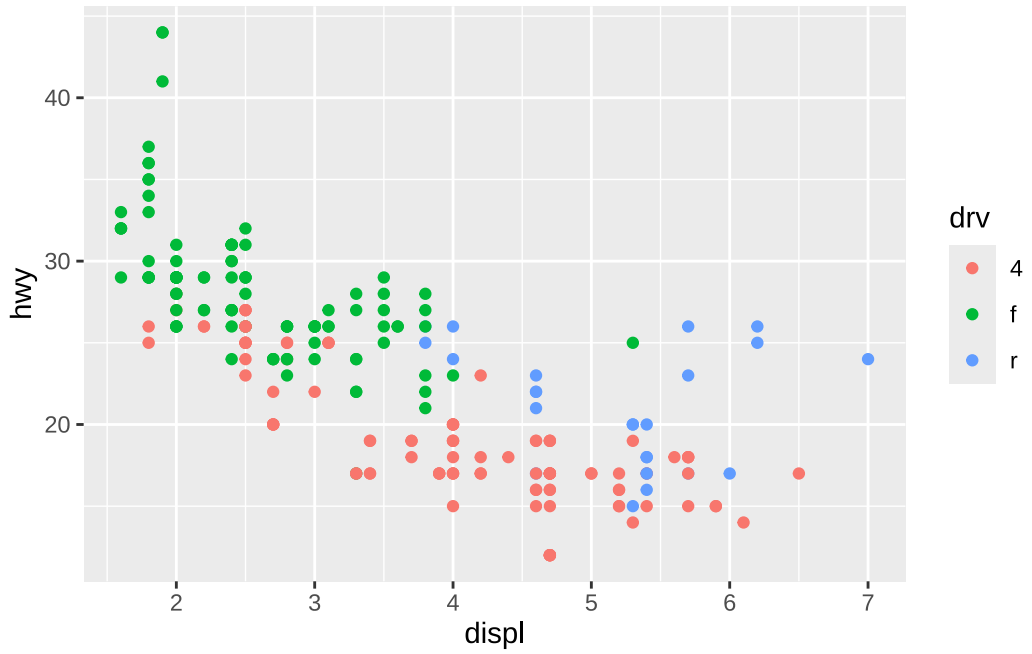
- 将颜色映射到分类变量 drv
- 对不同 drv 值的数据，按不同颜色分别绘图

```
ggplot(data = mpg, aes(x = displ, y = hwy, color = drv)) +
geom_point()
```



对 color 的操作可以放到 geom_point 中

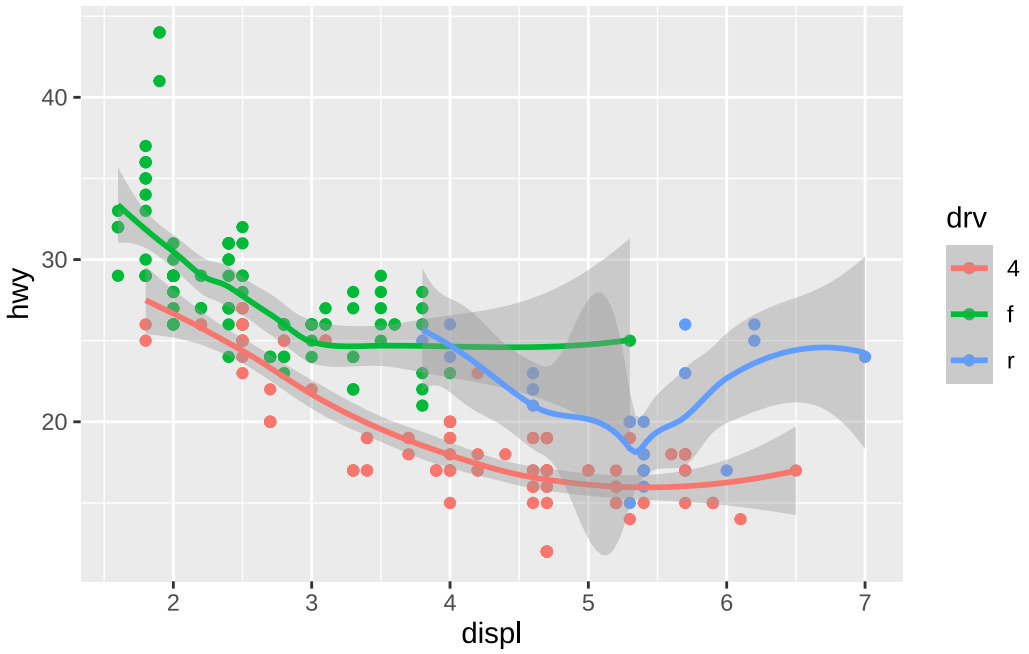
```
ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point(aes(color = drv))
```



- 分析区别

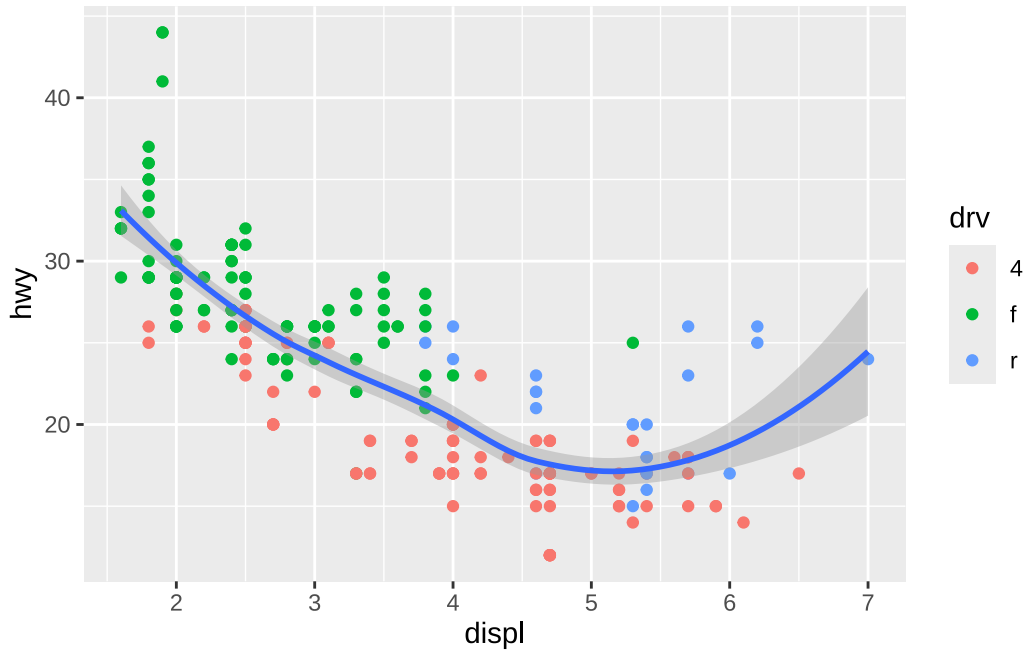
```
ggplot(data = mpg, aes(x = displ, y = hwy,color = drv)) +
geom_point() +
geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



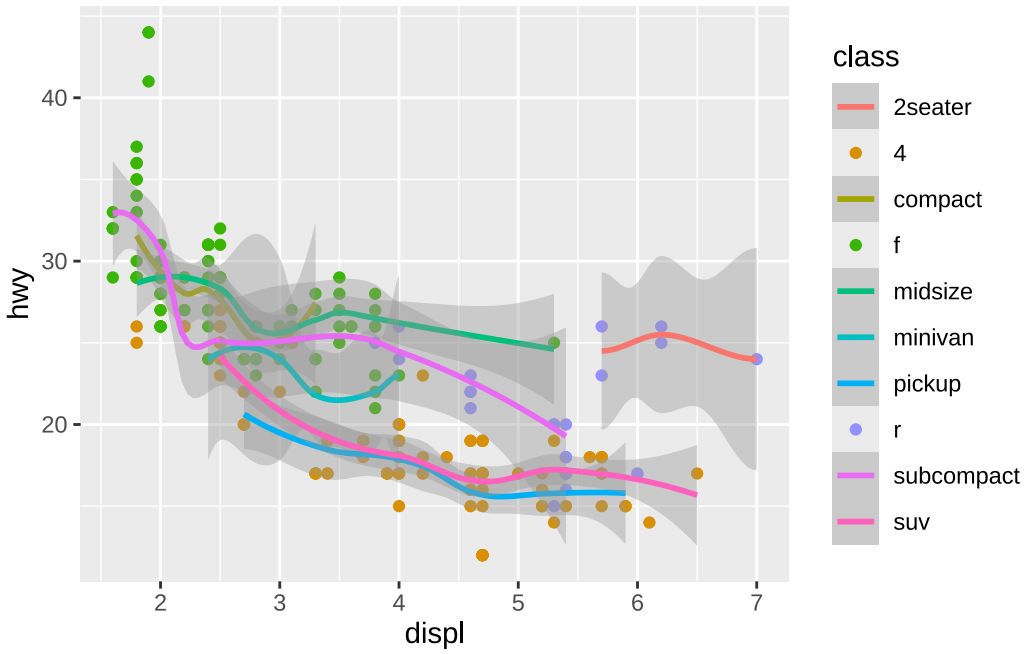

```
ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point(aes(color = drv)) +
geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
ggplot(data = mpg, aes(x = displ, y = hwy, color = class)) +
geom_point(aes(color = drv)) +
geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



几何对象 ggplot2 绘图系统中全局与局部的规定

- ggplot() 中的数据和映射是全局的，可供所有几何对象共用
- 位于几何对象中的数据和映射，是局部的，只供该几何对象使用
- 几何对象优先使用局部的，局部没有则用全局的

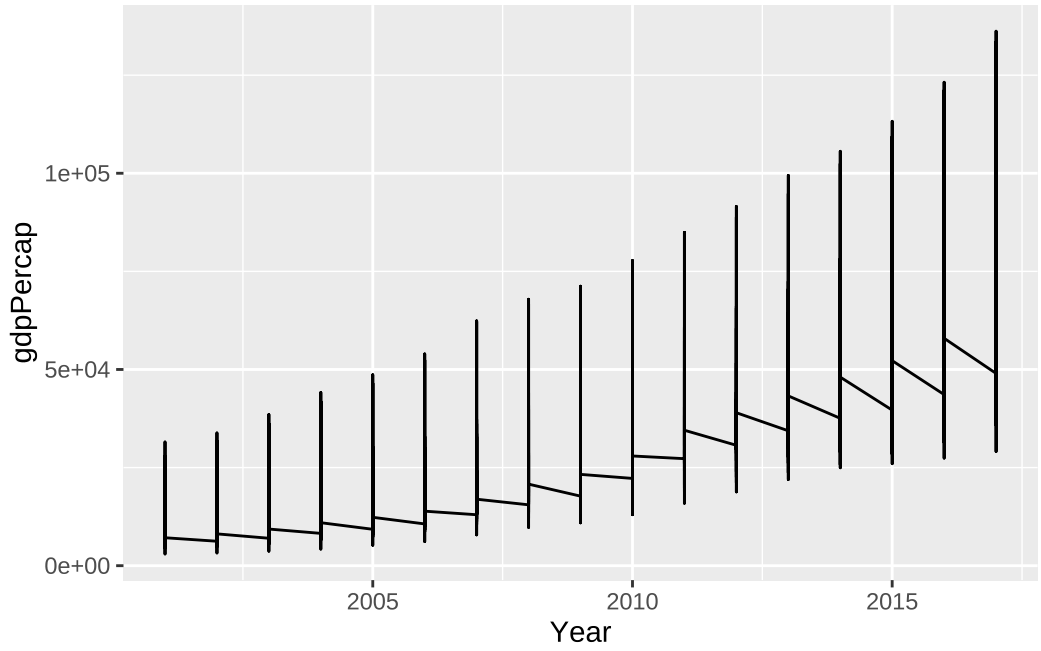
例：针对分省数据绘制人均 GDP 与年份之间的折线图，每个年份都对应 31 个省份人均 GDP 值

```
load("../slides2024/Lecture2/data/ecostats.rda")
ecostats
```

```
# A tibble: 527 x 7
   Region  Year Electricity Investment Consumption Population gdpPercap
   <chr>  <int>       <dbl>      <dbl>       <dbl>      <dbl>     <dbl>
 1 安徽    2001        360.       893.        2739       6128     5716.
 2 北京    2001        400.      1513.        9057       1385    27881.
 3 福建    2001        439.      1173.        4770       3445    11823.
 4 甘肃    2001        306.       460.        2099       2523     4461.
```

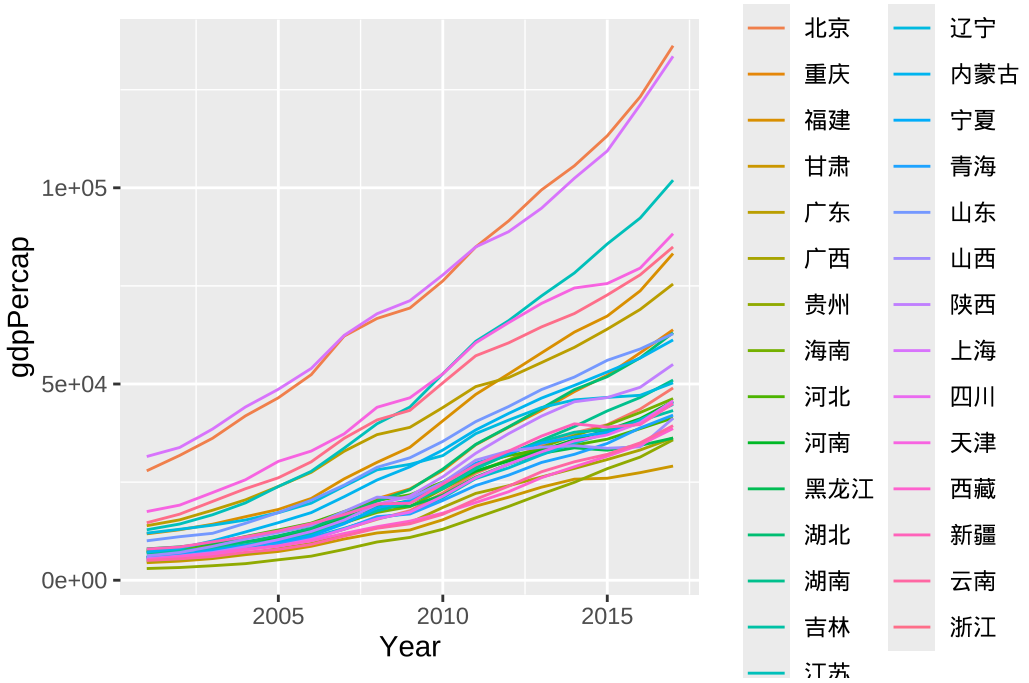
```
 5 广东     2001      1458.       3484.        5445       8733     13886.
 6 广西     2001       332.        656.        2572       4788      4760.
 7 贵州     2001       335.        536.        2178       3799      2983.
 8 海南     2001        43.0       213.        2971        796      7276.
 9 河北     2001       868.       1913.        2749       6699      7558.
10 河南     2001       808.       1544.        2381       9555      5791.
# i 517 more rows
```

```
ggplot(data = ecostats, aes(x = Year, y = gdpPercap)) +
geom_line()
```



- 用分组操作实现对不同省分的变量情况刻画

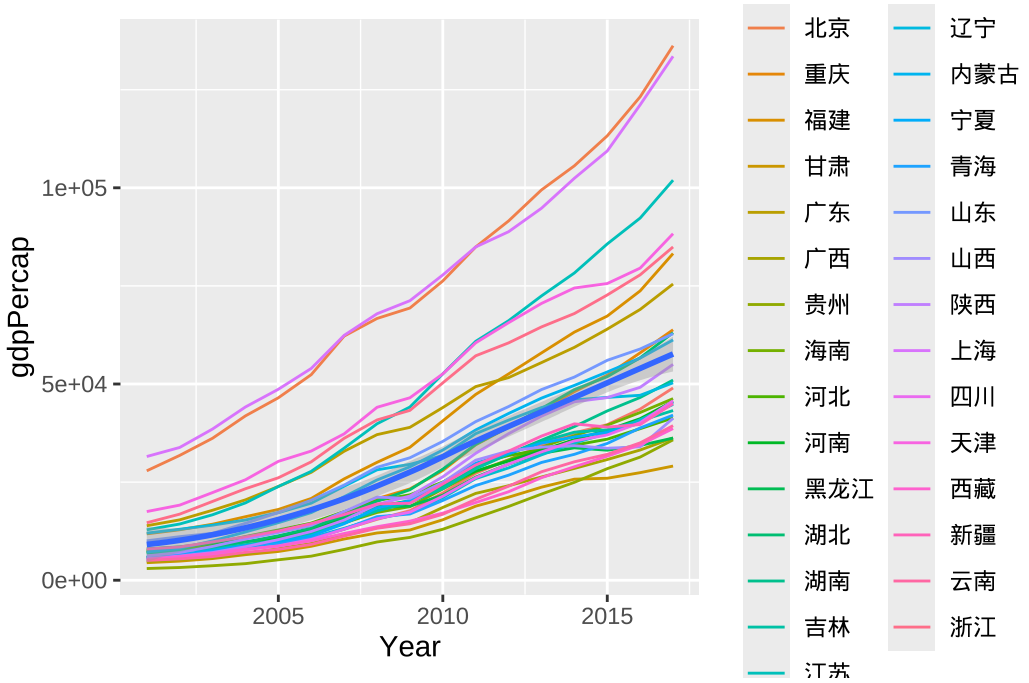
```
ggplot(data = ecostats, aes(x = Year, y = gdpPercap)) +
geom_line(aes(color = Region))
```



- 添加平滑线

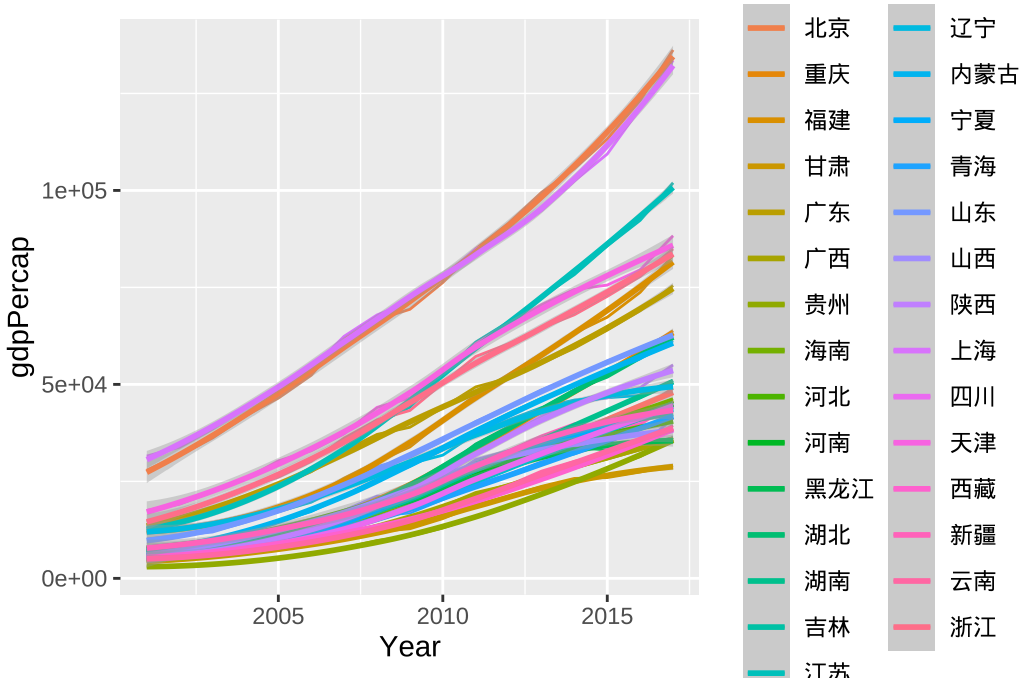
```
ggplot(data = ecostats, aes(x = Year, y = gdpPercap)) +
geom_line(aes(color = Region)) +
geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



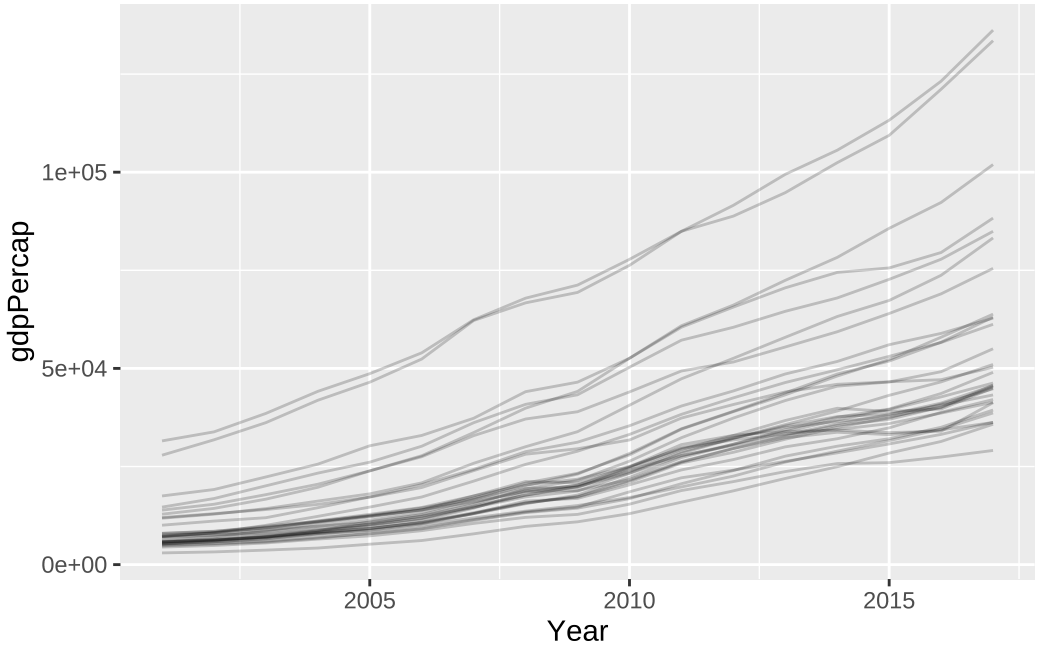

```
ggplot(data = ecostats, aes(x = Year, y = gdpPercap,color = Region)) +
geom_line() +
geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

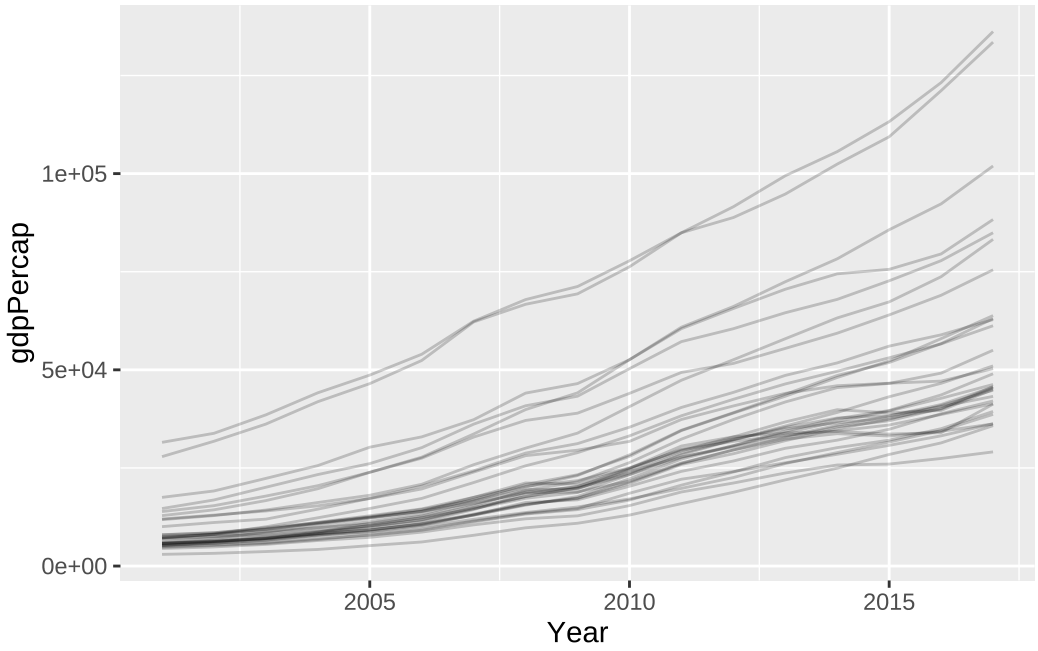


分组图形特征映射

```
ggplot(data = ecostats, aes(x = Year, y = gdpPercap)) +
geom_line(aes(group = Region), alpha = 0.2)
```



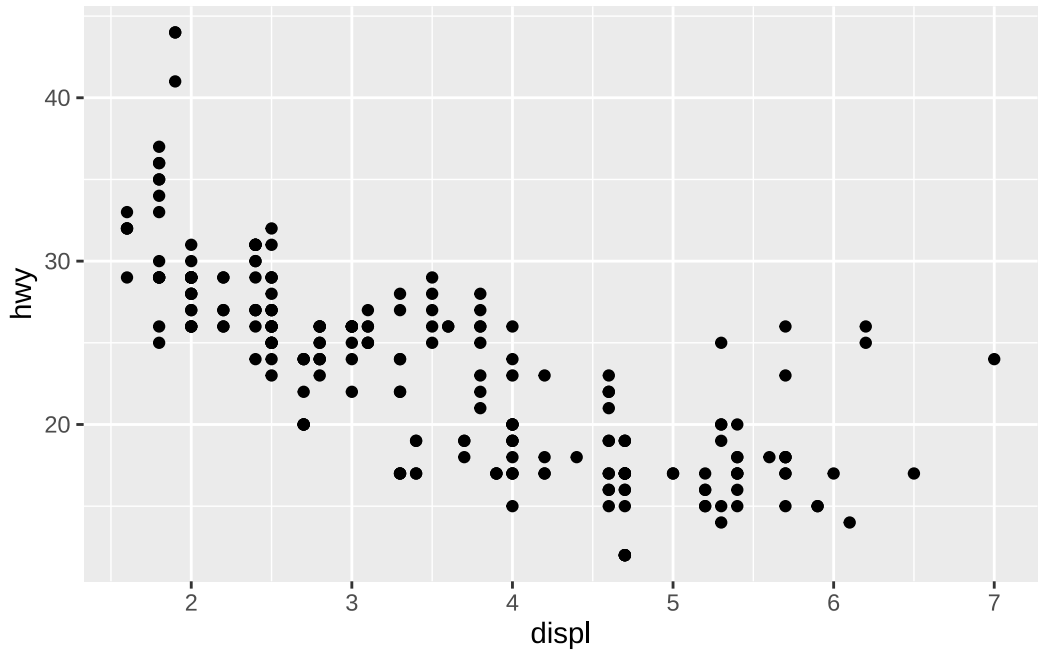
```
ggplot(data = ecostats, aes(x = Year, y = gdpPercap, group=Region)) +
geom_line(alpha = 0.2)
```



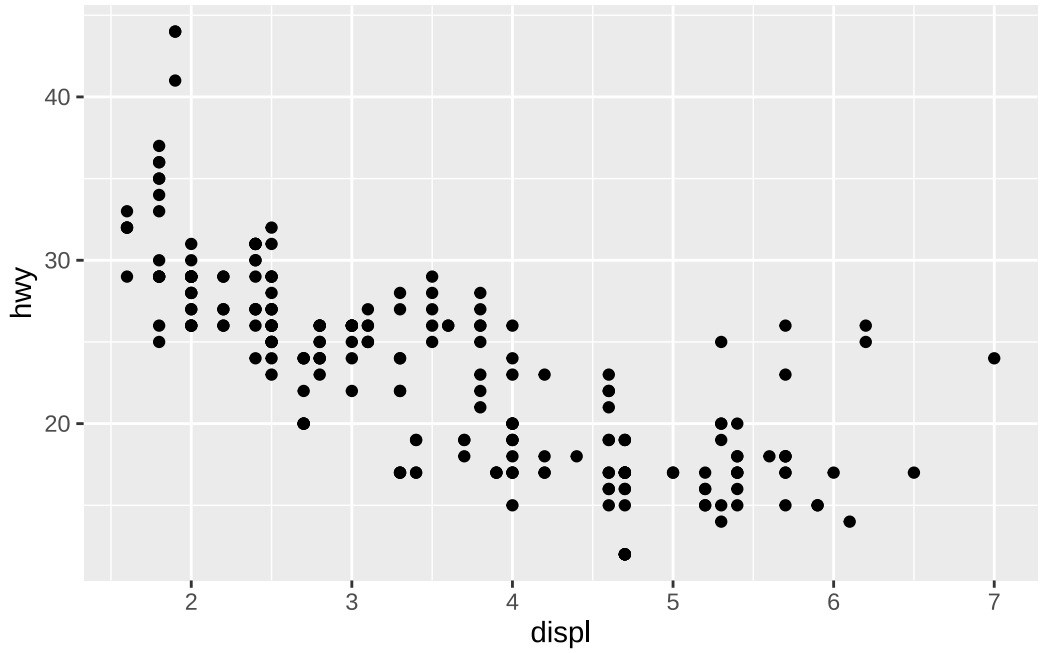
4.3 标度

4.3.1 修改坐标轴标签

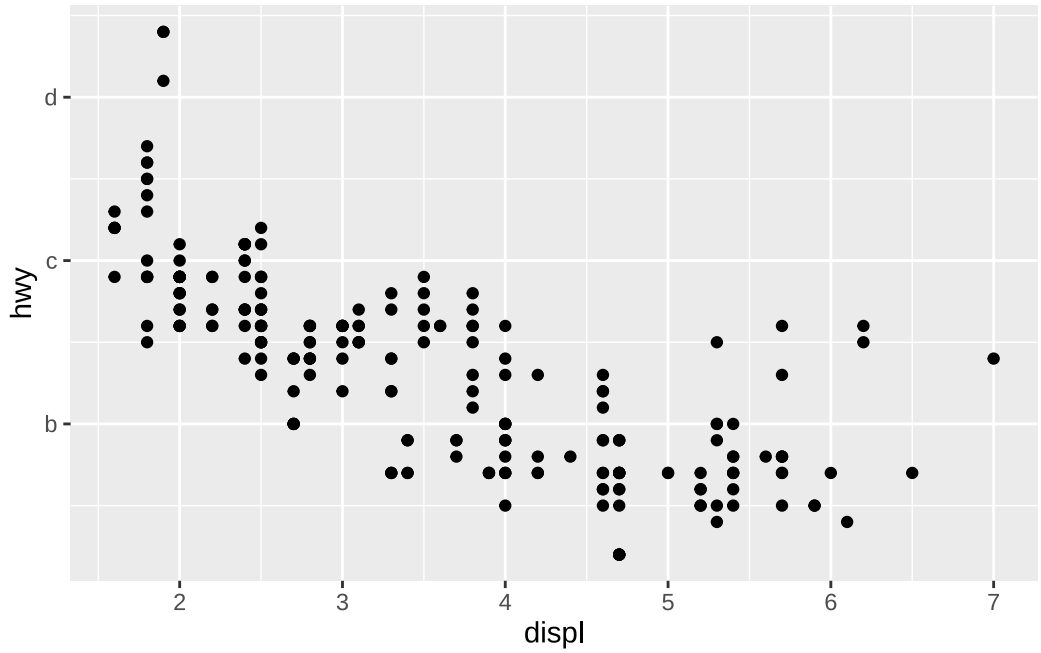
```
p1 <- ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point()
p1
```




```
p1 + scale_y_continuous(breaks = seq(10,40,by=10))
```



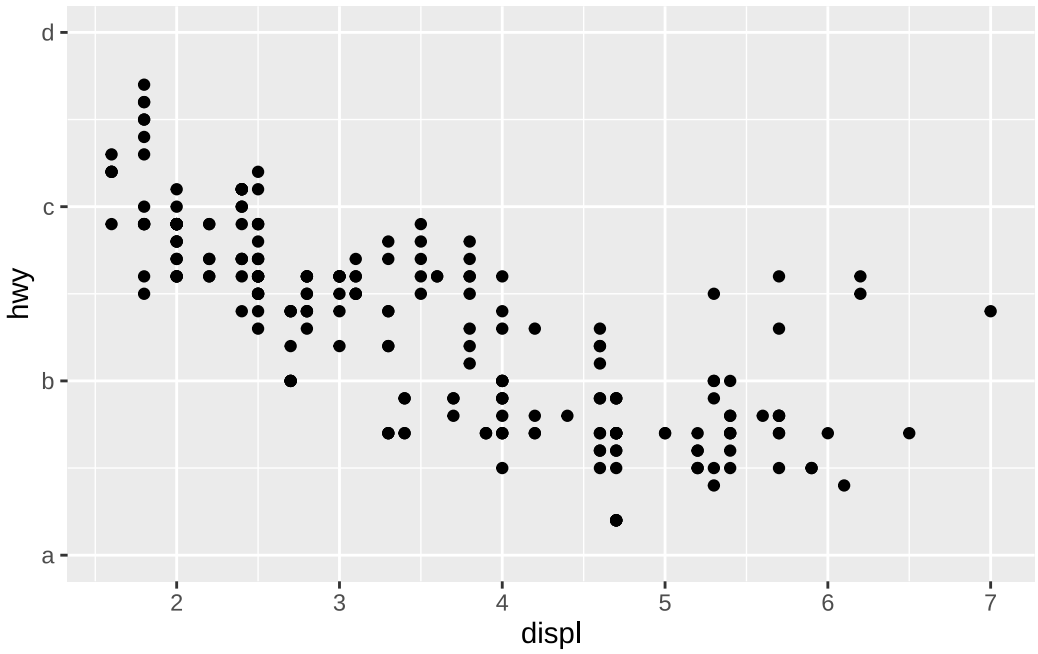
```
p1 + scale_y_continuous(breaks = seq(10,40,by=10), labels = letters[1:4])
```



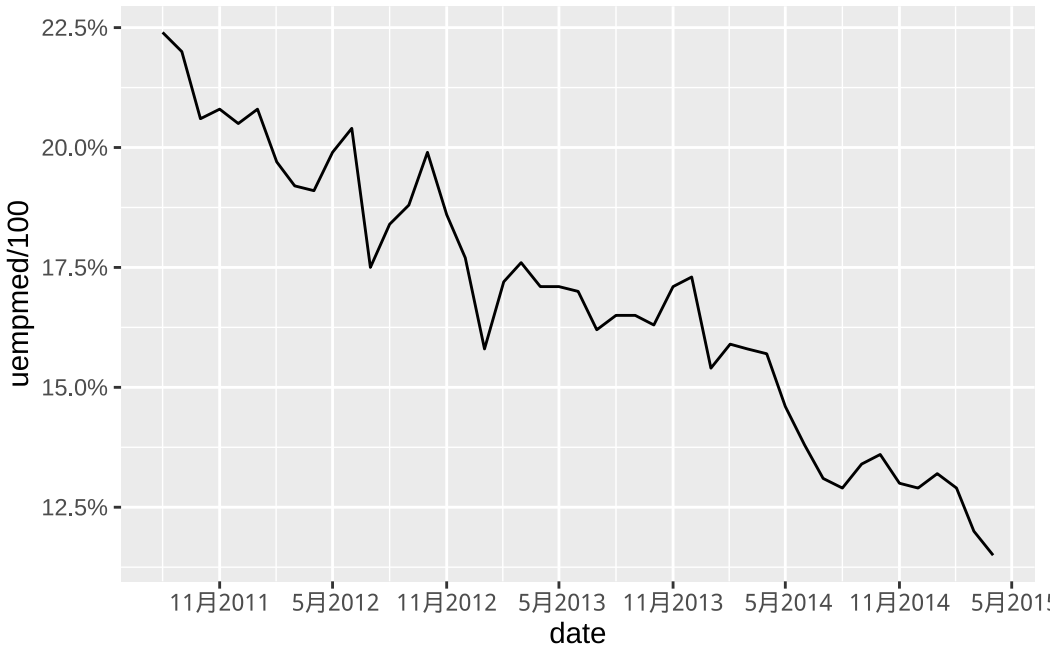
- 思考：为什么上图没有显示字母 a?

```
p1 + scale_y_continuous(limits = c(10, 40), breaks = seq(10,40,by=10), labels = letters[1:4])
```

```
Warning: Removed 3 rows containing missing values or values outside the scale range
(`geom_point()`).
```

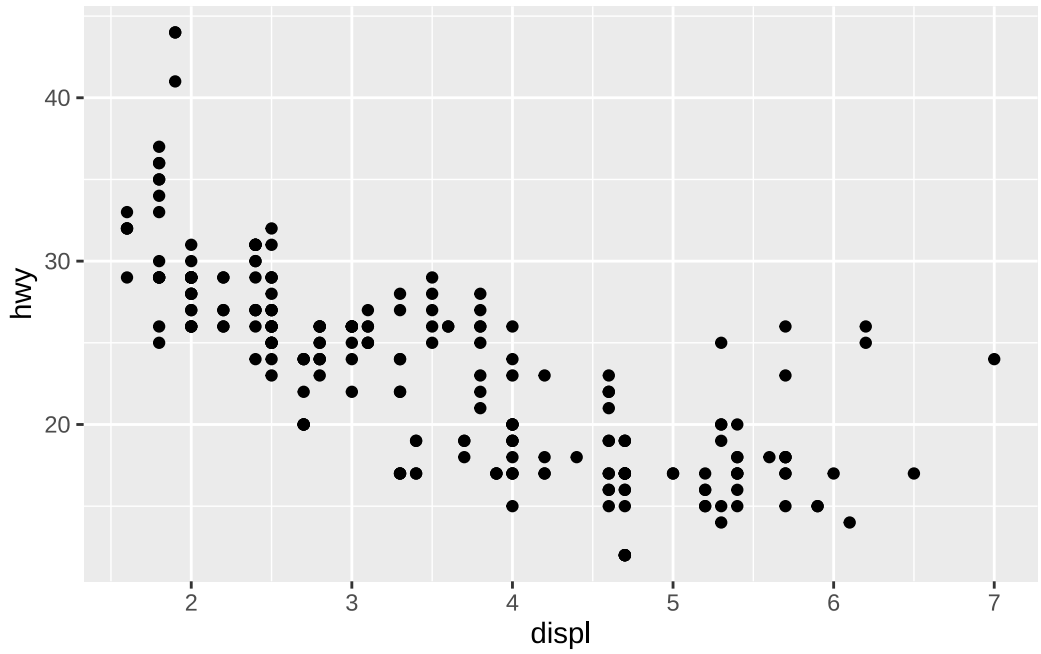



```
p <- ggplot(tail(economics,45), aes(date, uempmed/100)) +
geom_line()+
scale_x_date(date_breaks = "6 months", date_labels = "%b%Y") +
scale_y_continuous(labels = scales::percent)
p
```

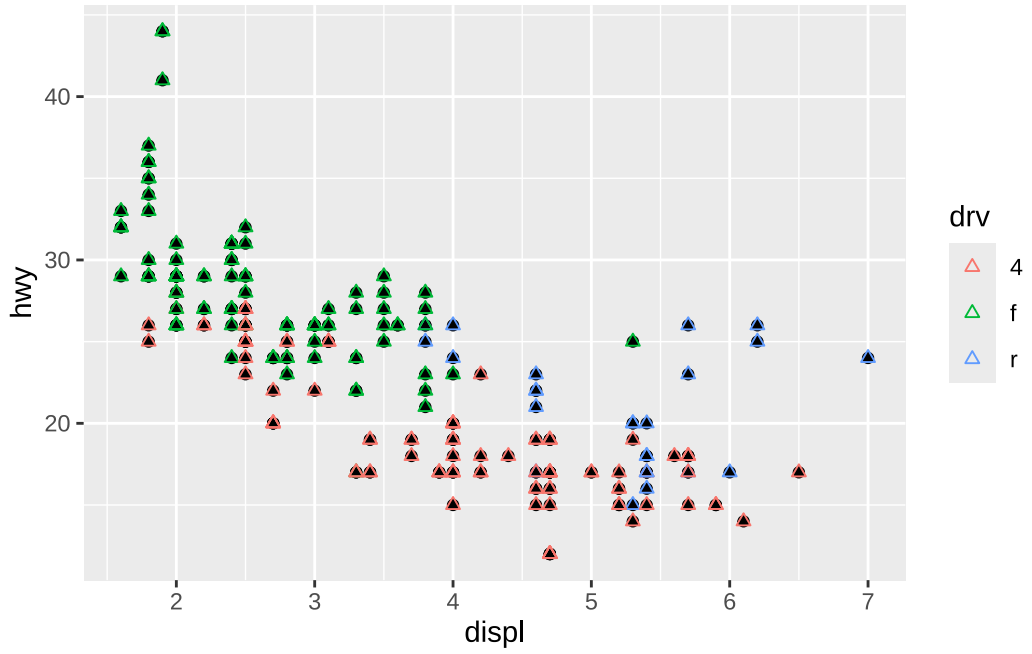


4.3.2 修改坐标轴名称

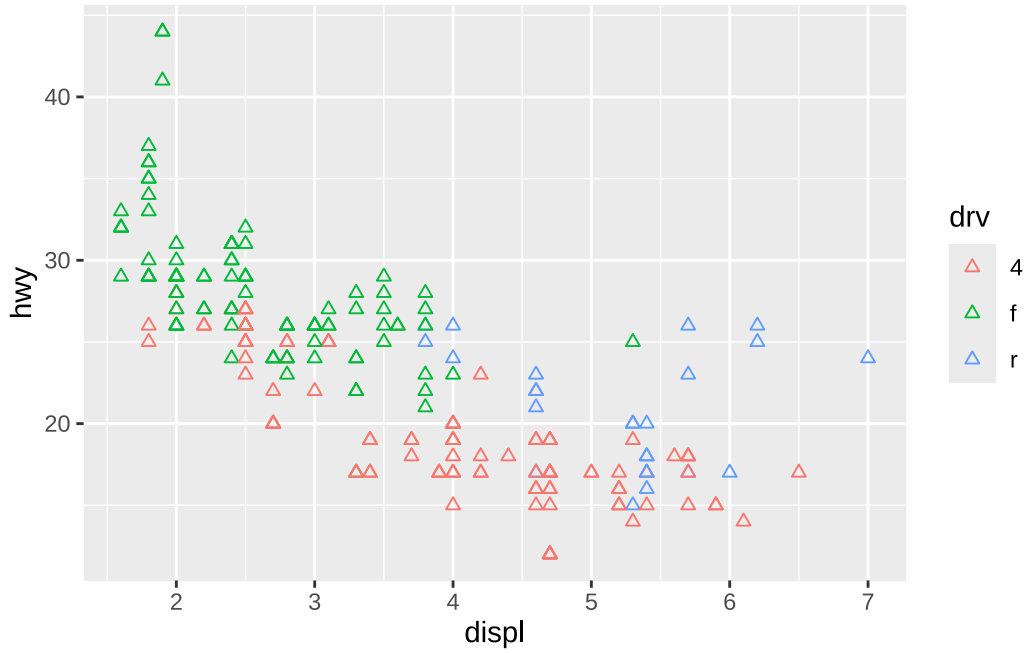
```
p1 <- ggplot(data = mpg, aes(x = displ, y = hwy))
p2 <- p1 + geom_point()
p2
```



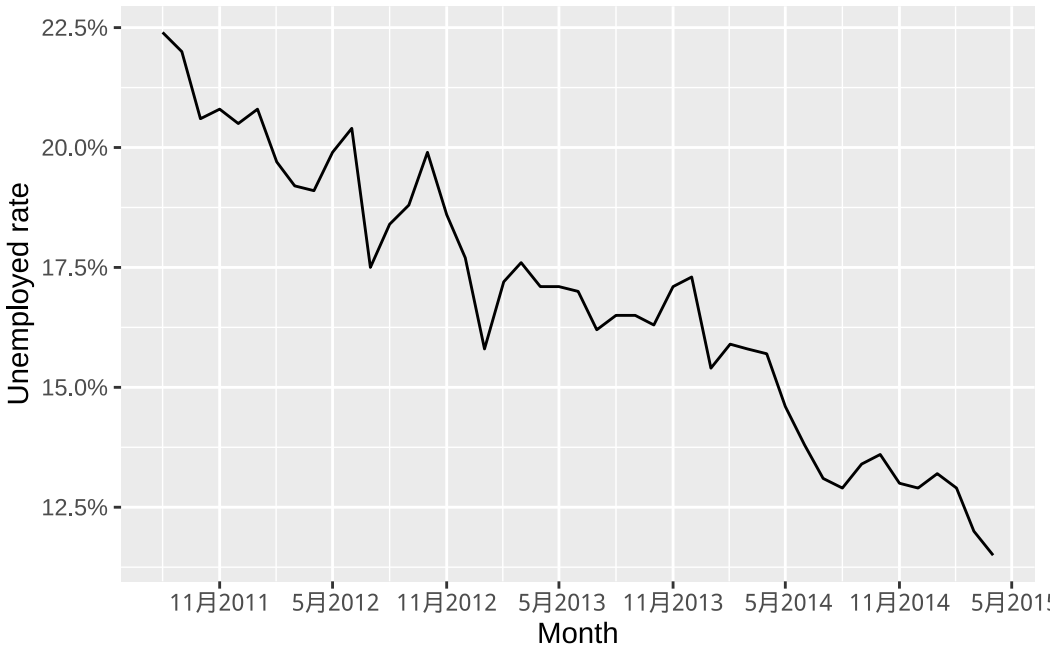

```
p2 +
geom_point(aes(color = drv), shape = 2)
```



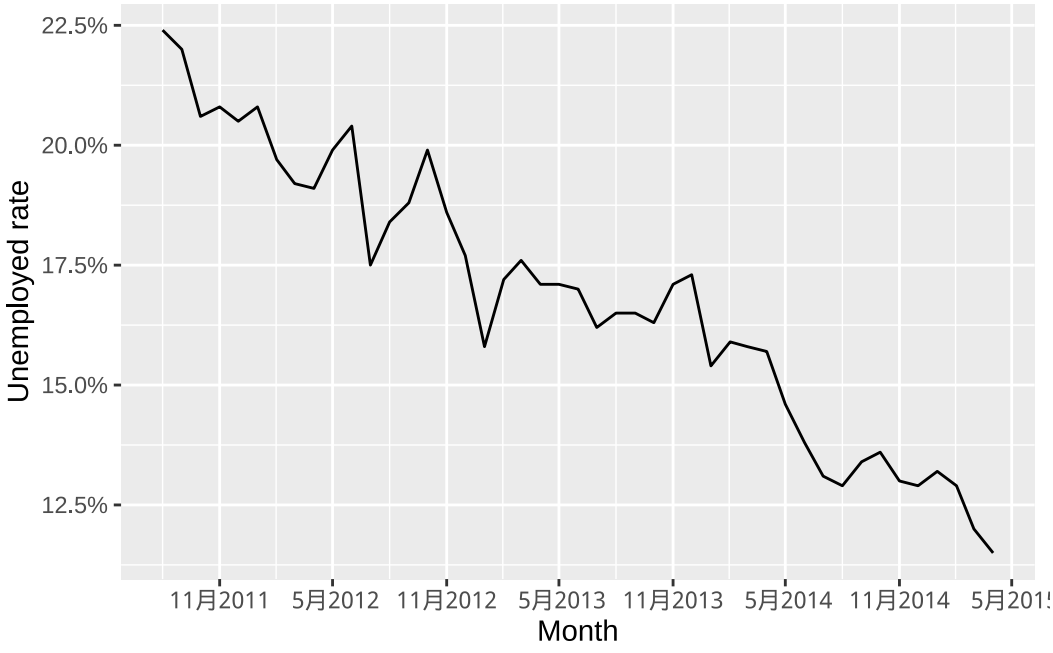
```
p1 +
geom_point(aes(color = drv), shape = 2)
```



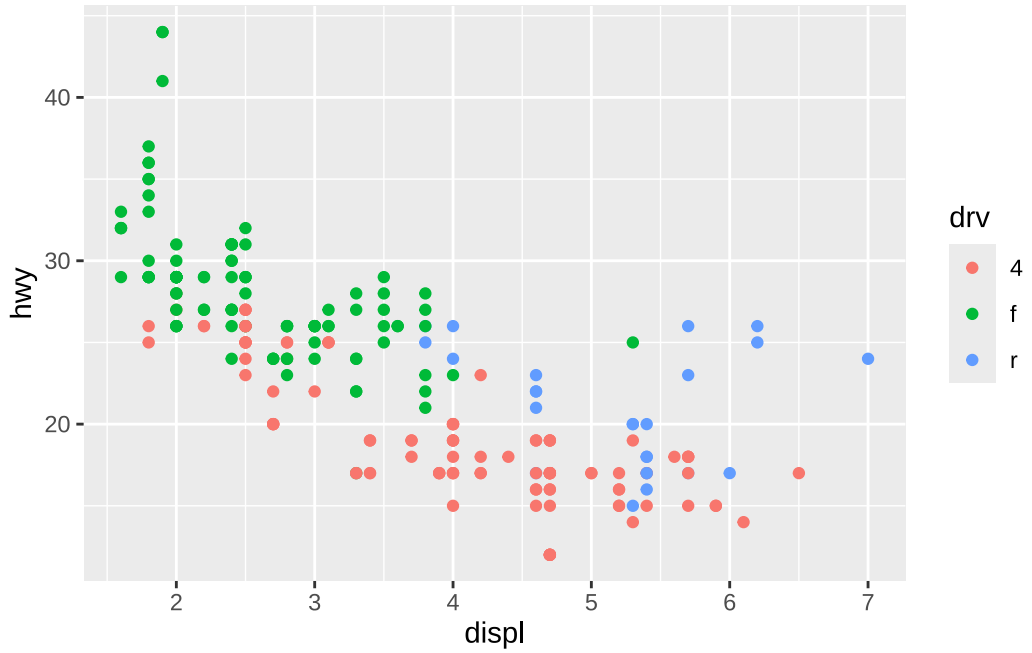

```
p1 <- p +
labs(x = "Month", y = "Unemployed rate")
p1
```




```
p2 <- p +
xlab("Month") +
ylab("Unemployed rate")
p2
```

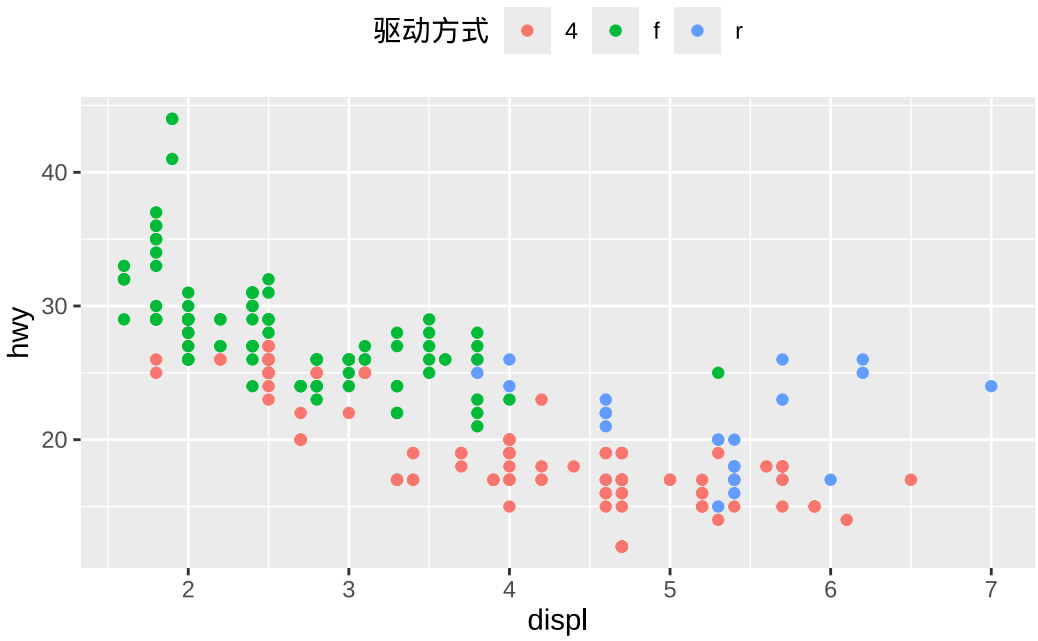


```
pMpg = ggplot(data = mpg, aes(x = displ, y = hwy))
p1 = pMpg + geom_point(aes(color = drv))
p1
```



- 若前面已使用 color 参数，则可以在 labs() 函数中使用参数 color 修改颜色的图例名称

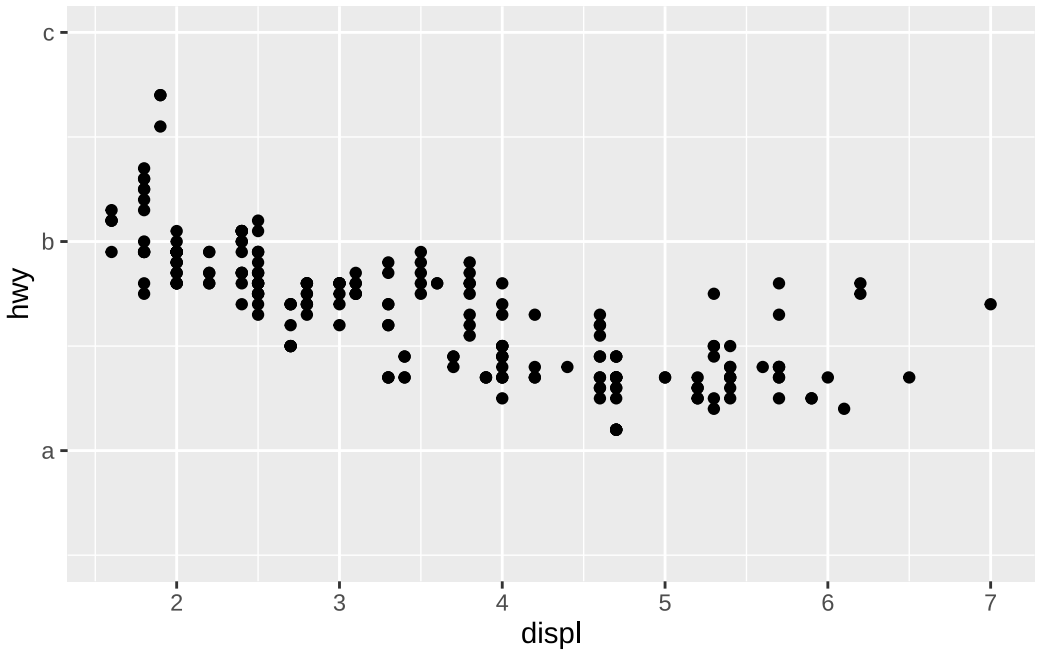
```
p2 = p1 +
labs(color = " 驱动方式") +
theme(legend.position = "top")
p2
```



4.3.3 设置坐标轴范围

- coord_cartesian 命令
- 笛卡尔坐标系就是直角坐标系和斜角坐标系的统称

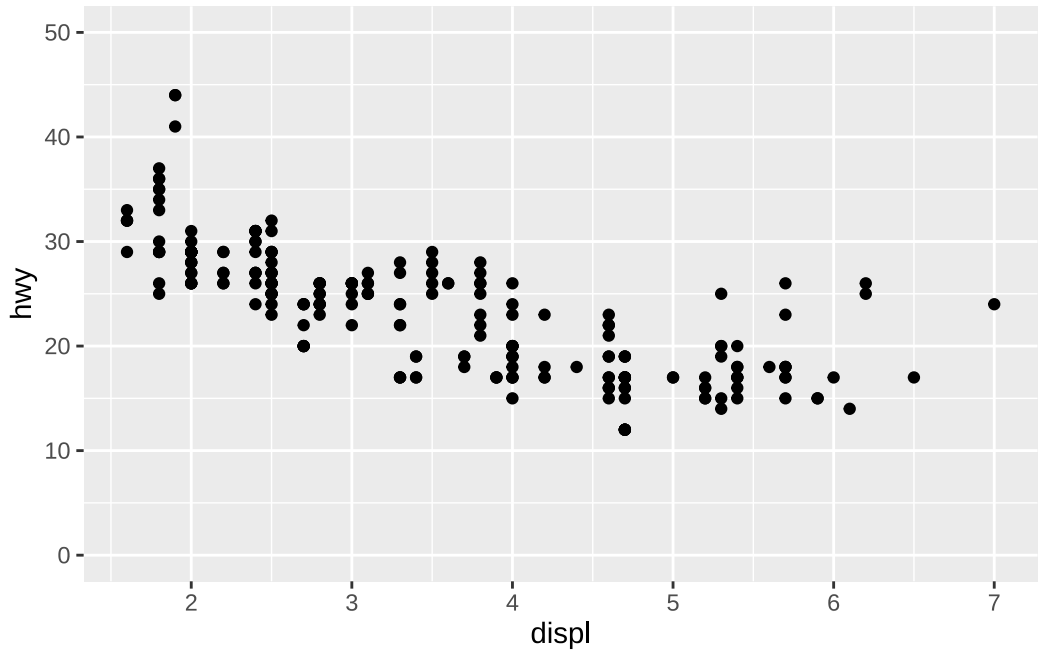
```
ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point() +
scale_y_continuous(breaks = seq(10,50,by=20), labels = letters[1:3])+
coord_cartesian(ylim = c(0,50))
```



ylim 后不能显示 scale 设置的 breaks 点的 label

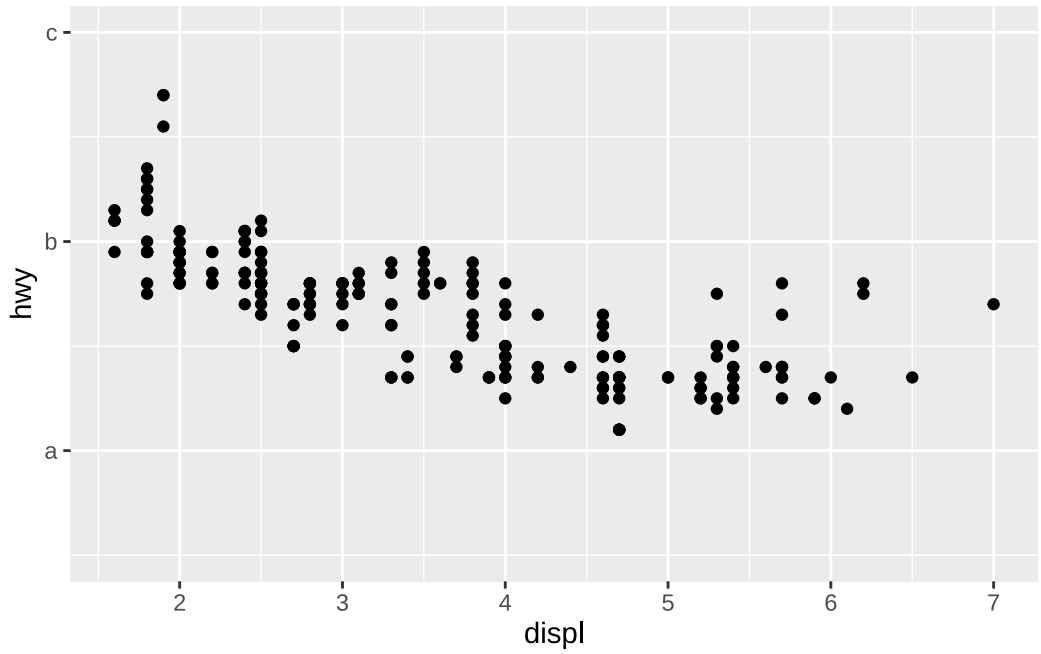
```
ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point() +
scale_y_continuous(breaks = seq(10,50,by=20), labels = letters[1:3])+
ylim(0,50)
```

```
Scale for y is already present.
Adding another scale for y, which will replace the existing scale.
```



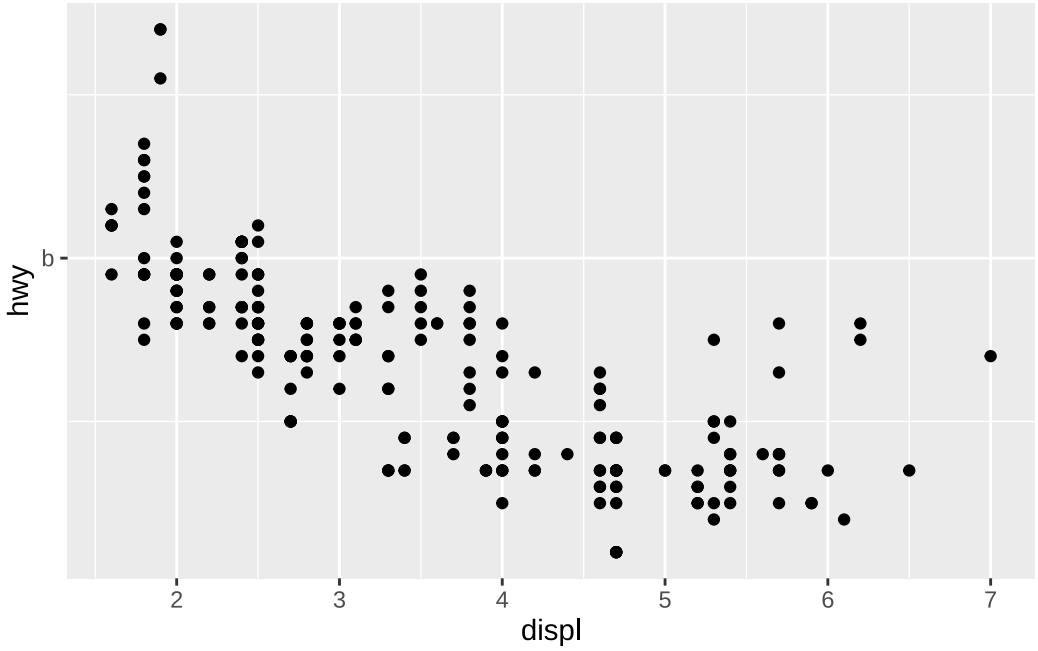
- 体会一下 scale 与 ylim 设置顺序的不同，结果的不同

```
ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point() +
coord_cartesian(ylim = c(0,50))+
scale_y_continuous(breaks = seq(10,50,by=20), labels = letters[1:3])
```




```
ggplot(data = mpg, aes(x = displ, y = hwy)) +
geom_point() +
ylim(0,50) +
scale_y_continuous(breaks = seq(10,50,by=20), labels = letters[1:3])
```

```
Scale for y is already present.
Adding another scale for y, which will replace the existing scale.
```

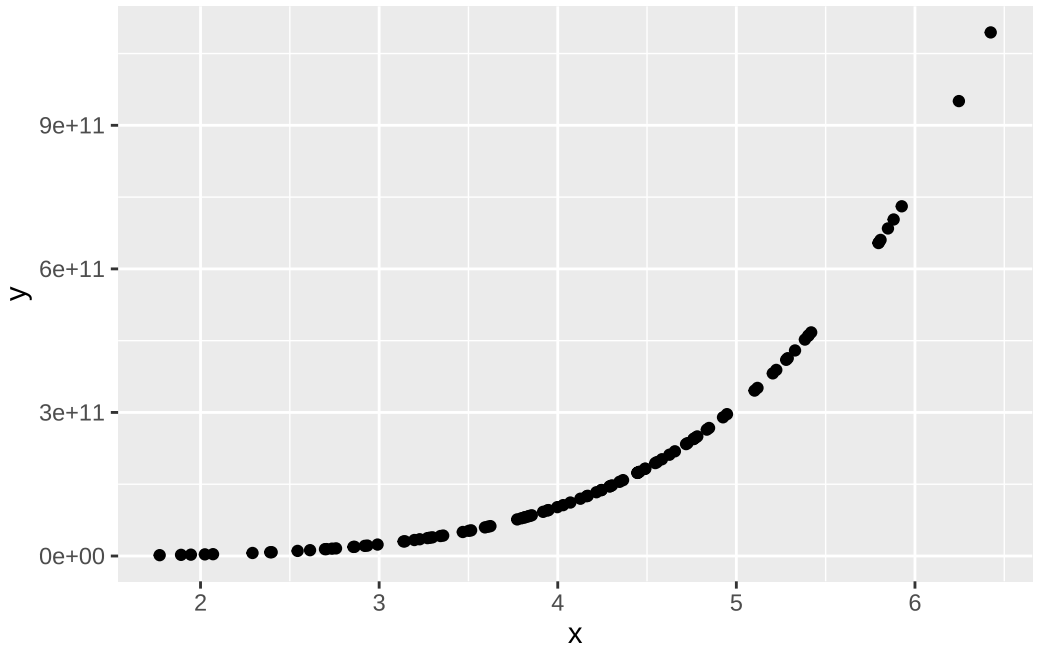


4.3.4 变换

```
n=100
x=rnorm(n,4,1)
y=10^(5*log10(x)+8)+rnorm(n,1,2)
```

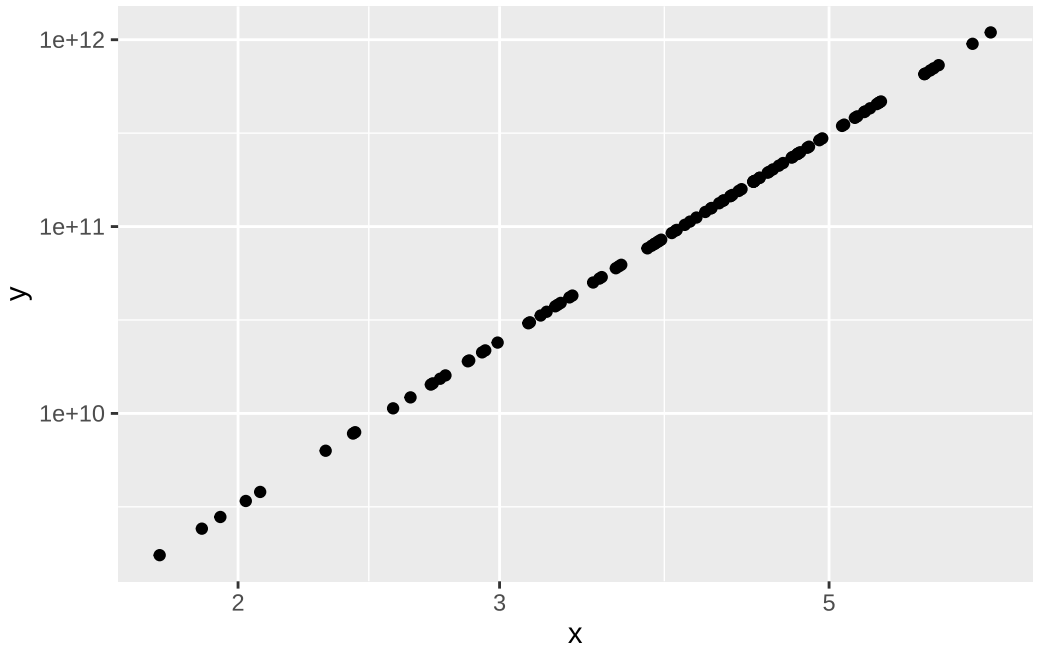
- 画散点图

```
dataxy = tibble(x,y)
p1 <- ggplot(data = dataxy, aes(x,y)) +
geom_point()
p1
```



对数变换之后作散点图

```
p2 <- p1 +
scale_x_log10() +
scale_y_log10()
p2
```



- 使用自定义的函数会报错

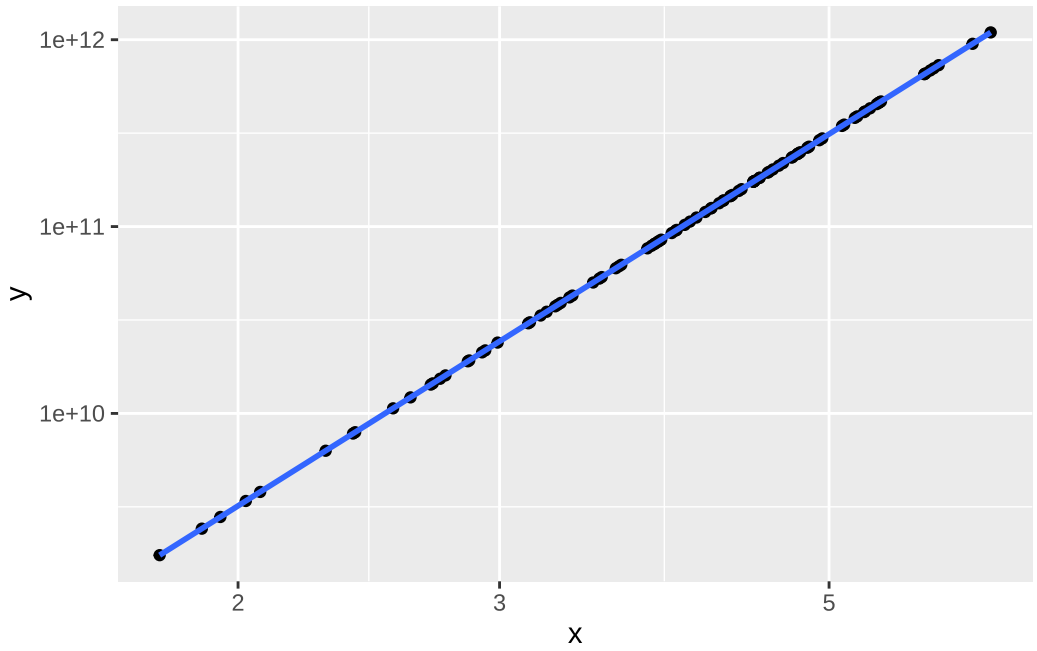
```
mylog <- function(x){
  return(log10(x))
}
```

```
p2 <- p1 +
scale_x_mylog() +
scale_y_mylog()
p2
```

进行回归分析

```
p2 <- p1 +
scale_x_log10() +
scale_y_log10() +
geom_smooth()
p2
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



使用 deepseek 代码

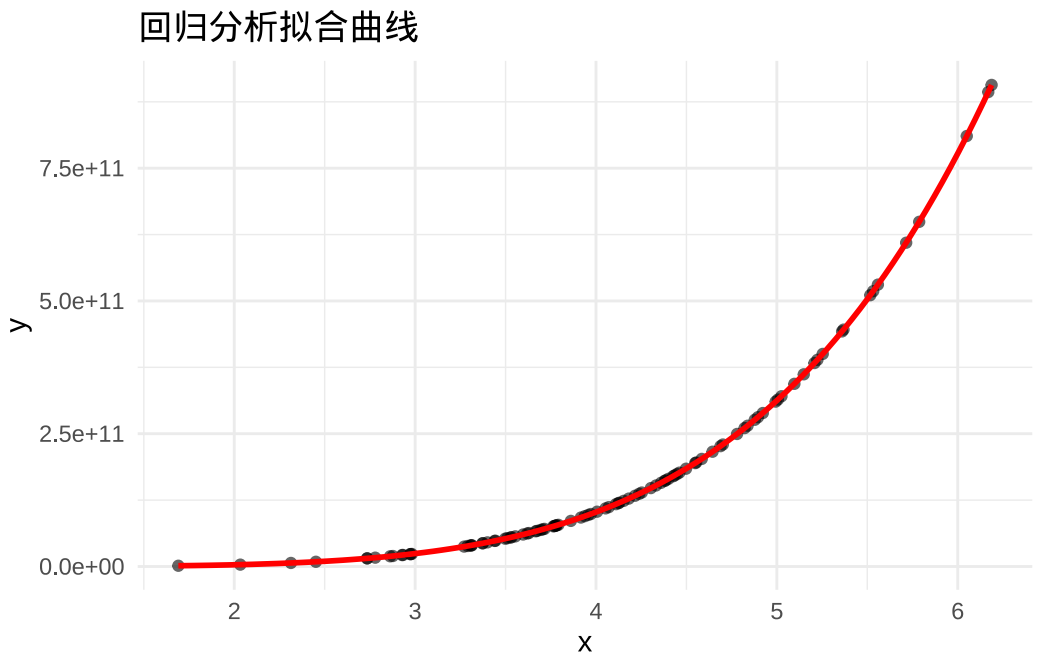
```r
# 生成数据
n <- 100
set.seed(123)  # 设置随机种子以确保可重复性
x <- rnorm(n, mean = 4, sd = 1)
y <- 10^(5 * log10(x) + 8) + rnorm(n, mean = 1, sd = 2)
df <- data.frame(x, y)

# 拟合对数转换后的线性模型
model <- lm(log10(y) ~ log10(x), data = df)

# 生成预测值
x_range <- seq(min(x), max(x), length.out = 100)
predicted_log_y <- predict(model, newdata = data.frame(x = x_range))
predicted_y <- 10^predicted_log_y
pred_df <- data.frame(x = x_range, y = predicted_y)

# 使用 ggplot2 绘图
```

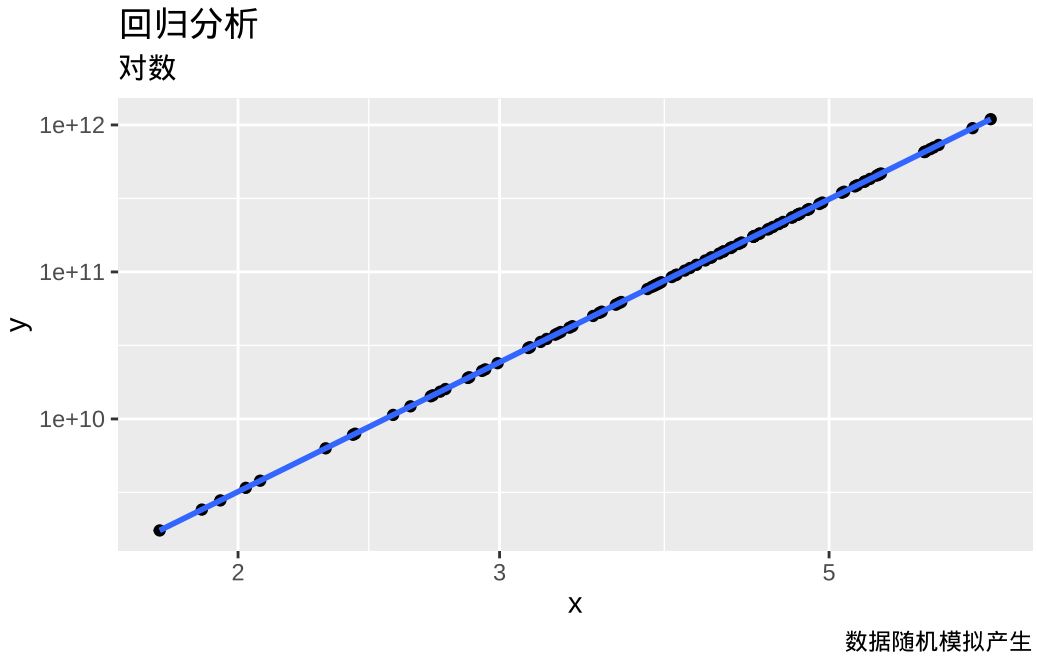
```r
library(ggplot2)
ggplot(df, aes(x = x, y = y)) +
  geom_point(alpha = 0.6) +  # 绘制原始数据点
  geom_line(data = pred_df, aes(x = x, y = y), color = "red", linewidth = 1) +  # 绘制拟合曲线
  labs(title = " 回归分析拟合曲线", x = "x", y = "y") +
  theme_minimal()
```



4.3.5 设置图形标题与副标题

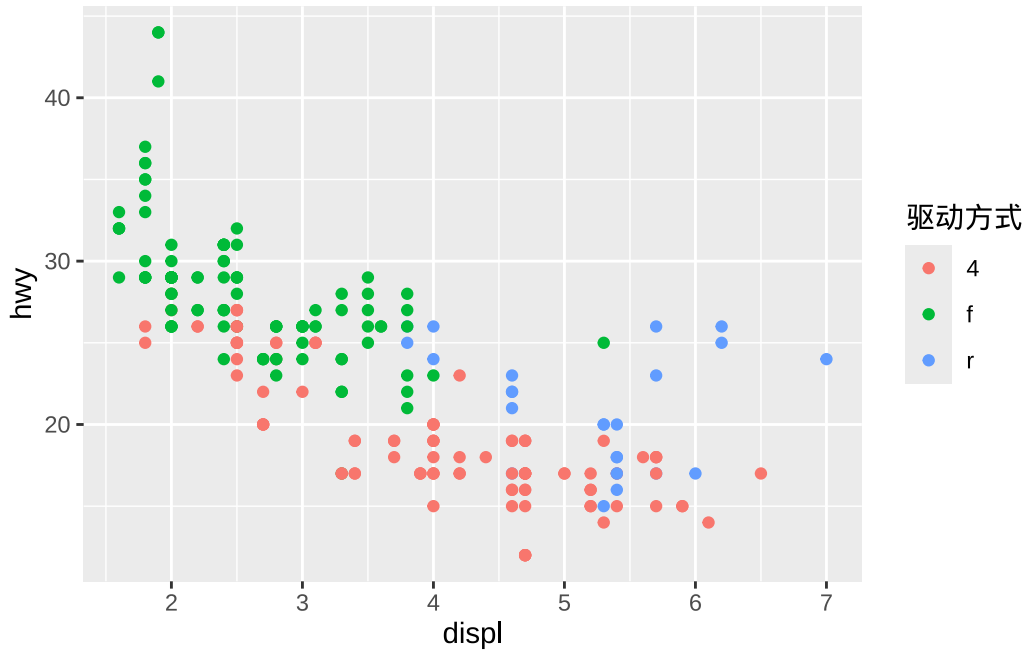
```r
p2 +
labs(title = " 回归分析",
subtitle = " 对数",
caption = " 数据随机模拟产生")
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

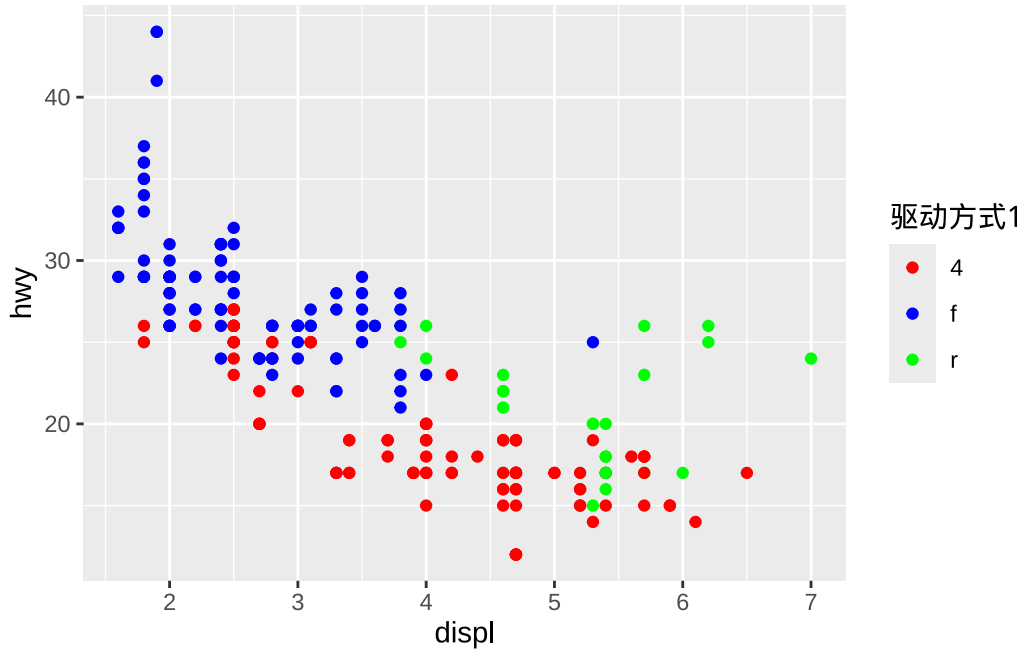


4.3.6 颜色配置

```
p1 <- ggplot(data = mpg, aes(x = displ, y = hwy, color = drv)) +
geom_point() +
labs(color = " 驱动方式")
p1
```

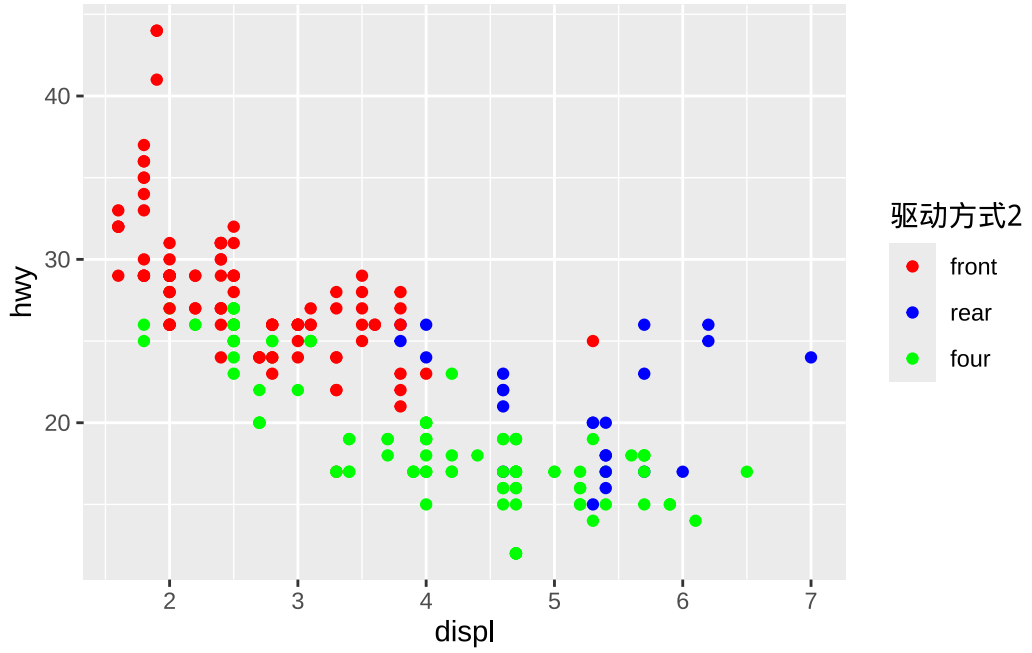



```
p1 +
scale_color_manual(" 驱动方式 1",
values = c("red", "blue", "green"),
labels = c("4", "f", "r"))
```

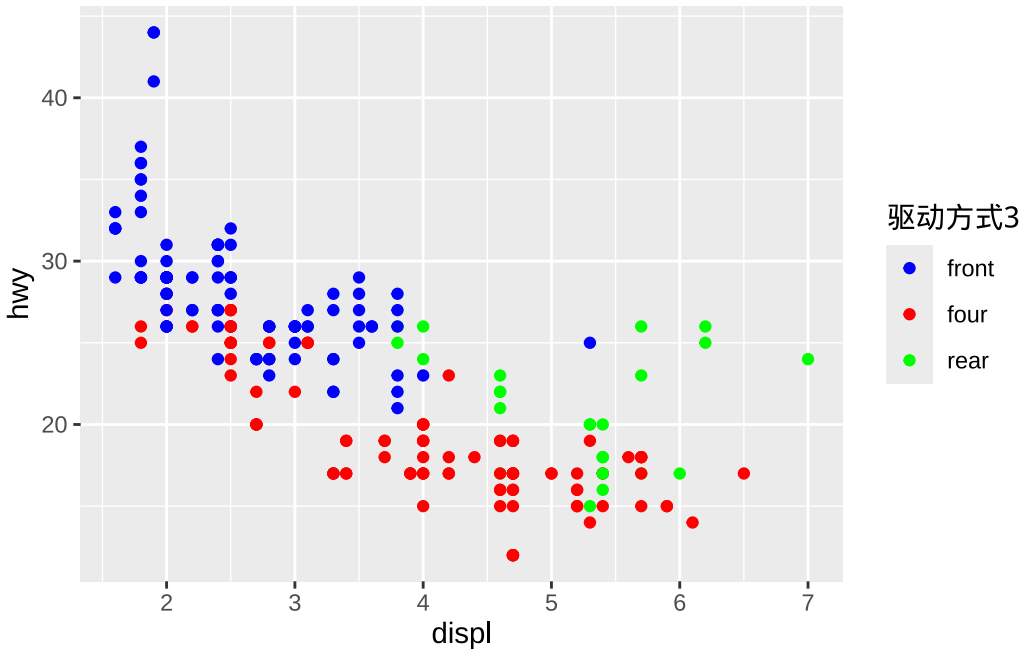


按什么顺序排的颜色?

```
p1 +
scale_color_manual(" 驱动方式 2",
breaks = c("f", "r", "4"),
values = c("red", "blue", "green"),
labels = c("front", "rear", "four"))
```



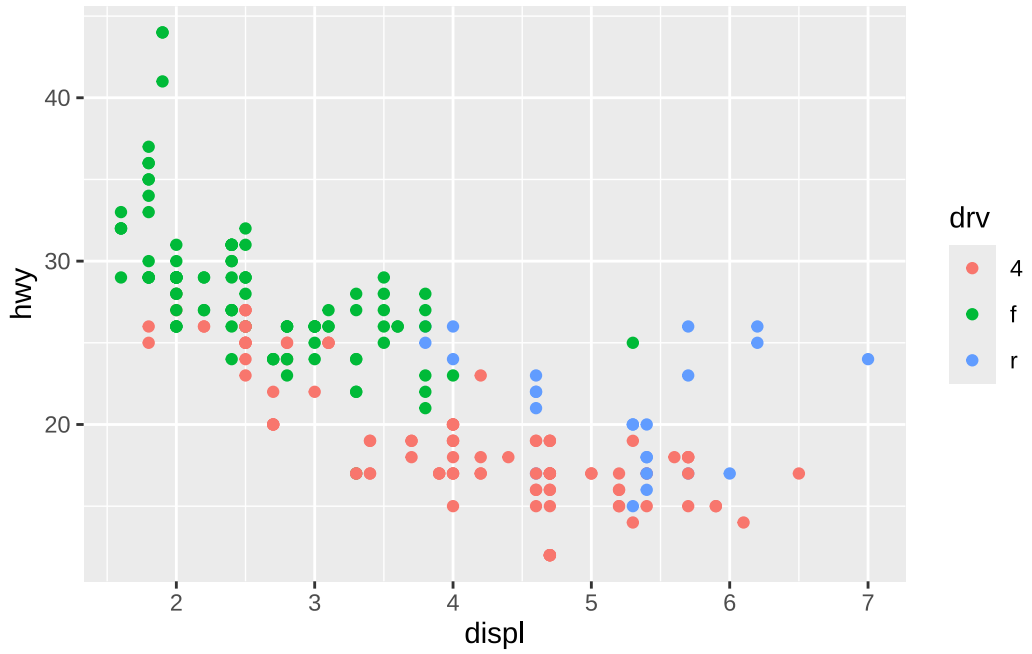

```
p1 +
scale_color_manual(" 驱动方式 3",
breaks = c("f", "4", "r"),
values = c("blue", "red", "green"),
labels = c("front", "four", "rear"))
```



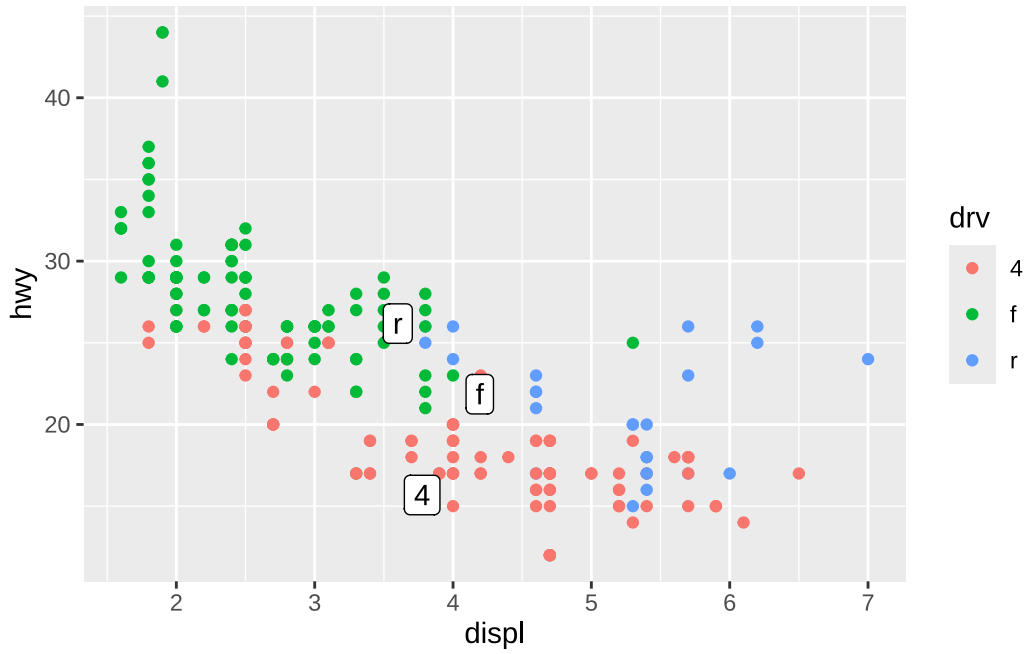
4.3.7 文字标注

- 从不同的 drv 中随机选一个，作标注

```
p1<-ggplot(mpg,aes(displ, hwy))+geom_point(aes(color = drv))
p1
```




```
library(ggrepel)
id1 = sample(which(mpg$drv=="r"),1)
id2 = sample(which(mpg$drv=="f"),1)
id3 = sample(which(mpg$drv=="4"),1)
dataMark = mpg[c(id1,id2,id3),]
p1 +
geom_label_repel(data = dataMark, aes(label = drv))
```



- 从不同 drv 中选取 hwy 最大的进行标注

  deepseek 代码

- 找出每一组 hwy 最大的

```
max_hwy <- mpg %>%
group_by(drv) %>%
slice_max(hwy, n = 1)
max_hwy
```

```
# A tibble: 7 x 11
# Groups:   drv [3]
  manufacturer model      displ  year   cyl trans  drv     cty   hwy fl    class
  <chr>        <chr>      <dbl> <int> <int> <chr>  <chr> <int> <int> <chr> <chr>
1 audi         a4 quattro   2    2008     4 manua~ 4        20    28 p     comp~
2 volkswagen   jetta        1.9  1999     4 manua~ f        33    44 d     comp~
3 volkswagen   new beetle   1.9  1999     4 manua~ f        35    44 d     subc~
4 chevrolet    corvette     5.7  1999     8 manua~ r        16    26 p     2sea~
5 chevrolet    corvette     6.2  2008     8 manua~ r        16    26 p     2sea~
```

```
6 ford         mustang      3.8  1999     6 manua~ r        18    26 r     subc~
7 ford         mustang      4    2008     6 manua~ r        17    26 r     subc~
```

ungroup 函数

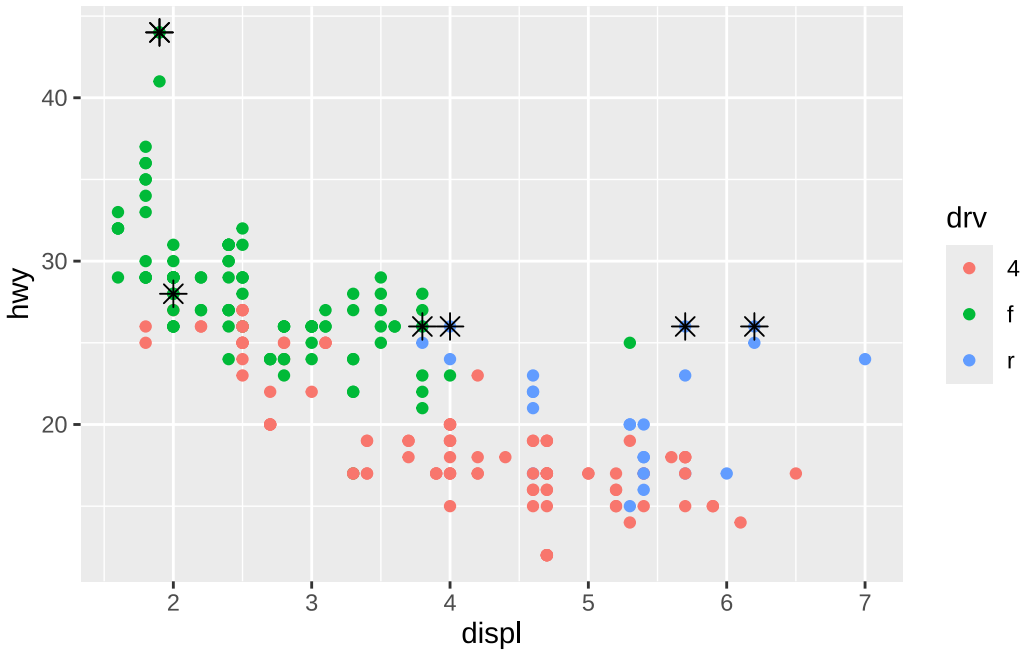
```
max_hwy_un <- max_hwy %>%
ungroup()
```

```
max_hwy %>%
group_by(class) %>%
slice_max(hwy,n=1)
```

```
# A tibble: 4 x 11
# Groups:   class [3]
  manufacturer model      displ  year   cyl trans  drv     cty   hwy fl    class
  <chr>        <chr>      <dbl> <int> <int> <chr>  <chr> <int> <int> <chr> <chr>
1 chevrolet    corvette     5.7  1999     8 manua~ r        16    26 p     2sea~
2 chevrolet    corvette     6.2  2008     8 manua~ r        16    26 p     2sea~
3 volkswagen   jetta        1.9  1999     4 manua~ f        33    44 d     comp~
4 volkswagen   new beetle   1.9  1999     4 manua~ f        35    44 d     subc~
```

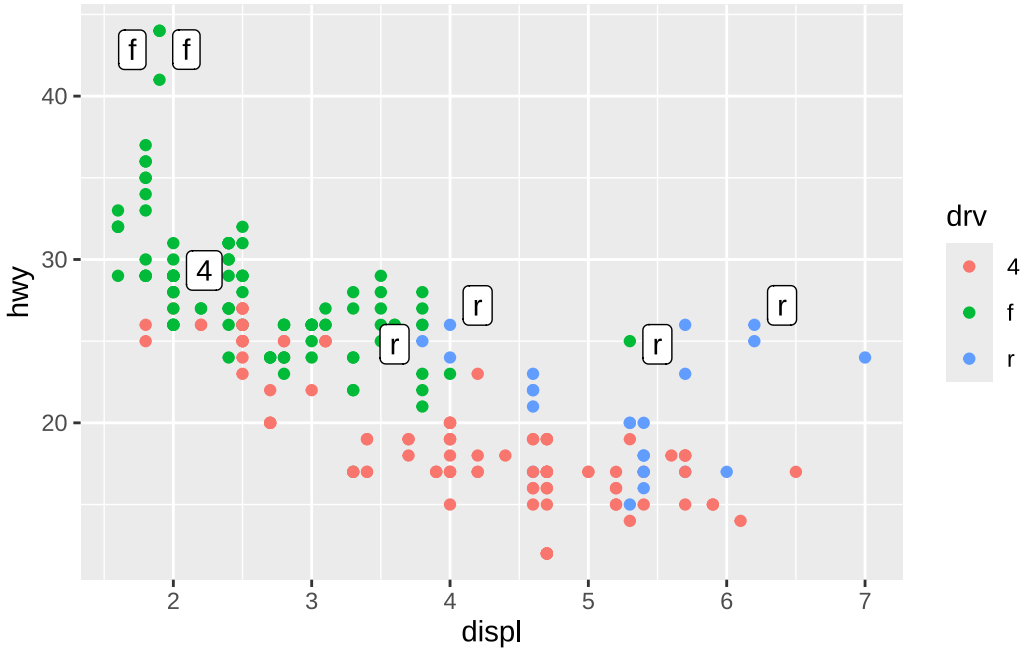
- 打标

```
p1 +
geom_point(data = max_hwy, shape = 8, size = 3)
```



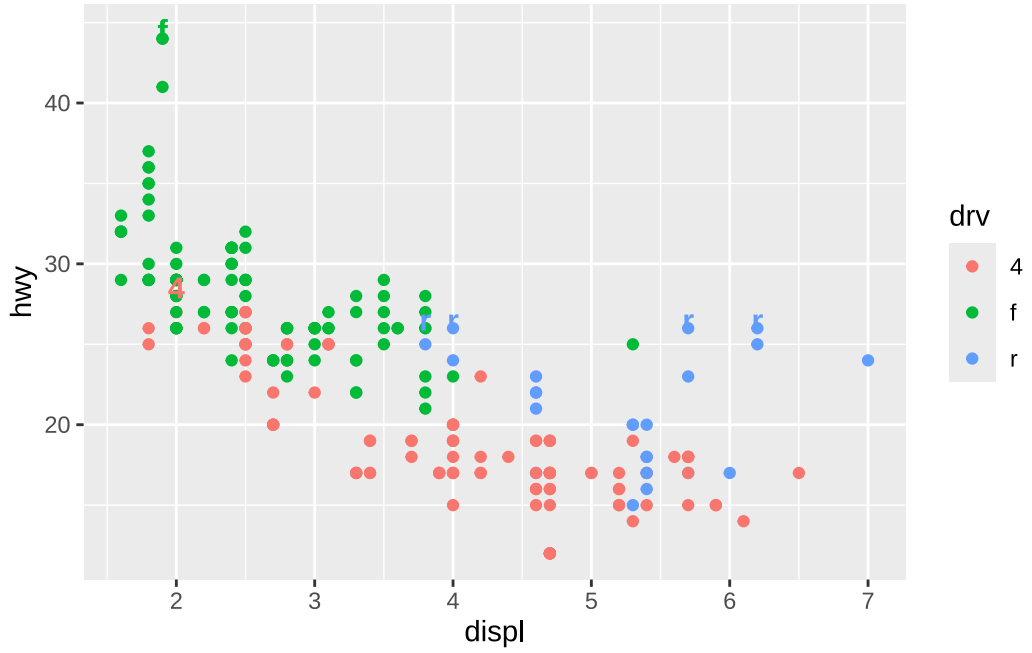
使用 ggrepel 包的 geom_label_repel 函数

```
p1 +
geom_label_repel(data = max_hwy, aes(label =drv))
```



使用 geom_text 函数

```
p1 +
geom_text(
  data = max_hwy,
  aes(label = drv, color = drv),  # 颜色与 drv 映射一致
  size = 4,
  nudge_y = 0.5,
  fontface = "bold",
  show.legend = FALSE  # 隐藏文本的图例（避免重复）
)
```



filter 函数

```
max_hwy1 <- mpg %>%
  group_by(drv) %>%
  filter(hwy == max(hwy))
max_hwy1
```

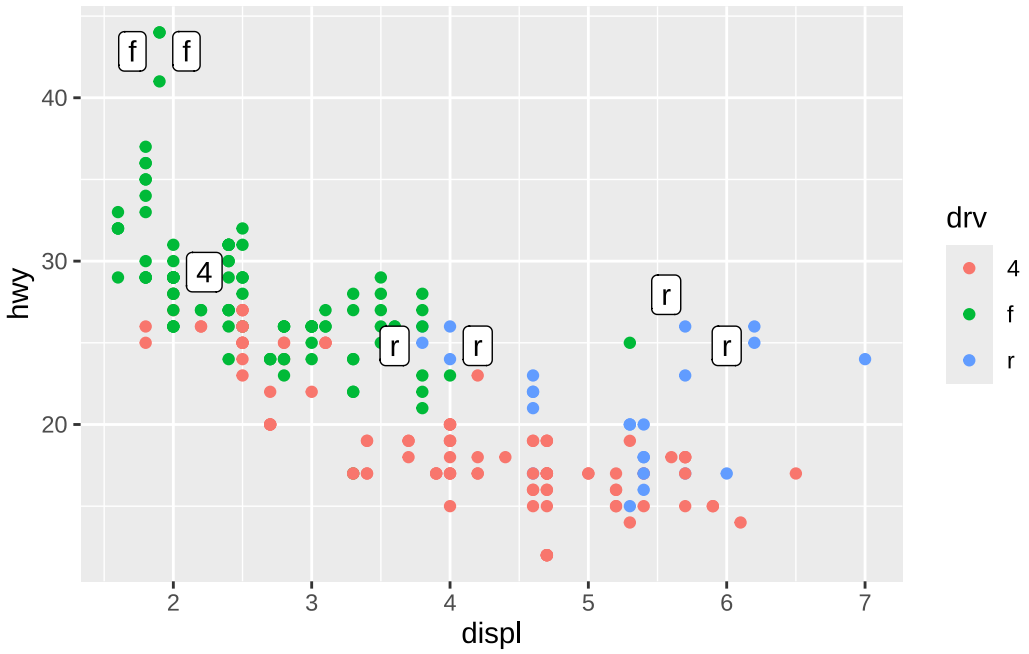
```
# A tibble: 7 x 11
```

```
# Groups:   drv [3]
  manufacturer model      displ  year   cyl trans  drv     cty   hwy fl    class
  <chr>        <chr>      <dbl> <int> <int> <chr>  <chr> <int> <int> <chr> <chr>
1 audi         a4 quattro   2    2008     4 manua~ 4        20    28 p     comp~
2 chevrolet    corvette     5.7  1999     8 manua~ r        16    26 p     2sea~
3 chevrolet    corvette     6.2  2008     8 manua~ r        16    26 p     2sea~
4 ford         mustang      3.8  1999     6 manua~ r        18    26 r     subc~
5 ford         mustang      4    2008     6 manua~ r        17    26 r     subc~
6 volkswagen   jetta        1.9  1999     4 manua~ f        33    44 d     comp~
7 volkswagen   new beetle   1.9  1999     4 manua~ f        35    44 d     subc~
```

```
mpg %>%
  filter(hwy == max(hwy)) %>%
  group_by(drv)
```

```
# A tibble: 2 x 11
# Groups:   drv [1]
  manufacturer model      displ  year   cyl trans  drv     cty   hwy fl    class
  <chr>        <chr>      <dbl> <int> <int> <chr>  <chr> <int> <int> <chr> <chr>
1 volkswagen   jetta        1.9  1999     4 manua~ f        33    44 d     comp~
2 volkswagen   new beetle   1.9  1999     4 manua~ f        35    44 d     subc~
```

```
maxValueR = max(mpg[which(mpg$drv=="r"),]$hwy)
maxValueF = max(mpg[which(mpg$drv=="f"),]$hwy)
maxValue4 = max(mpg[which(mpg$drv=="4"),]$hwy)
id1 = which(mpg$drv=="r" & mpg$hwy == maxValueR)
id2 = which(mpg$drv=="f" & mpg$hwy == maxValueF)
id3 = which(mpg$drv=="4" & mpg$hwy == maxValue4)
dataMark = mpg[c(id1,id2,id3),]
p1 +
geom_label_repel(data = dataMark, aes(label = drv))
```



利用分组统计操作

```
library(magrittr)
```

```
载入程辑包：'magrittr'

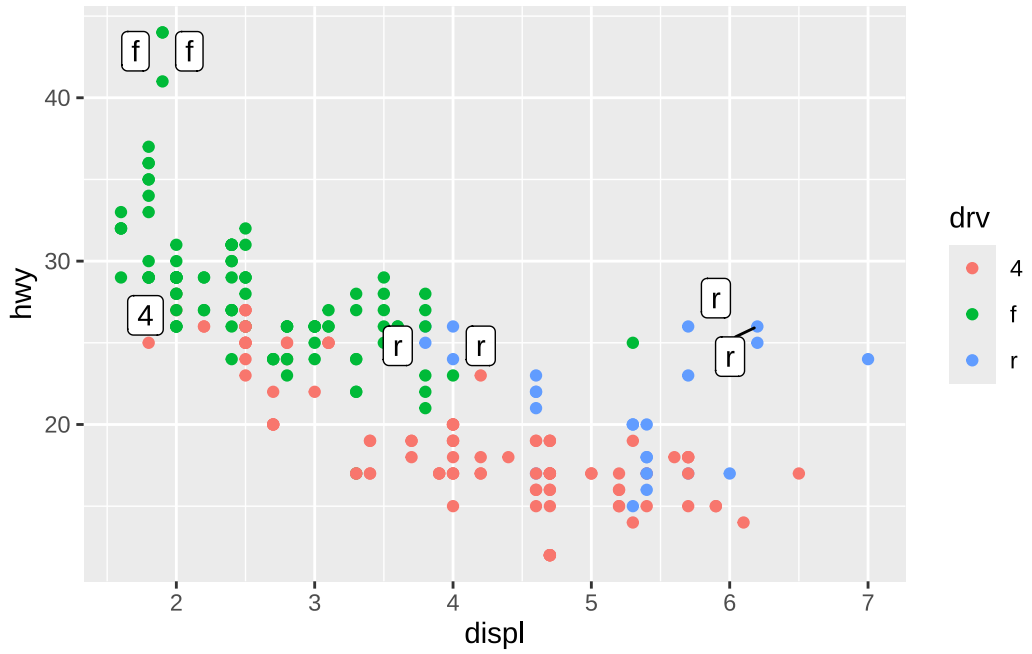
The following object is masked from 'package:purrr':

    set_names

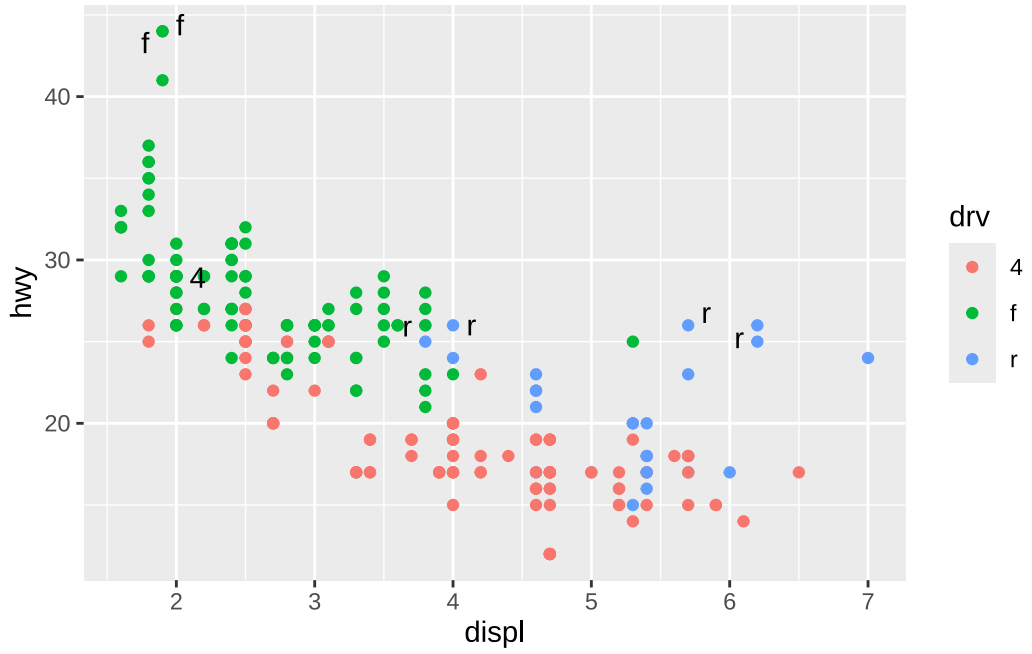
The following object is masked from 'package:tidyr':

    extract
```

```
best_in_drv=mpg %>%
group_by(drv)%>%
slice_max(hwy,n =1)
p1 +
geom_label_repel(data = best_in_drv, aes(label = drv))
```

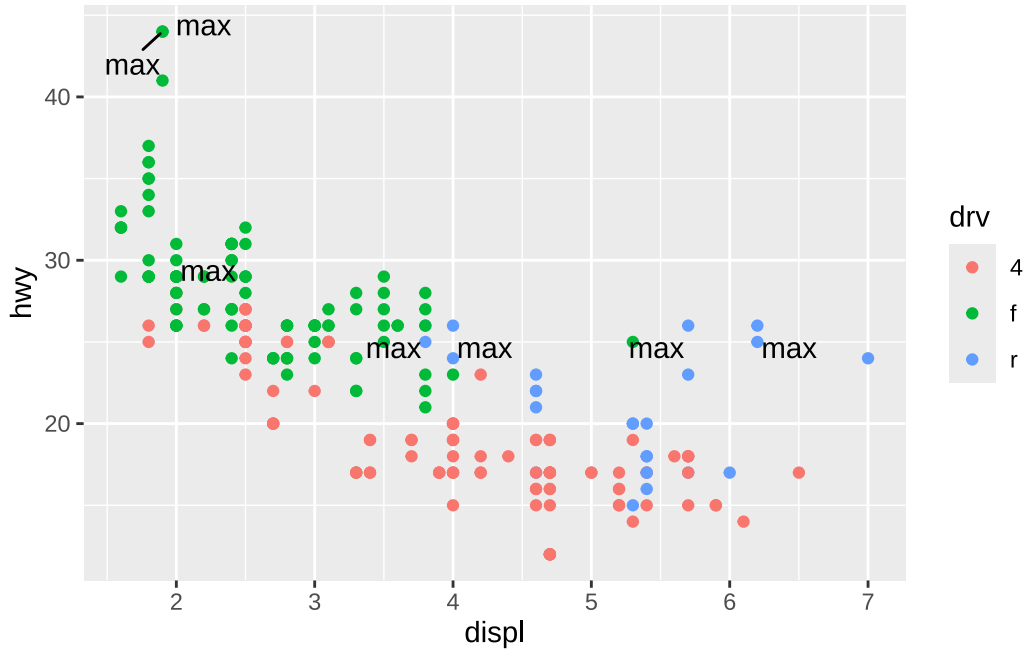



```
p1+
geom_text_repel(data =best_in_drv, aes(label= drv))
```



注意映射与赋值的区别

```
p1+
geom_text_repel(data =best_in_drv, label= "max")
```



4.3.8 统计变换

每个 class 均值的柱状图

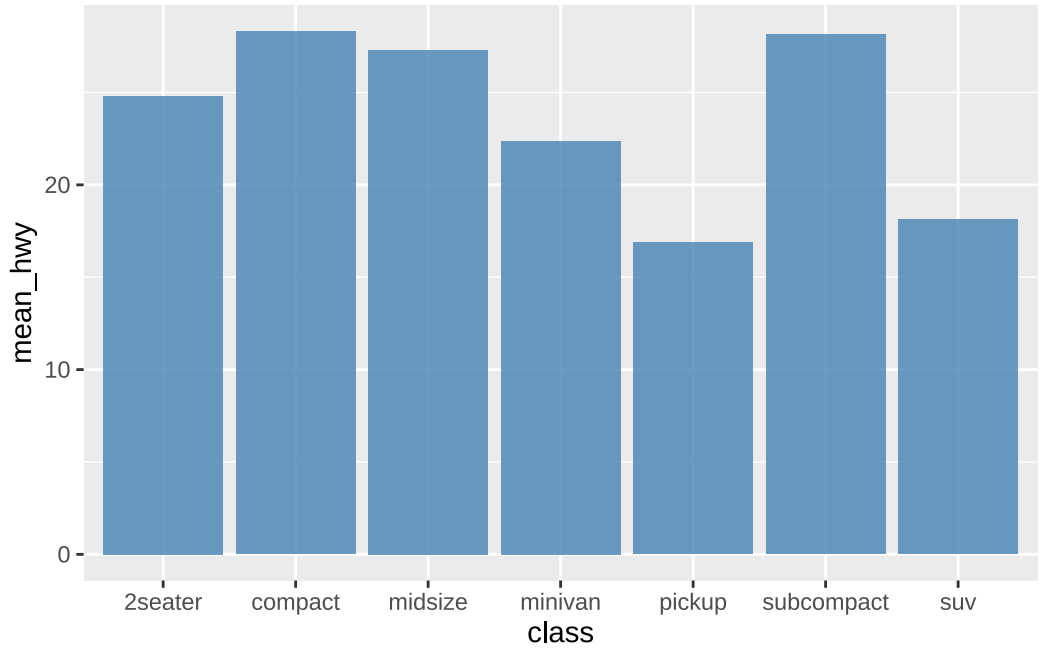
- deepseek 代码

```
library(ggplot2)
library(dplyr)

# 计算每个 class 的 hwy 均值
class_hwy <- mpg %>%
  group_by(class) %>%
  summarise(mean_hwy = mean(hwy, na.rm = TRUE))

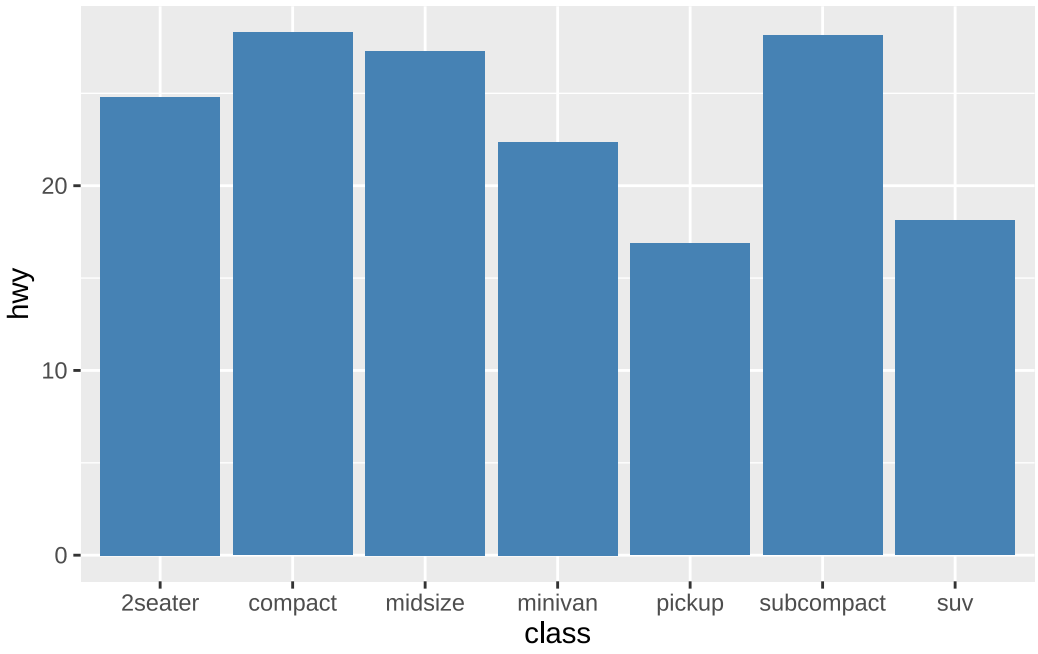
# 绘制柱状图
```

```r
ggplot(class_hwy, aes(x = class, y = mean_hwy)) +
  geom_col(fill = "steelblue", alpha = 0.8)
```



- 限定关键字：使用 stat_summary 计算均值并绘制柱状图

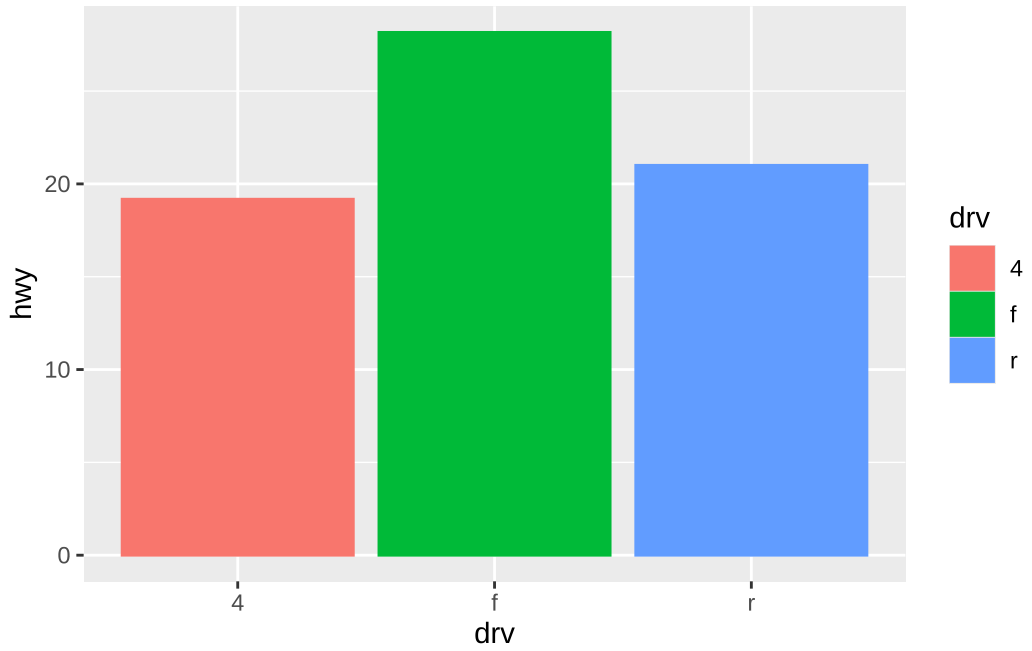
```r
ggplot(mpg, aes(x = class, y = hwy)) +
  # 使用 stat_summary 计算均值并绘制柱子
  stat_summary(
    fun = mean,               # 计算均值
    geom = "bar",             # 指定几何对象为柱状图
    fill = "steelblue"
  )
```



以 drv 为横轴，对 hwy 数据进行汇总

画出每个 drv 对应的均值

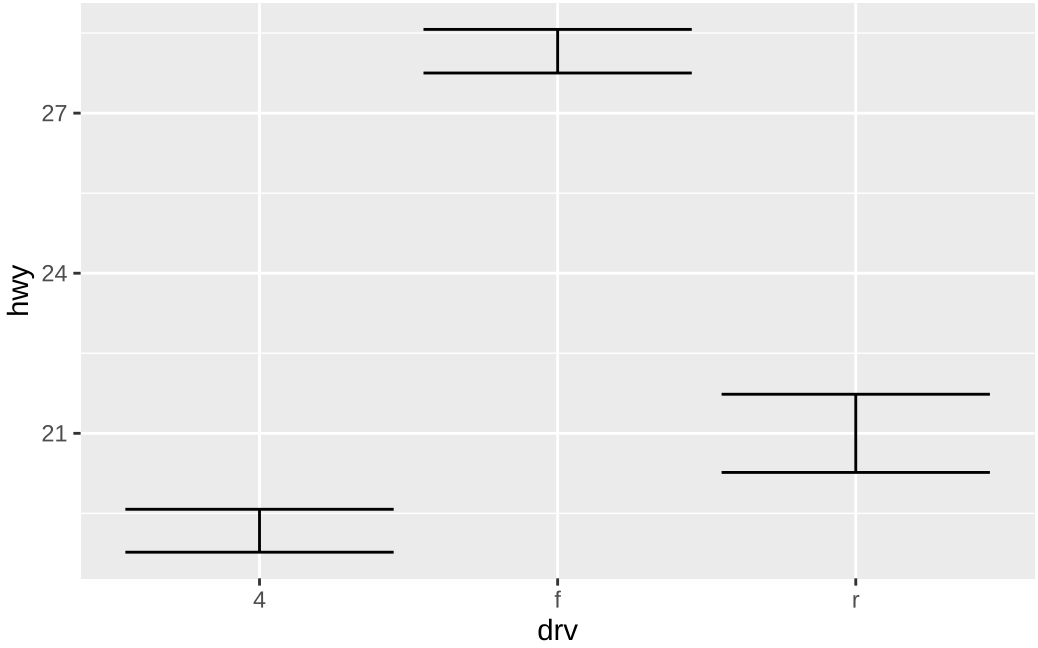
```
p1 <- ggplot(data = mpg, aes(x = drv, y = hwy))
p2 <- p1 +
stat_summary(fun.data = mean_se, geom = "bar", aes(color = drv, fill = drv))
p2
```



- fun.data 参数可以省略，默认参数是 mean_se

画出每个 drv 的误差线

```
p3 <- p1 +
stat_summary(fun.data = mean_se, geom = "errorbar")
p3
```

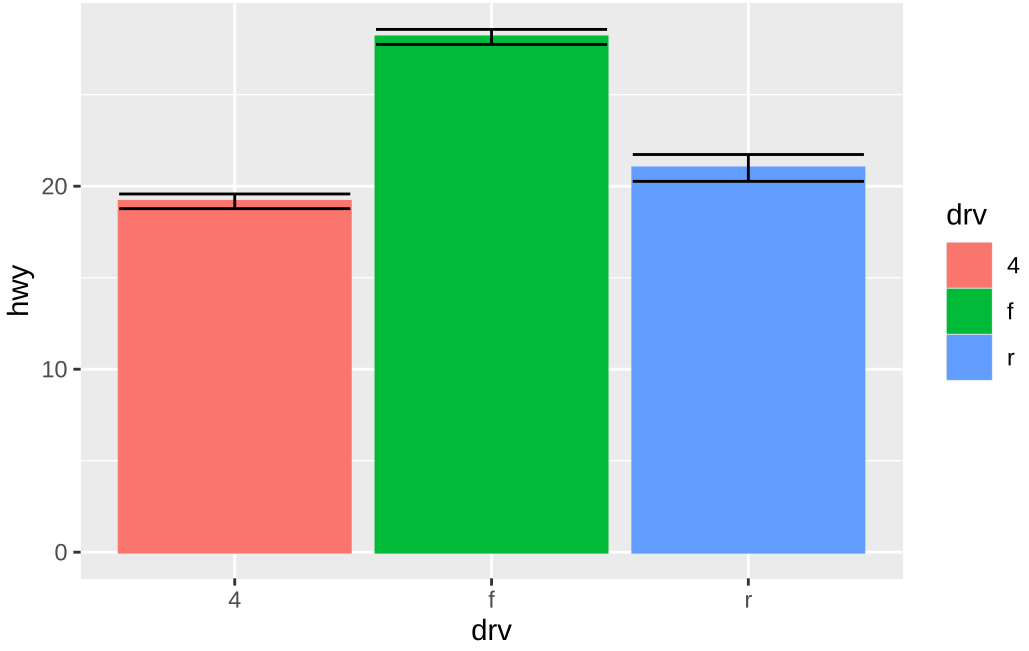


参数 fun.data

- 将数据变形，这个函数默认是 mean_se()
- 返回的是数据框，这个数据框将用于 geom 参数画图
- 如果 fun.data 返回的数据框包含了所需要的美学映射，图形就会显示出来

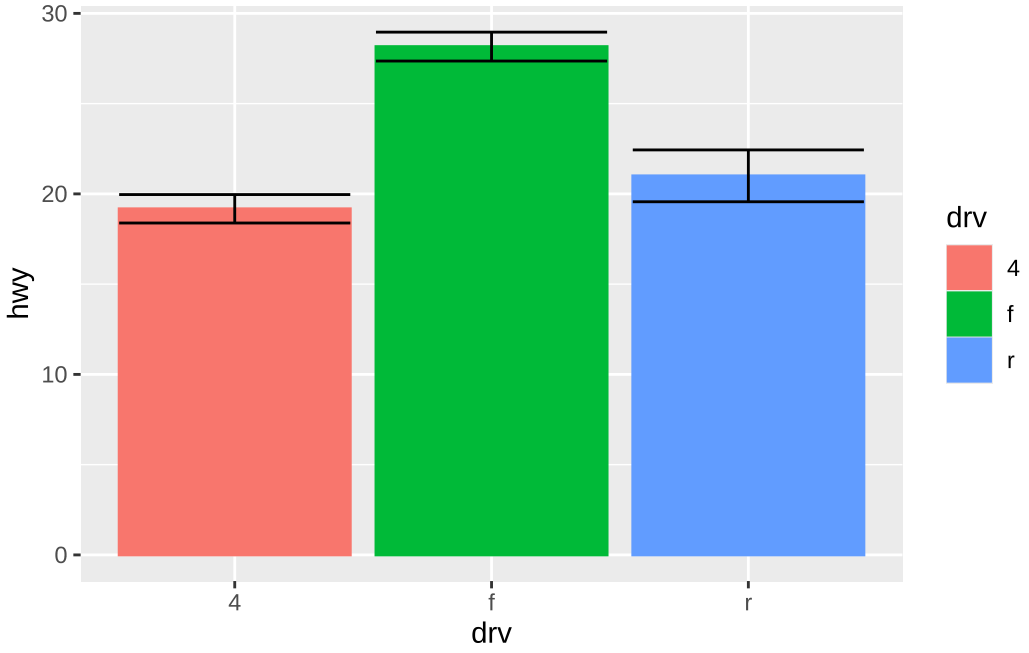
将柱状图与误差线放到一起

```
p1 +
stat_summary(fun = mean, geom = "bar", aes(color = drv, fill = drv)) +
stat_summary(fun.data = mean_se, geom = "errorbar")
```



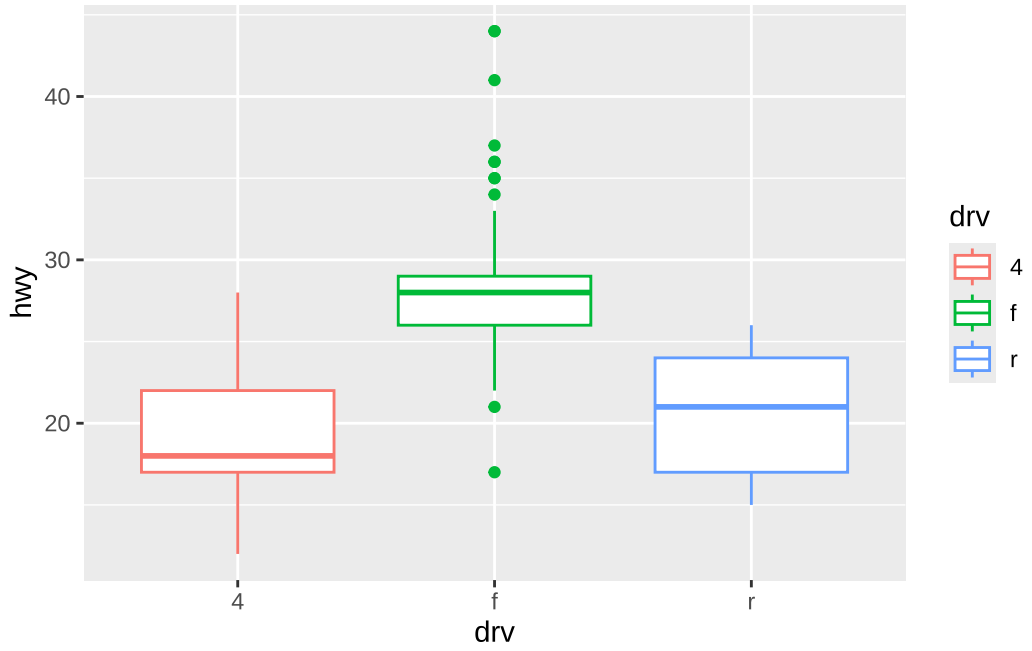
95% 置信区间

```
p1 +
stat_summary(fun = mean, geom = "bar", aes(color = drv, fill = drv)) +
stat_summary(fun.data = ~mean_se(., mult = 1.96), geom = "errorbar")
```



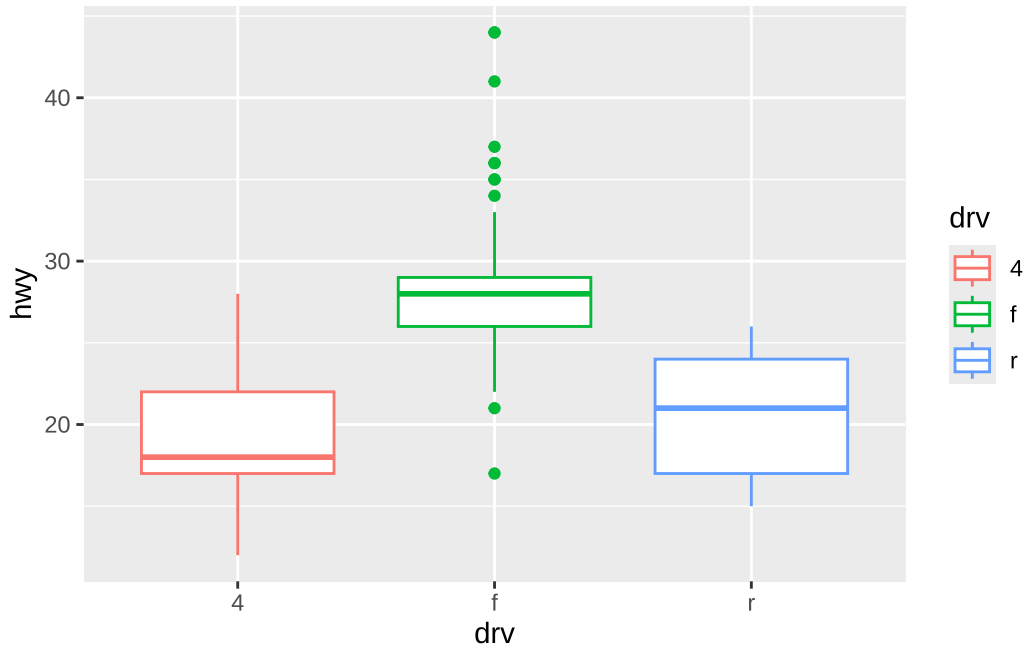
- 箱线图

```
p1  +
stat_boxplot(geom = "boxplot", aes(color = drv))
```

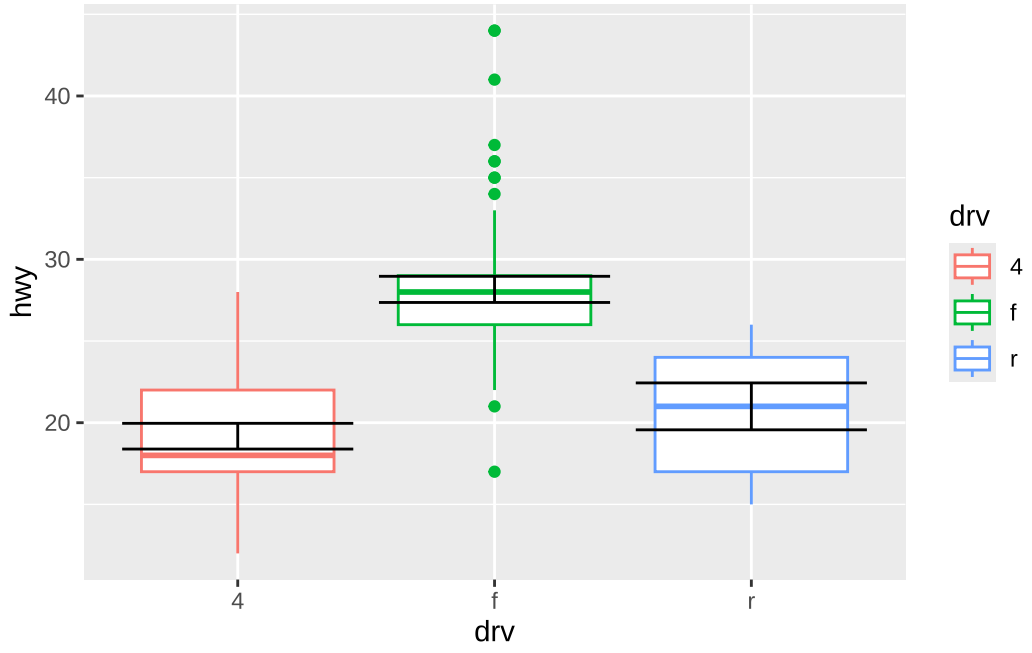


- 可在几何对象函数 geom_*() 中直接创建

```
p1 +
geom_boxplot(aes(color = drv))
```



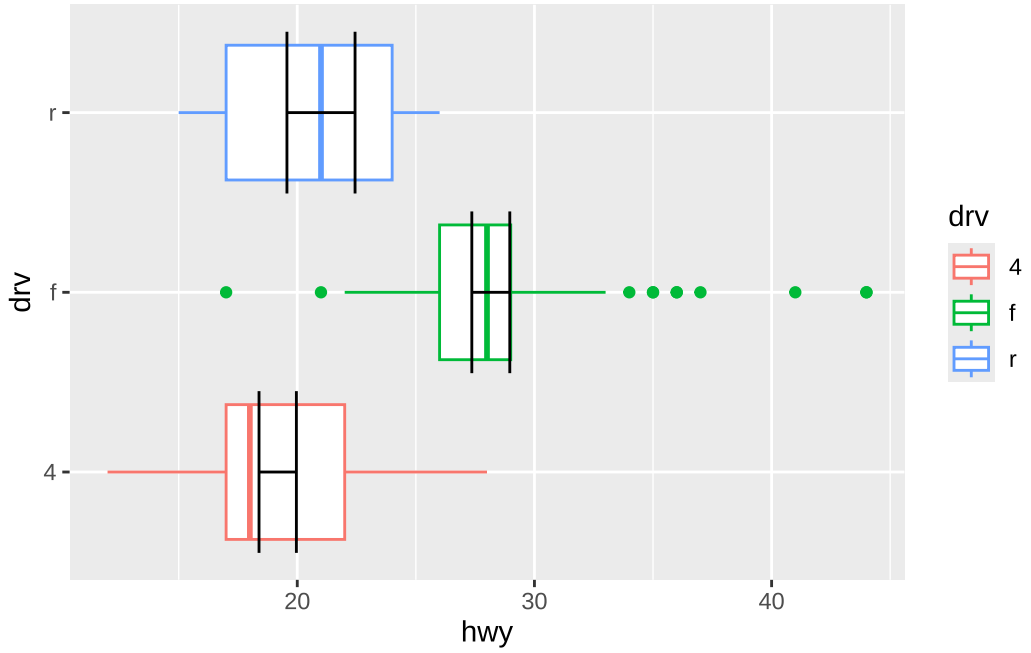

```
p1 +
geom_boxplot(aes(color = drv)) +
stat_summary(fun.data = ~mean_se(., mult = 1.96), geom = "errorbar")
```



4.3.9 坐标系

- 坐标轴翻转

```
p1 +
geom_boxplot(aes(color = drv)) +
stat_summary(fun.data = ~mean_se(., mult = 1.96), geom = "errorbar") +
coord_flip()
```



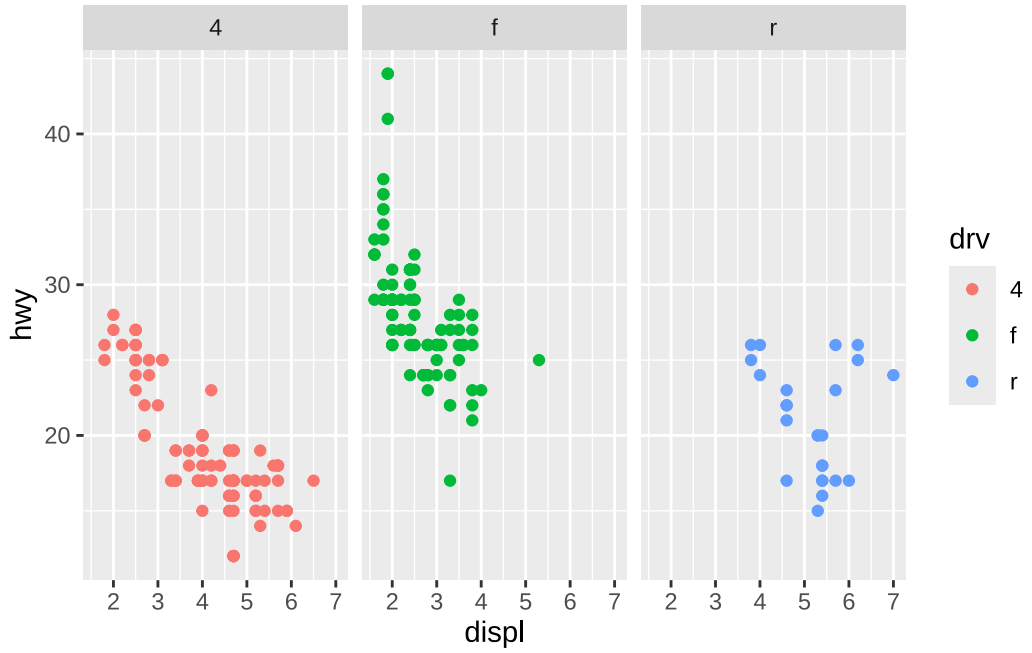
4.4 泛函式编程

mpg 数据集，按 drv 分组，画出各组 displ 与 hwy 的散点图

- 使用分面

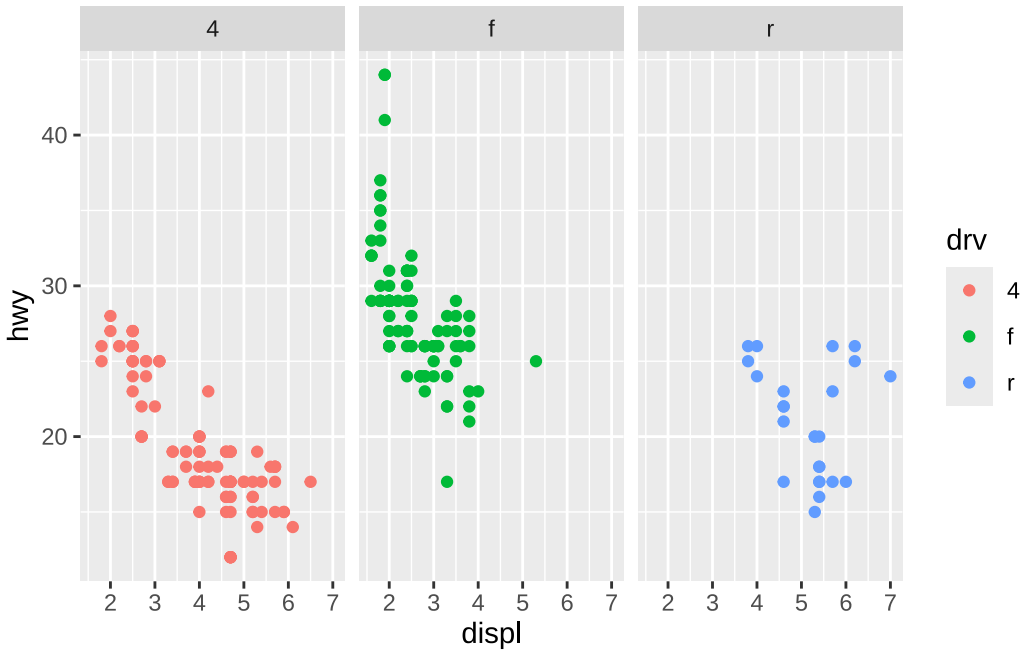
  先画图再分面

```
# 按 drv 分组绘制散点图（分面展示）
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point(
    aes(color = drv)  # 按 drv 着色（可选）
  ) +
  facet_wrap(~ drv, ncol = 3)   # 按 drv 分面，3 列布局
```



先分面再画图

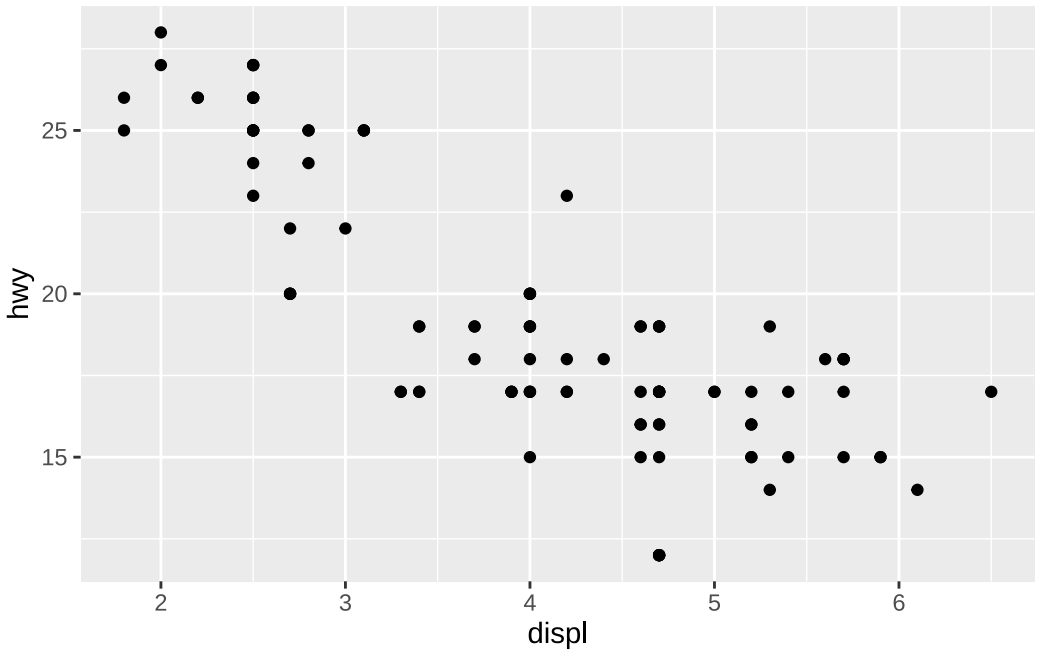
```
ggplot(mpg, aes(x = displ, y = hwy)) +
  facet_wrap(~ drv, ncol = 3)+   # 按 drv 分面，3 列布局
  geom_point(
    aes(color = drv)  # 按 drv 着色（可选）
  )
```



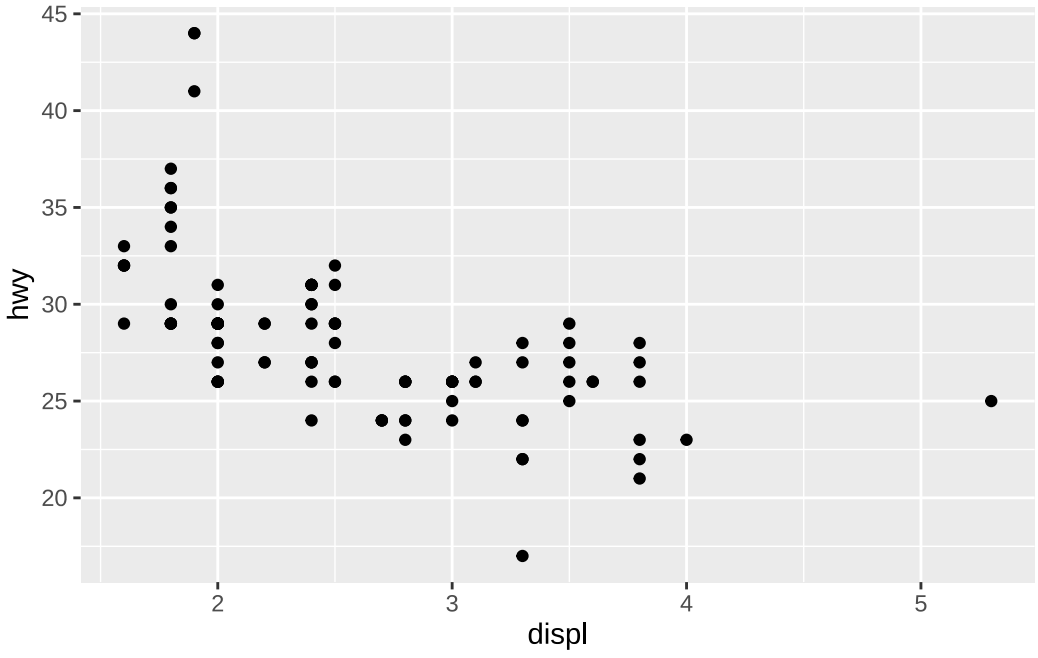
- 使用 map 函数

```
mpg  %>%
group_split(drv) %>%
map(~ggplot(data=.x, aes(x = displ, y = hwy))+geom_point())
```

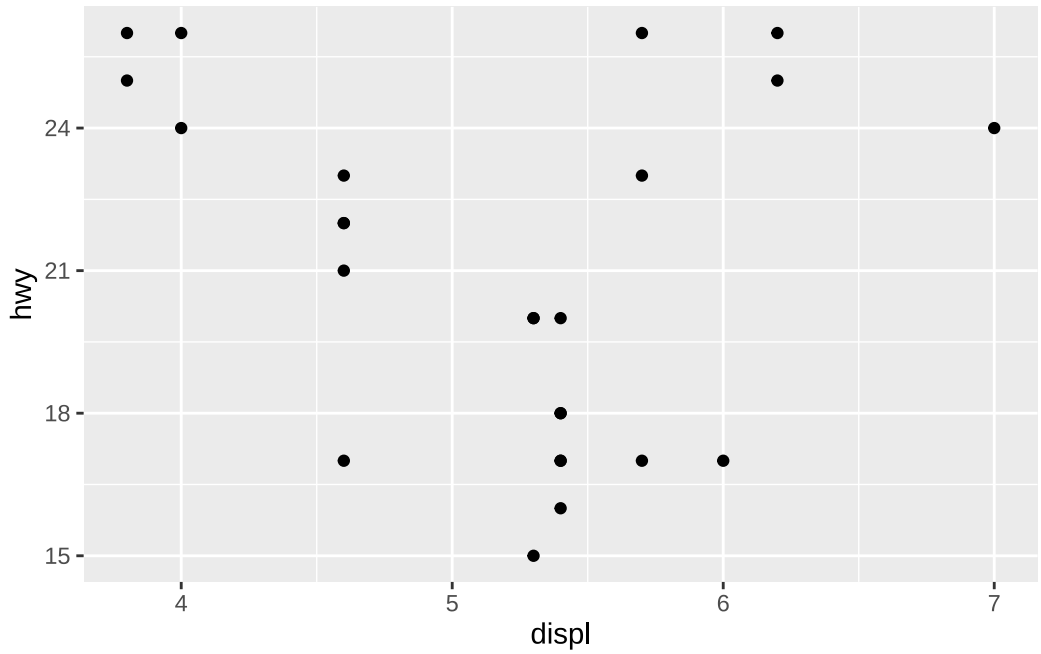
```
[[1]]
```




```
[[2]]
```



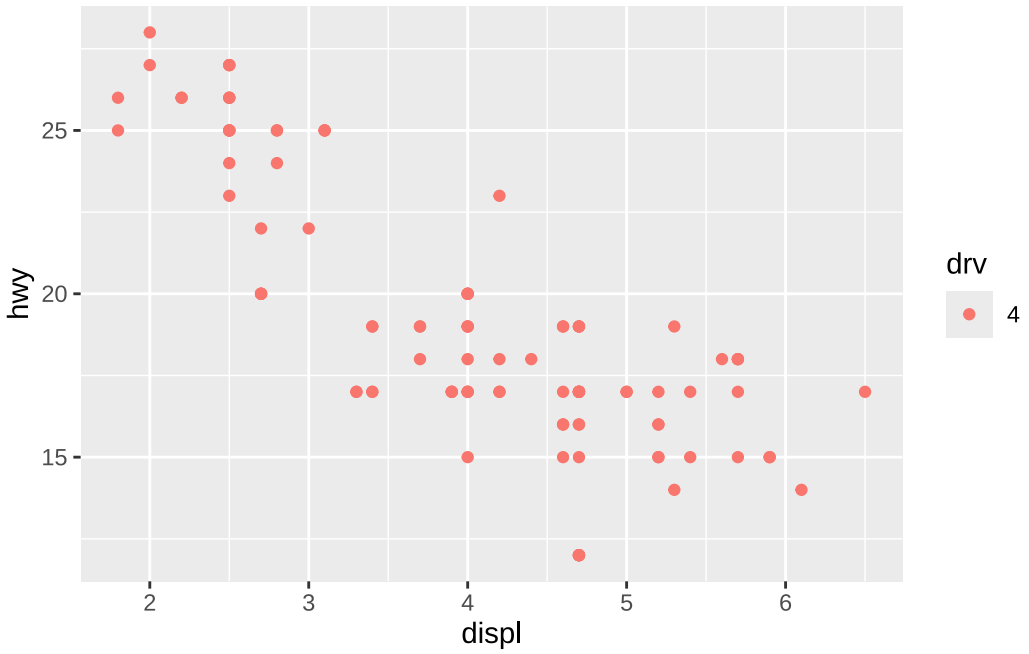
```
[[3]]
```



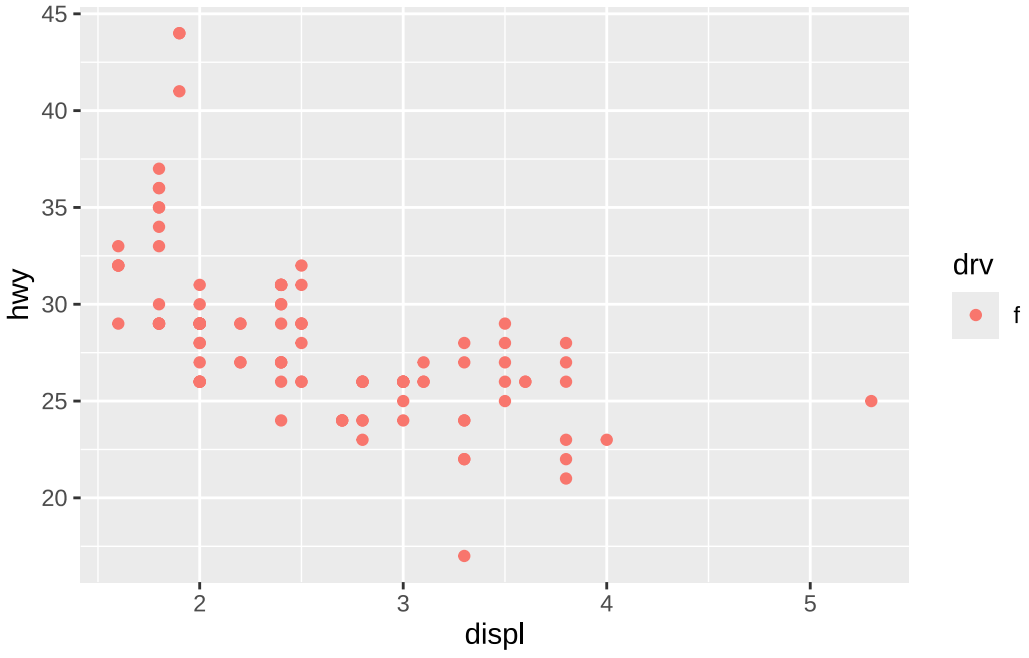
如何在三幅图中，对不同的 drv 类型，使用不同的颜色

```
mpg  %>%
group_split(drv) %>%
map(~ggplot(data=.x, aes(x = displ, y = hwy))+
geom_point(aes(color = drv)))
```

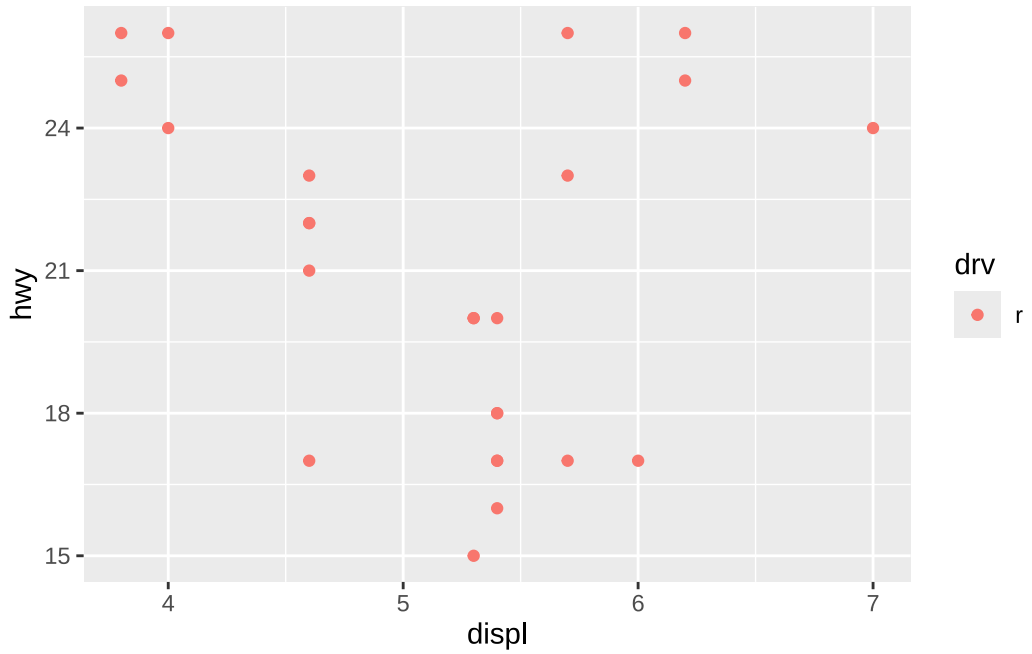
```
[[1]]
```


hwy
25
20
15
2
3
4
5
6
displ
drv
4

```
[[2]]
```


hwy
45
40
35
30
25
20
2
3
4
5
displ
drv
f

```
[[3]]
```

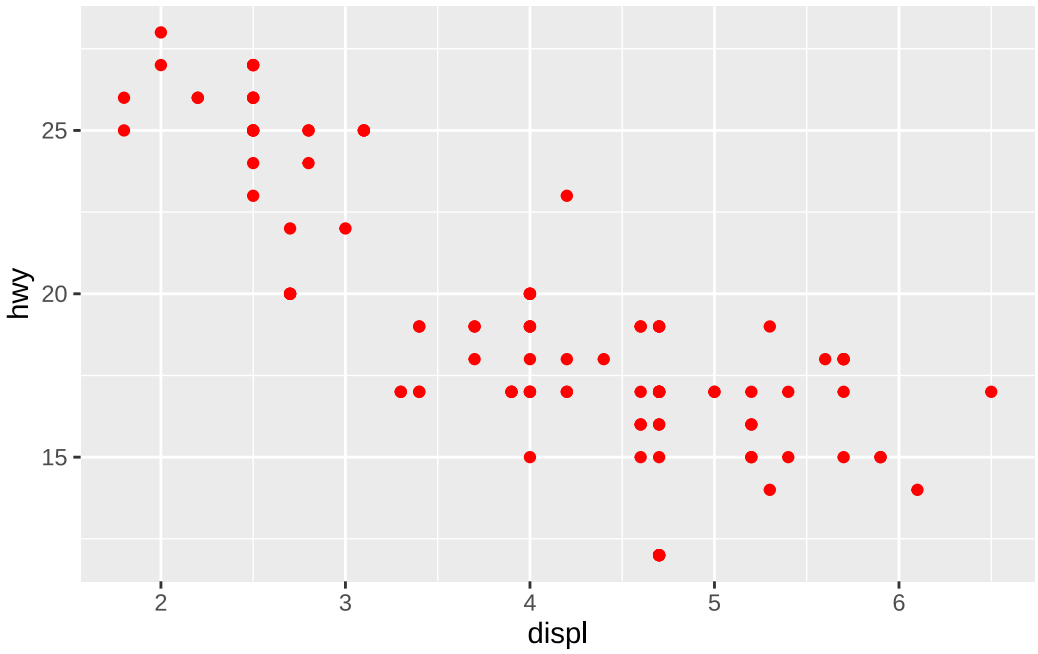


为什么都是红色？怎样实现颜色不同分配？

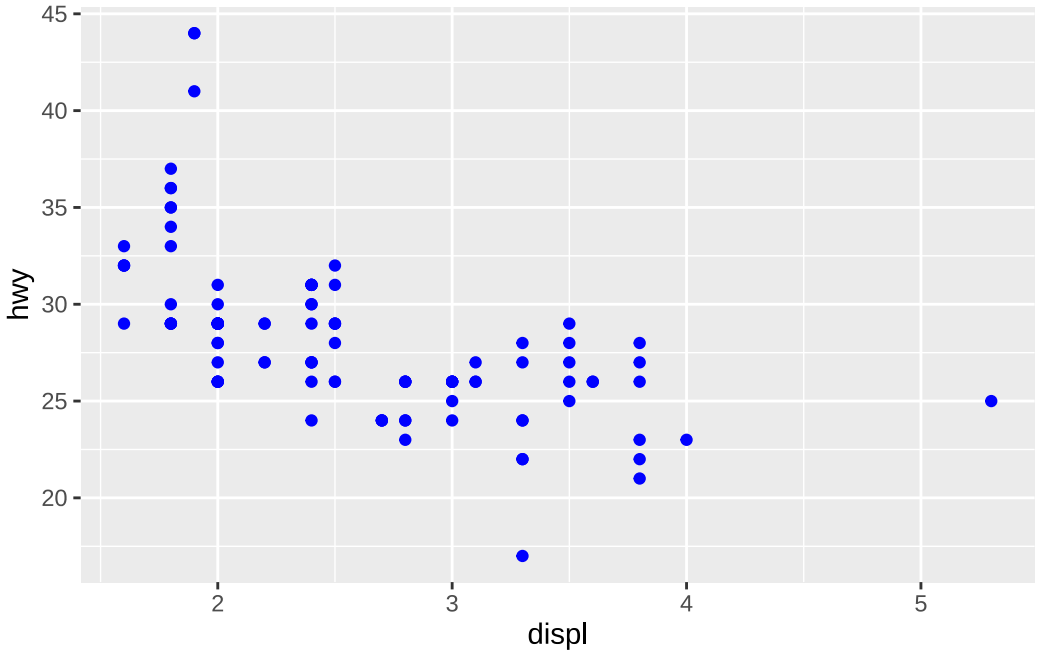
- 手动设置颜色

```
drv_colors <- c("4" = "red", "f" = "blue", "r" = "green")
# 按 drv 分组并分割数据
mpg %>%
  group_split(drv) %>%
  # 对每个子数据集绘图
  map(~ {
    drv_type <- unique(.x$drv)  # 提取当前组的 drv 值
    ggplot(.x, aes(x = displ, y = hwy)) +
      geom_point(color = drv_colors[drv_type])   # 设置颜色和透明度
  })
```

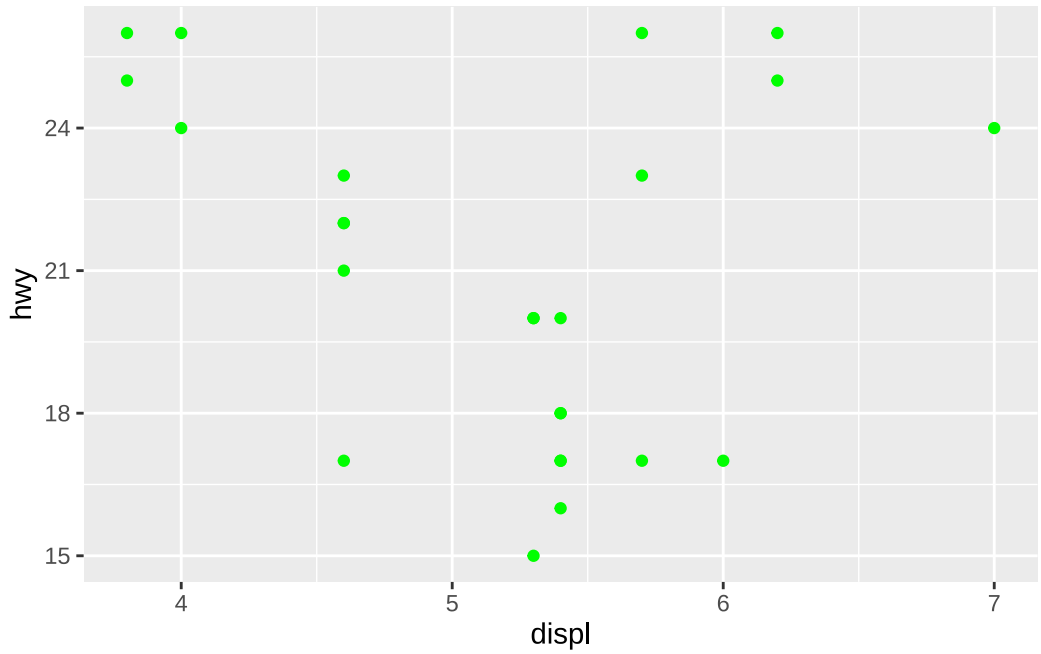
```
[[1]]
```




```
[[2]]
```



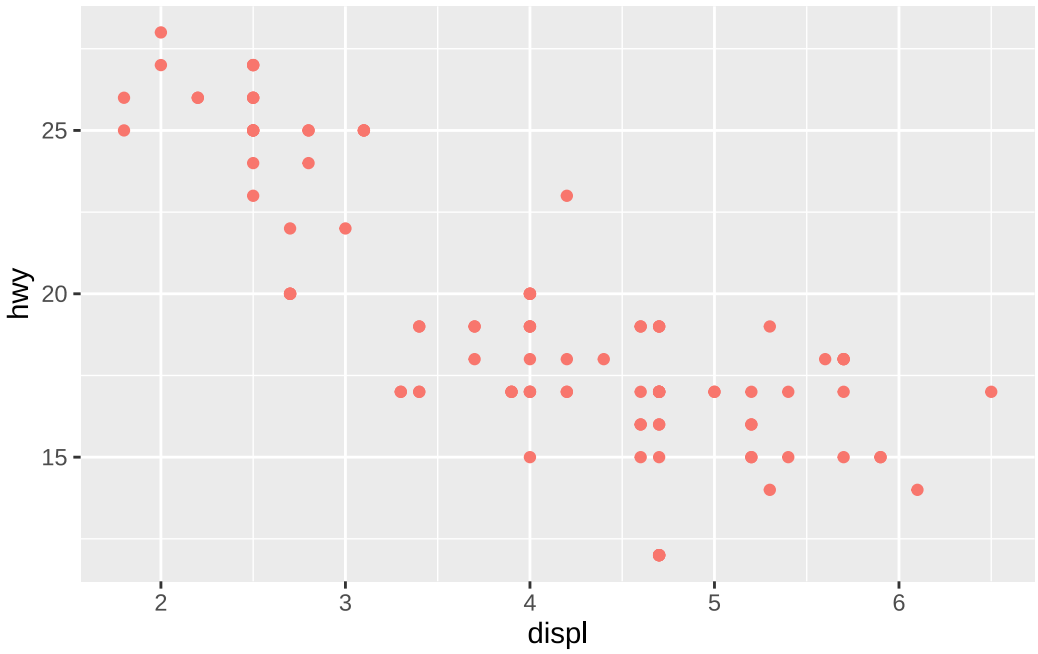
```
[[3]]
```



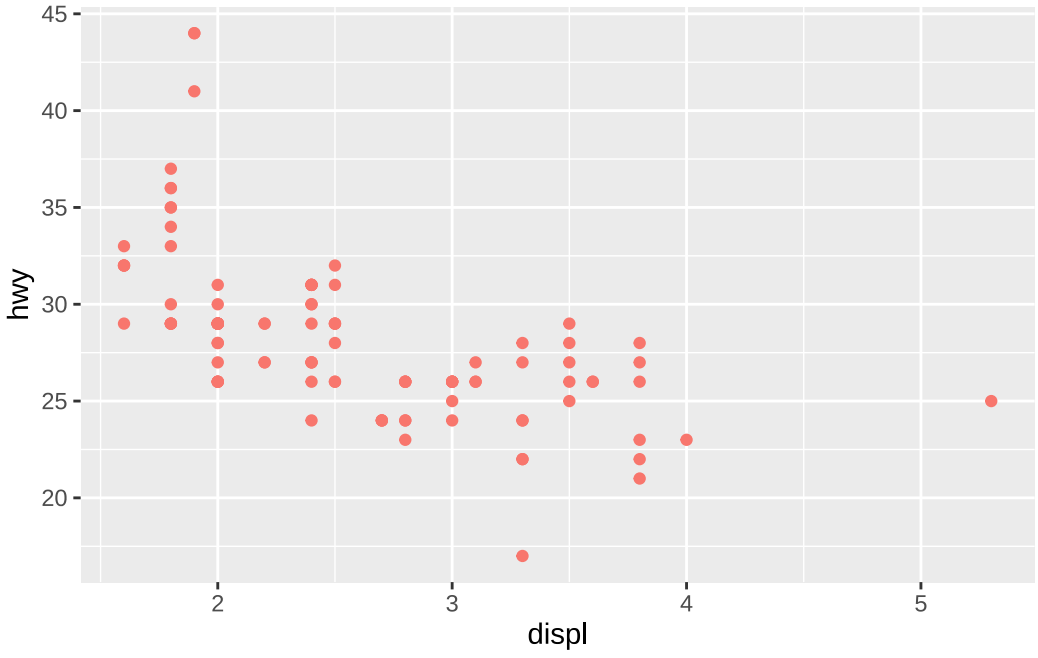
- 利用调色板

```
mpg %>%
  group_split(drv) %>%
  # 对每个子数据集绘图
  map(~{
    drv_type <- unique(.x$drv)  # 提取当前组的 drv 值
    ggplot(.x, aes(x = displ, y = hwy)) +
      geom_point(aes(color = drv_type)) +
         scale_color_manual(values = scales::hue_pal()(3)) +
         guides(color = "none")
  })
```

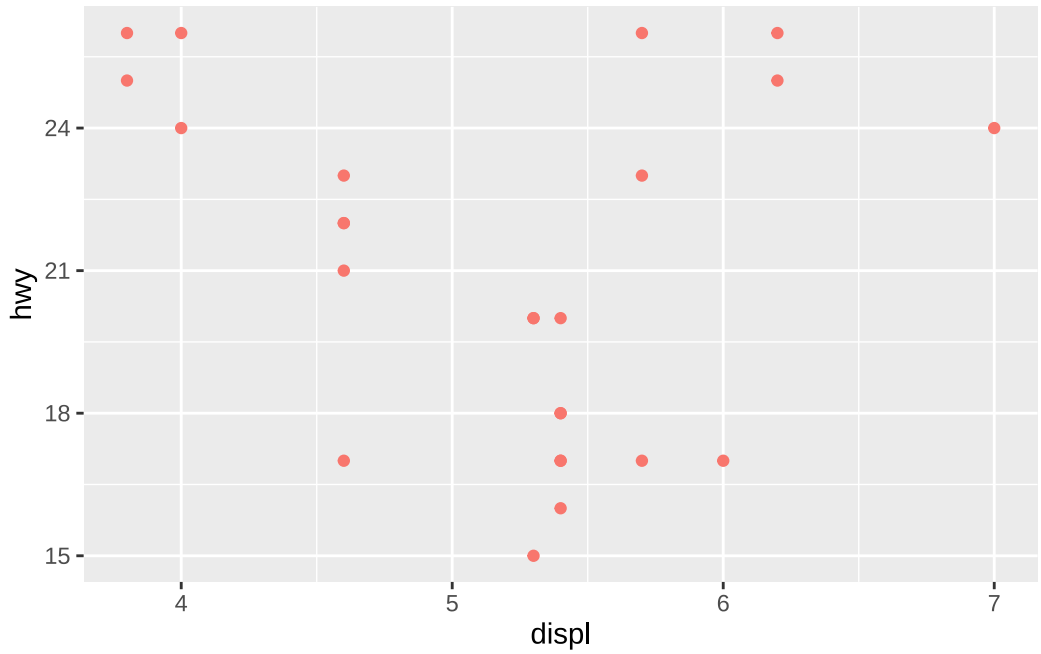
```
[[1]]
```




```
[[2]]
```



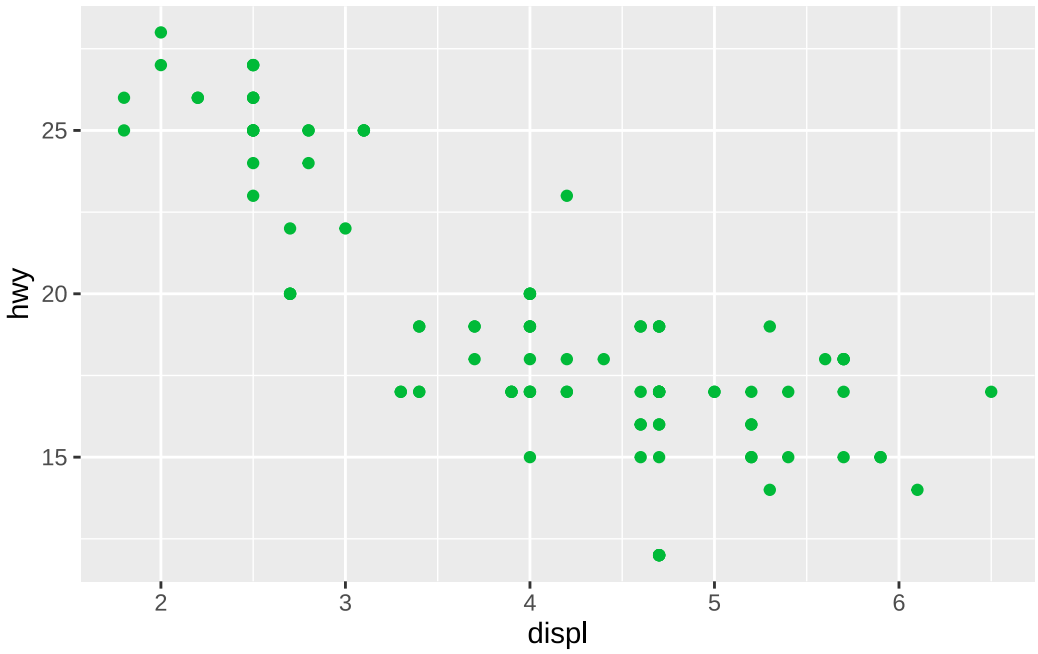
```
[[3]]
```



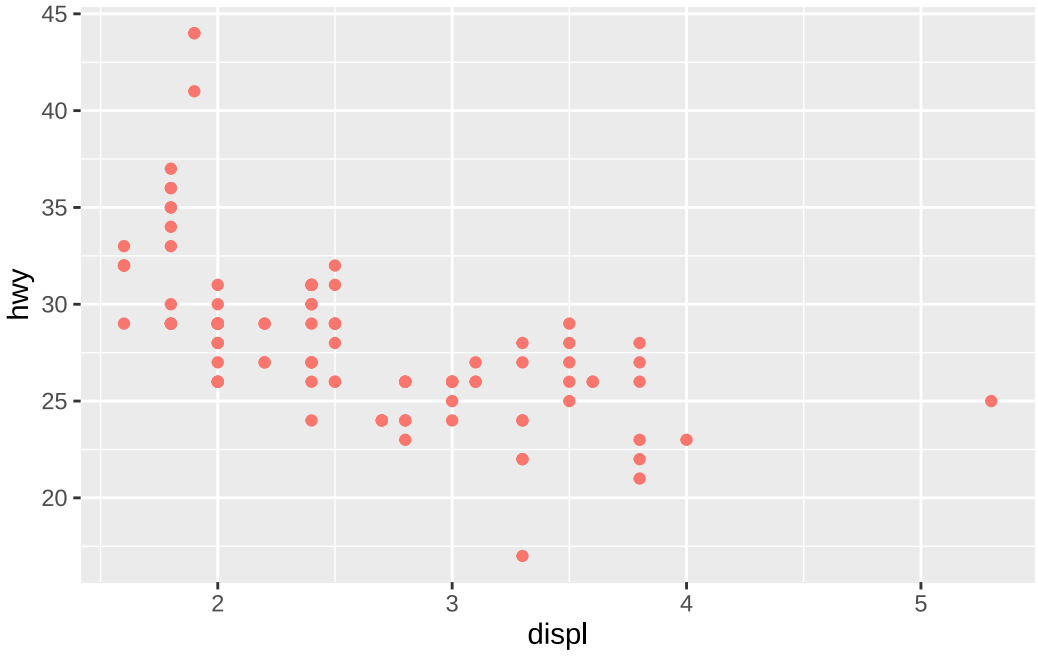
- 上面的代码为啥还都是红色?

```
drv_types <- unique(mpg$drv)
drv_colors <- scales::hue_pal()(3)
mpg %>%
  group_split(drv) %>%
  # 对每个子数据集绘图
  map(~{
    drv_type <- unique(.x$drv)
    current_color <- drv_colors[drv_types == drv_type]
    ggplot(.x, aes(x = displ, y = hwy)) +
      geom_point(color = current_color) +
         guides(color = "none")
  })
```

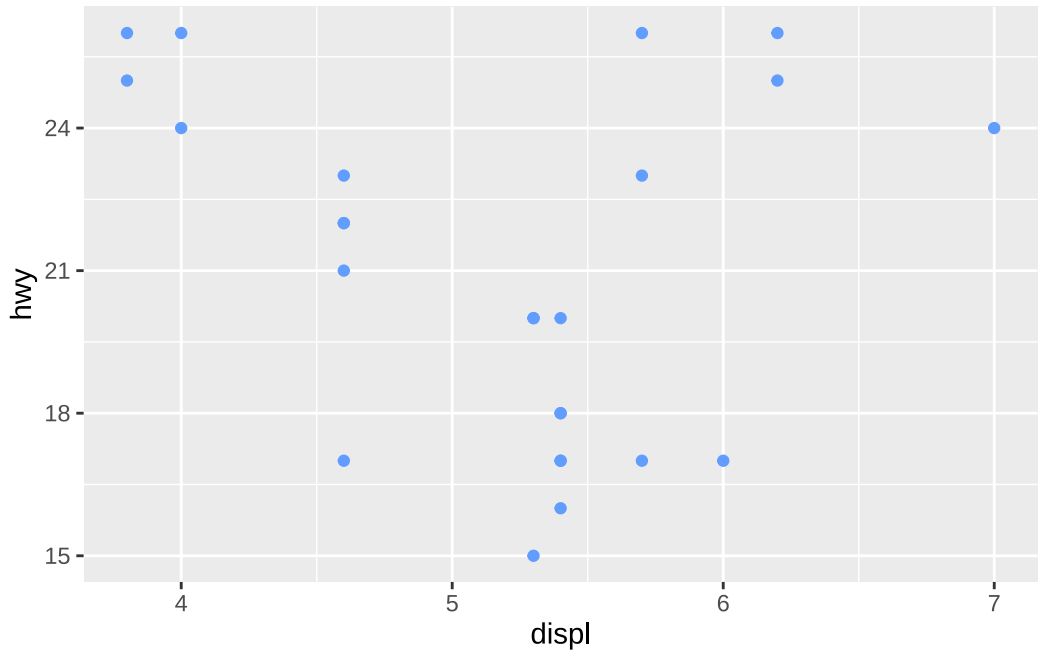
```
[[1]]
```




```
[[2]]
```



```
[[3]]
```



4.5 分组统计

统计 mpg 数据集中不同 drv 的数量

```
mpg %>%
group_by(by = drv) %>%
summarise(DrvNum = n())
```

```
# A tibble: 3 x 2
  by    DrvNum
  <chr>  <int>
1 4        103
2 f        106
3 r         25
```

4.6 数据重塑造

```
# 创建数据集
data <- tibble(
  observation = c("Rich(Sam)", "Win(Ash)", "Bil(Jules)"),
  A_count = c(7, 10, 5),
  B_count = c(2, 5, 8),
  A_dbh = c(100, 80, 95),
  B_dbh = c(110, 87, 90)
)

# 处理 'observation' 列，拆分成 'location' 和 'person'
cleaned_data <- data %>%
  separate(observation, into = c("location", "person"), sep = "\\(", extra = "merge") %>%
  mutate(person = gsub("\\)", "", person)) # 去掉右括号

# 将 A 和 B 拆分成单独的 'type' 列，将 count 和 dbh 拆分成单独列
long_data <- cleaned_data %>%
  pivot_longer(cols = starts_with("A_") | starts_with("B_"),
               names_to = c("type", "variable"),
                names_pattern = "(.)_(.)")
              # %>%
  # pivot_wider(names_from = "variable", values_from = "value")

# 查看结果
long_data
```

```
# A tibble: 12 x 5
   location person type  variable value
   <chr>    <chr>  <chr> <chr>    <dbl>
 1 Rich     Sam    A     c            7
 2 Rich     Sam    A     d          100
 3 Rich     Sam    B     c            2
 4 Rich     Sam    B     d          110
```

```
 5 Win      Ash    A     c           10
 6 Win      Ash    A     d           80
 7 Win      Ash    B     c            5
 8 Win      Ash    B     d           87
 9 Bil      Jules  A     c            5
10 Bil      Jules  A     d           95
11 Bil      Jules  B     c            8
12 Bil      Jules  B     d           90
```

```
long_data1 <- long_data %>%
pivot_wider(names_from = "variable", values_from = "value")
long_data1
```

```
# A tibble: 6 x 5
  location person type      c     d
  <chr>    <chr>  <chr> <dbl> <dbl>
1 Rich     Sam    A         7   100
2 Rich     Sam    B         2   110
3 Win      Ash    A        10    80
4 Win      Ash    B         5    87
5 Bil      Jules  A         5    95
6 Bil      Jules  B         8    90
```

5 案例 1：钻石数据分析

6 案例：钻石数据集分析

```
library(magrittr)
library(tidyverse)
```

```
Warning: 程辑包'ggplot2'是用R版本4.3.3 来建造的
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.3     v readr     2.1.4
v forcats   1.0.0     v stringr   1.5.0
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.0
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::combine()   masks gridExtra::combine()
x tidyr::extract()   masks magrittr::extract()
x dplyr::filter()    masks stats::filter()
x dplyr::lag()       masks stats::lag()
x purrr::set_names() masks magrittr::set_names()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become e
```

```
library(modelr)
```

```
head(diamonds)
```

```
# A tibble: 6 x 10
  carat cut       color clarity depth table price     x     y     z
```

```
  <dbl> <ord>     <ord> <ord>   <dbl> <dbl> <int> <dbl> <dbl> <dbl>
1  0.23 Ideal     E     SI2      61.5    55   326  3.95  3.98  2.43
2  0.21 Premium   E     SI1      59.8    61   326  3.89  3.84  2.31
3  0.23 Good      E     VS1      56.9    65   327  4.05  4.07  2.31
4  0.29 Premium   I     VS2      62.4    58   334  4.2   4.23  2.63
5  0.31 Good      J     SI2      63.3    58   335  4.34  4.35  2.75
6  0.24 Very Good J     VVS2     62.8    57   336  3.94  3.96  2.48
```

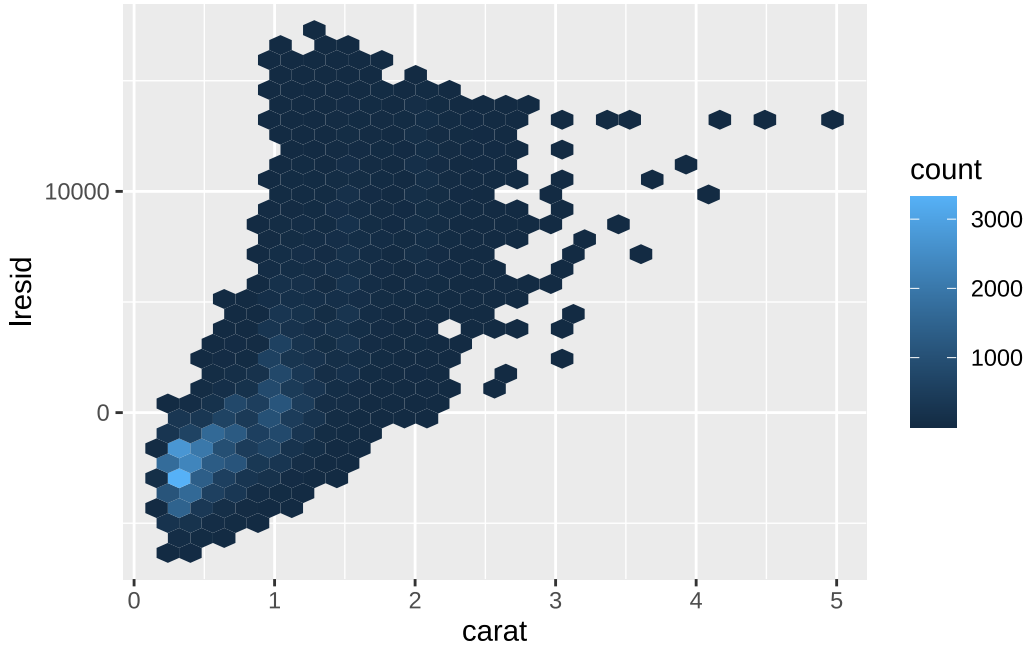
6.1 回归分析

- 线性回归

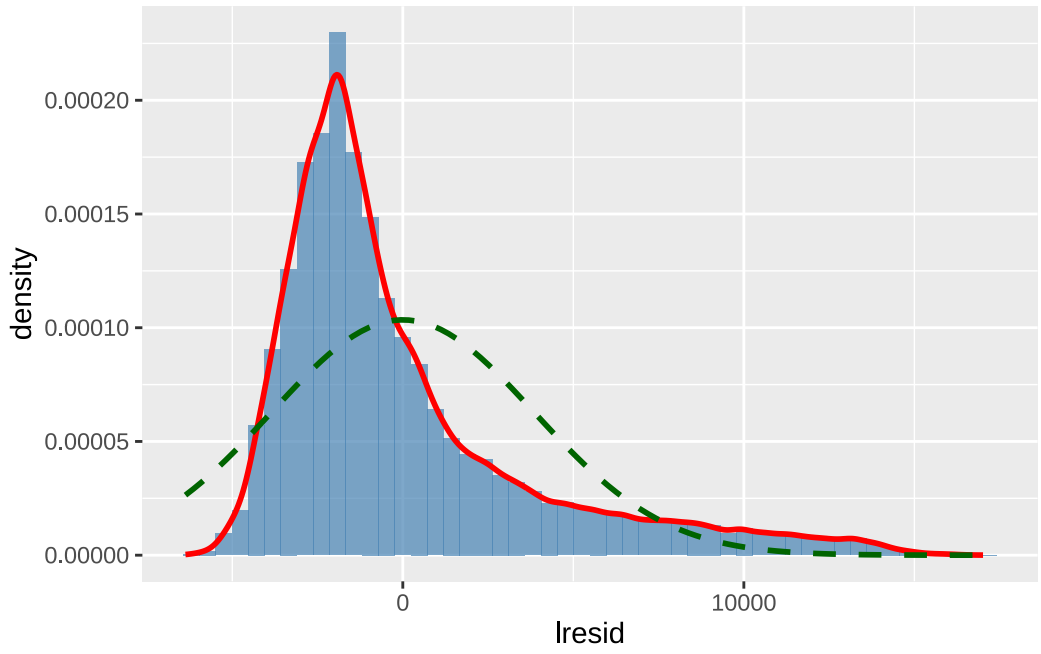
```
model <- lm(price ~ cut + color + clarity, data = diamonds)

diamondsNew <- diamonds  %>%
add_residuals(model, "lresid")

ggplot(diamondsNew, aes(x= carat, y = lresid)) +
geom_hex()
```



```r
ggplot(diamondsNew, aes(x = lresid)) +
# geom_density()
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 50,
    fill = "steelblue",
    alpha = 0.7
  ) +
  geom_density(color = "red", linewidth = 1) +
  # 添加正态分布参考曲线
  stat_function(
    fun = dnorm,
    args = list(
      mean = mean(diamondsNew$lresid),
      sd = sd(diamondsNew$lresid)
    ),
    color = "darkgreen",
    linewidth = 1,
    linetype = "dashed"
  )
```

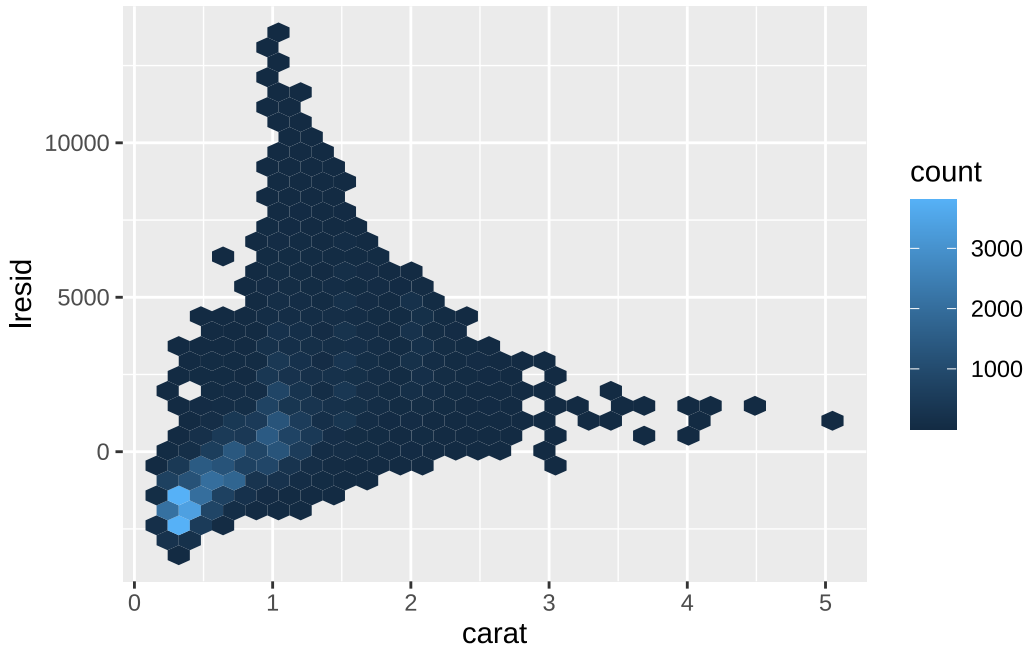



```
diamondsPri <- diamonds  %>%
mutate(PricePer = price/carat)

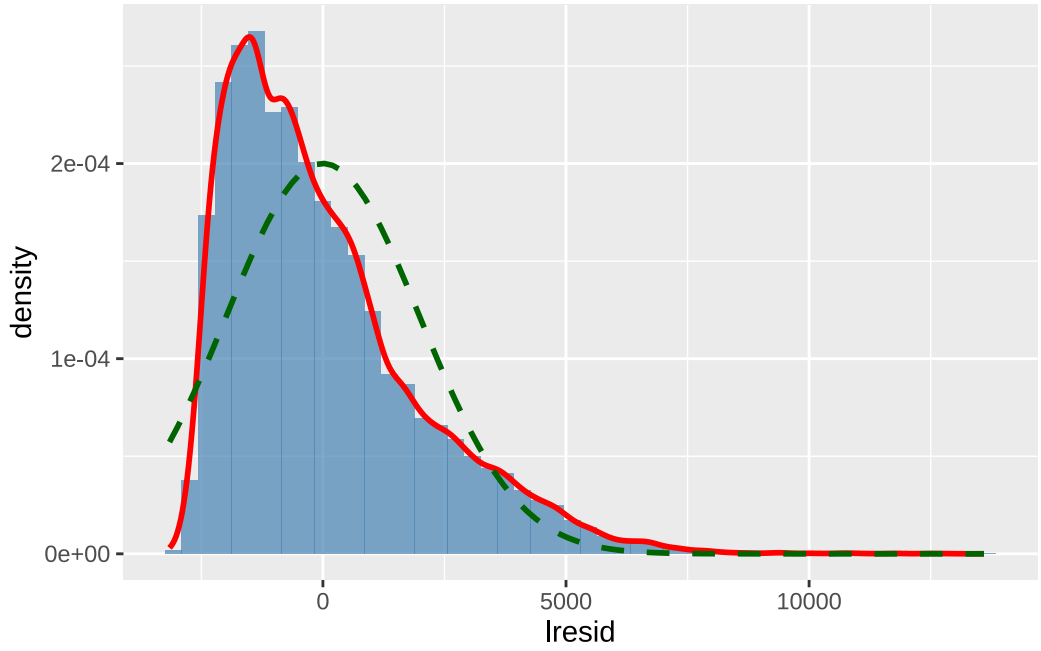
model <- lm(PricePer ~ cut + color + clarity, data = diamondsPri)

diamondsNew <- diamondsPri  %>%
add_residuals(model, "lresid")

ggplot(diamondsNew, aes(x= carat, y = lresid)) +
geom_hex()
```




```
ggplot(diamondsNew, aes(x = lresid)) +
# geom_density()
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 50,
    fill = "steelblue",
    alpha = 0.7
  ) +
  geom_density(color = "red", linewidth = 1) +
  # 添加正态分布参考曲线
  stat_function(
    fun = dnorm,
    args = list(
      mean = mean(diamondsNew$lresid),
      sd = sd(diamondsNew$lresid)
    ),
    color = "darkgreen",
    linewidth = 1,
    linetype = "dashed"
  )
```



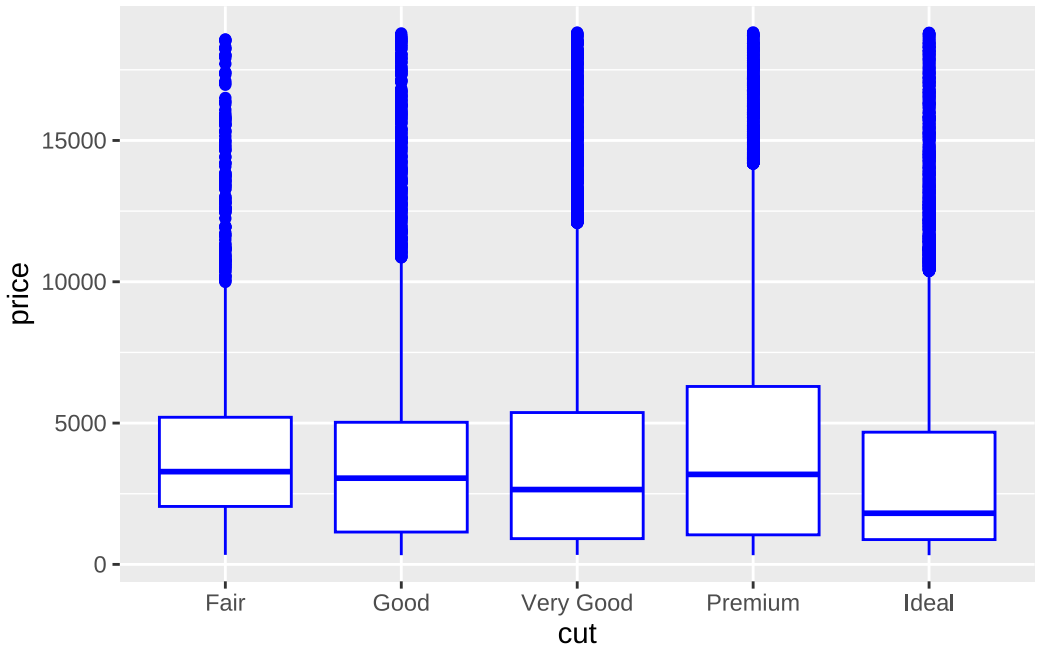
- 非线性回归

6.2 寻找影响因素

6.2.1 描述性统计发现反常识现象

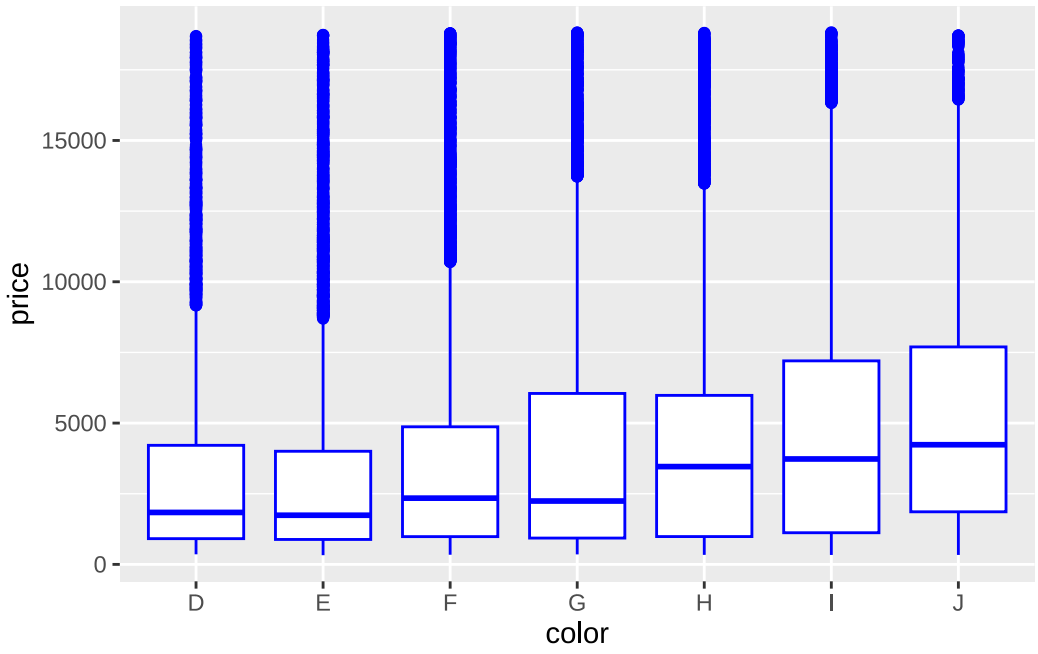
不同 cut, price 线箱图的变化

```
diamonds  %>%
ggplot(aes(x = cut, y = price))  +
geom_boxplot(color = "blue")
```



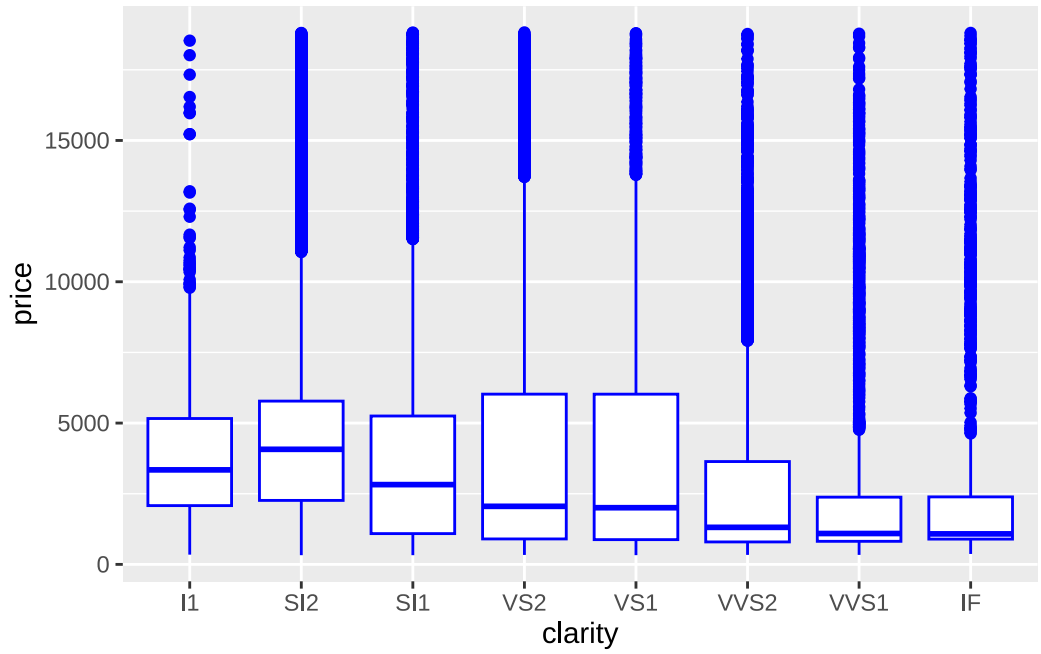
不同 color, price 线箱图的变化

```
diamonds  %>%
ggplot(aes(x = color, y = price))  +
geom_boxplot(color = "blue")
```



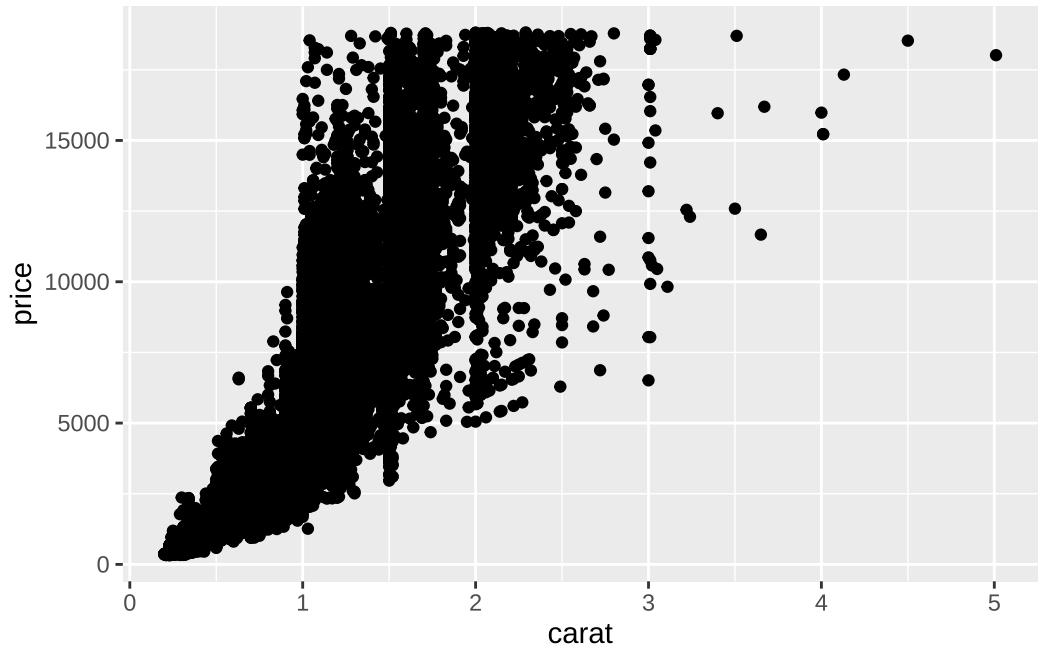
不同 clarity, price 线箱图的变化

```
diamonds  %>%
ggplot(aes(x = clarity, y = price))  +
geom_boxplot(color = "blue")
```



6.2.2 价格趋势分析

```
diamonds %>%
ggplot(aes(x = carat, y = price)) +
geom_point()
```




```
library(ggrepel)  # 用于避免标签重叠

# 设置分析范围 (0.9-1.1 克拉)
diamonds_1ct <- diamonds %>%
  filter(between(carat, 0.9, 1.1)) %>%
  mutate(
    # 创建精确到百分位的克拉分组
    carat_centi = round(carat, 2),
    # 标记关键锚定点
    is_anchor = if_else(carat_centi == 1.00, "1.00 克拉", " 其他"),
    # 特别标记 0.99 克拉点
    is_99 = if_else(carat_centi == 0.99, "0.99 克拉", " 其他")
  )

# 计算每个 0.01 克拉分组的汇总数据
price_summary <- diamonds_1ct %>%
  group_by(carat_centi) %>%
  summarise(
    avg_price = mean(price),
```

```
    median_price = median(price),
    n = n(),
    .groups = "drop"
  ) %>%
  # 计算与 1 克拉的价格比率
  mutate(
    price_ratio = avg_price / avg_price[carat_centi == 1.00],
    price_ratio_label = scales::percent(price_ratio, accuracy = 1)
  )

# 计算 0.99 克拉与 1.00 克拉的价格差异
price_gap <- price_summary %>%
  filter(carat_centi %in% c(0.99, 1.00)) %>%
  select(carat_centi, avg_price) %>%
  pivot_wider(
    names_from = carat_centi,
    values_from = avg_price,
    names_prefix = "price_"
  ) %>%
  mutate(
    price_difference = price_1 - price_0.99,
    price_ratio = price_0.99 / price_1
  )

# 创建阶梯变化可视化
ggplot(price_summary, aes(x = carat_centi, y = avg_price)) +
  # 绘制价格阶梯线
  geom_step(direction = "hv", color = "steelblue", size = 1.2) +
  # 添加数据点
  geom_point(data = filter(price_summary, carat_centi == 1.00),
             color = "red", size = 4, shape = 18) +
  geom_point(data = filter(price_summary, carat_centi == 0.99),
             color = "darkgreen", size = 4, shape = 18) +
  # 添加价格比率标签
  geom_text_repel(
```

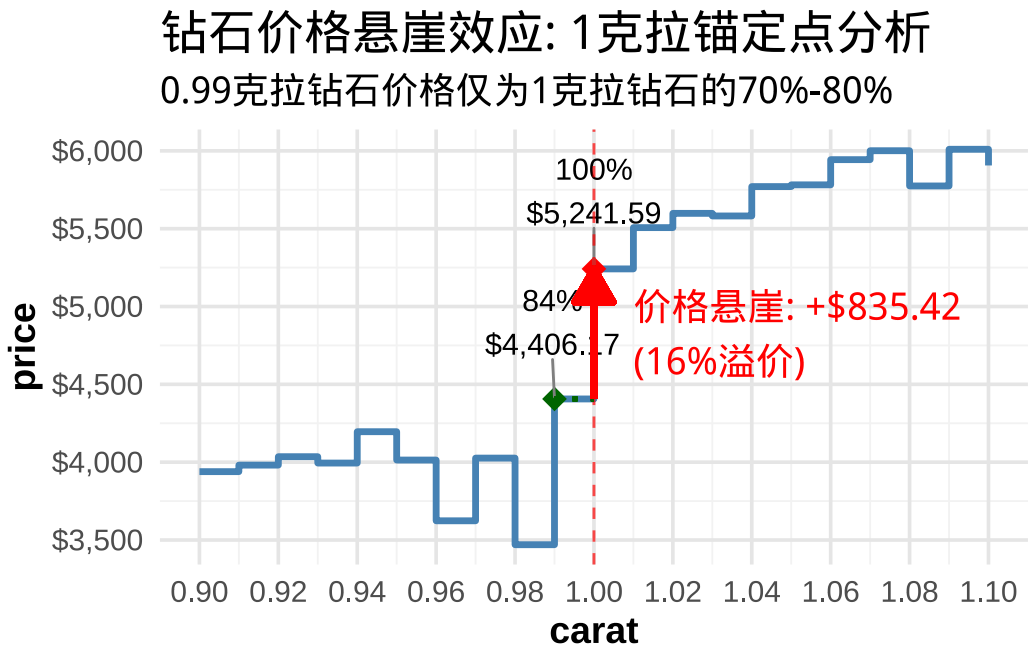
```
  data = filter(price_summary, carat_centi %in% c(0.99, 1.00)),
  aes(label = paste0(price_ratio_label, "\n", scales::dollar(avg_price))),
  nudge_y = 500,
  segment.color = "grey50",
  box.padding = 0.5,
  size = 4
) +
# 添加参考线
geom_vline(xintercept = 1.00, linetype = "dashed", color = "red", alpha = 0.7) +
# 添加悬崖效应箭头
geom_segment(
  aes(x = 0.99, xend = 1.00, y = price_gap$price_0.99, yend = price_gap$price_0.99),
  color = "darkgreen", linetype = "dotted", size = 0.8
) +
geom_segment(
  aes(x = 1.00, xend = 1.00, y = price_gap$price_0.99, yend = price_gap$price_1),
  color = "red", arrow = arrow(type = "closed", length = unit(0.2, "inches")), size = 1.2
) +
# 添加悬崖标签
annotate("text", x = 1.01, y = (price_gap$price_0.99 + price_gap$price_1)/2,
         label = paste0(" 价格悬崖: +", scales::dollar(price_gap$price_difference),
                        "\n(", scales::percent(1 - price_gap$price_ratio), " 溢价)"),
         hjust = 0, color = "red", size = 5) +
# 设置坐标轴和标签
scale_x_continuous(breaks = seq(0.90, 1.10, by = 0.02),
                   minor_breaks = seq(0.90, 1.10, by = 0.01)) +
scale_y_continuous(labels = scales::dollar_format()) +
labs(title = " 钻石价格悬崖效应: 1 克拉锚定点分析",
     subtitle = "0.99 克拉钻石价格仅为 1 克拉钻石的 70%-80%",
     x = "carat",
     y = "price") +
theme_minimal(base_size = 14) +
theme(
  plot.title = element_text(face = "bold", size = 18),
  axis.title = element_text(face = "bold"),
```

```
    panel.grid.major = element_line(color = "grey90"),
    panel.grid.minor = element_line(color = "grey95")
  )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

```
Warning in geom_segment(aes(x = 0.99, xend = 1, y = price_gap$price_0.99, : All aesthetics have
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

```
Warning in geom_segment(aes(x = 1, xend = 1, y = price_gap$price_0.99, yend = price_gap$price_1)
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```




```
# 打印关键统计结果
cat("==== 1 克拉价格悬崖关键统计 ====\n")
```

```
==== 1克拉价格悬崖关键统计 ====
```

```r
cat("0.99 克拉平均价格:", scales::dollar(price_gap$price_0.99), "\n")
```

0.99克拉平均价格: $4,406.17

```r
cat("1.00 克拉平均价格:", scales::dollar(price_gap$price_1), "\n")
```

1.00克拉平均价格: $5,241.59

```r
cat(" 价格差异:", scales::dollar(price_gap$price_difference), "\n")
```

价格差异: $835.42

```r
cat(" 价格比率:", scales::percent(price_gap$price_ratio),
    "(0.99 克拉价格是 1 克拉的", scales::percent(price_gap$price_ratio), ")\n")
```

价格比率: 84% (0.99克拉价格是1克拉的 84%)

```r
cat(" 样本数量 - 0.99 克拉:", price_summary$n[price_summary$carat_centi == 0.99], "\n")
```

样本数量 - 0.99克拉: 23

```r
cat(" 样本数量 - 1.00 克拉:", price_summary$n[price_summary$carat_centi == 1.00], "\n")
```

样本数量 - 1.00克拉: 1558

```r
# 设置分析区间和关键锚点
analysis_range <- c(0.9, 1.1)
anchor_points <- c(1, 1.5, 2)

# 数据预处理: 过滤区间并创建锚定标签
diamonds_analysis <- diamonds %>%
  filter(between(carat, analysis_range[1], analysis_range[2])) %>%
```

```
  mutate(
    # 创建锚定分类：将钻石标记为整数锚点附近（±0.01）
    anchor_group = case_when(
      between(carat, 0.99, 1.01) ~ "1.0ct 锚点区",
      between(carat, 1.49, 1.51) ~ "1.5ct 锚点区",
      between(carat, 1.99, 2.01) ~ "2.0ct 锚点区",
      TRUE ~ " 非锚点区"
    ),
    # 创建精确到百分位的克拉分组
    carat_group = round(carat, 2)
  )

# 1. 价格曲线可视化（0.9-1.1 克拉区间）
ggplot(diamonds_analysis, aes(carat, price)) +
  geom_point(aes(color = anchor_group), alpha = 0.3, size = 1.5) +
  geom_smooth(method = "loess", span = 0.2, color = "darkblue", se = FALSE) +
  geom_vline(xintercept = anchor_points, linetype = "dashed", color = "red") +
  labs(title = " 钻石价格悬崖效应分析（0.9-1.1 克拉区间）",
       subtitle = " 红色虚线表示消费者心理锚定点",
       x = " 克拉重量", y = " 价格（美元）",
       color = " 关键锚定区域") +
  scale_x_continuous(breaks = seq(0.9, 1.1, 0.02)) +
  theme_minimal() +
  theme(legend.position = "top")
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: pseudoinverse used at 0.899

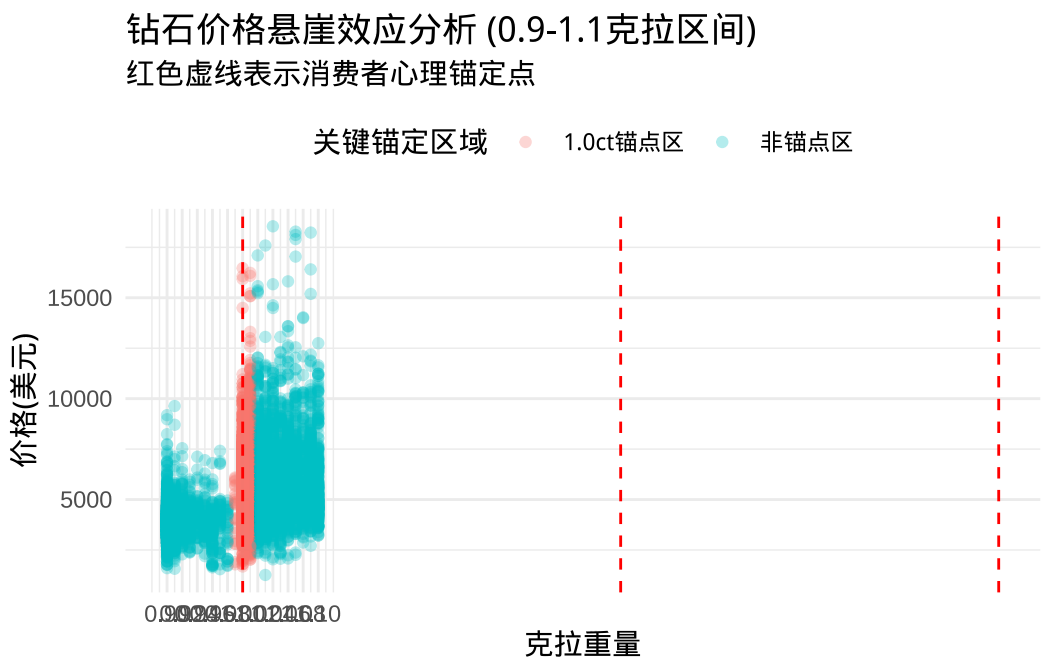
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: neighborhood radius 0.021

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: reciprocal condition number 1.0147e-14
```

```
Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: There are other near singularities as well. 0.0001

Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
: zero-width neighborhood. make span bigger

Warning: Failed to fit group -1.
Caused by error in `predLoess()`:
! 外接函数调用时不能有NA/NaN/Inf(arg5)
```




```
# 2. 关键锚定点价格对比分析
anchor_comparison <- diamonds_analysis %>%
  filter(anchor_group != " 非锚点区") %>%
  group_by(anchor_group, carat_group) %>%
  summarise(
    avg_price = mean(price),
    n = n(),
    .groups = "drop"
  ) %>%
```

```
  # 计算锚点两侧价格比率
  group_by(anchor_group) %>%
  mutate(
    price_ratio = avg_price / max(avg_price)
  )

# 打印关键价格悬崖数据
anchor_comparison %>%
  select(anchor_group, carat_group, avg_price, price_ratio) %>%
  arrange(anchor_group, carat_group) %>%
  filter(price_ratio < 1) %>%  # 筛选低于锚点的价格
  mutate(price_ratio = scales::percent(price_ratio, accuracy = 1))
```

```
# A tibble: 2 x 4
# Groups:   anchor_group [1]
  anchor_group carat_group avg_price price_ratio
  <chr>              <dbl>     <dbl> <chr>
1 1.0ct锚点区         0.99     4406. 80%
2 1.0ct锚点区         1        5242. 95%
```

```
# 3. 1 克拉锚定点详细分析（0.99ct vs 1.00ct）
one_carat_gap <- diamonds_analysis %>%
  filter(carat_group %in% c(0.99, 1.00)) %>%
  group_by(carat_group) %>%
  summarise(
    avg_price = mean(price),
    median_price = median(price),
    n = n()
  ) %>%
  mutate(
    price_ratio = avg_price / max(avg_price),
    price_gap = max(avg_price) - avg_price
  )
```

```
# 输出 1 克拉悬崖结果
cat("\n==== 1 克拉价格悬崖分析 ====\n")
```

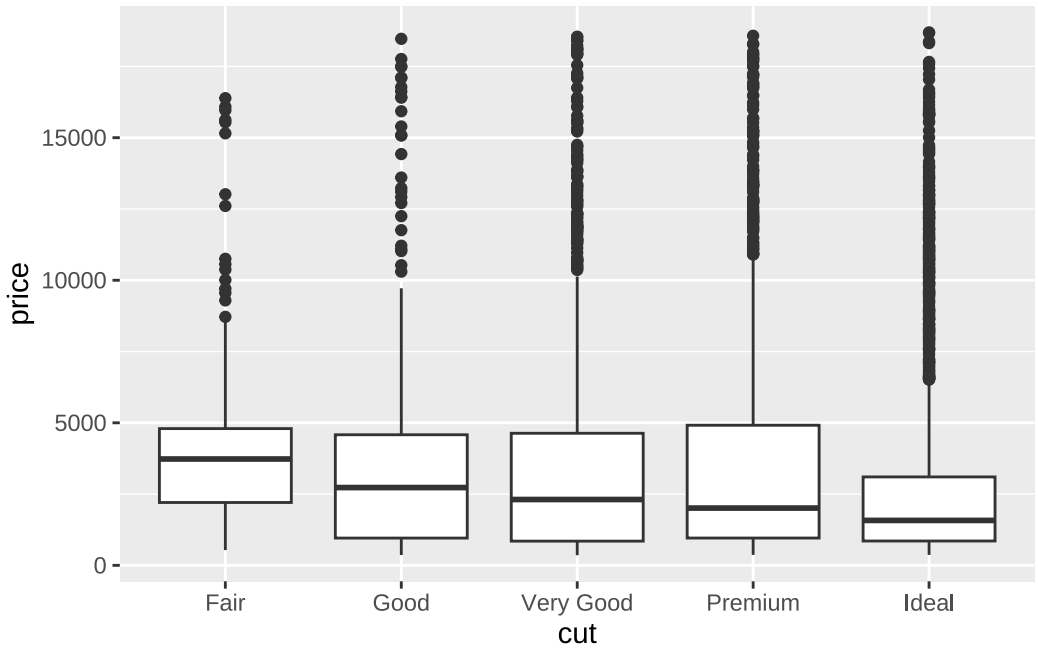
```
==== 1克拉价格悬崖分析 ====
```

```
one_carat_gap %>%
  mutate(across(where(is.numeric), ~scales::comma(.x, accuracy = 1))) %>%
  knitr::kable()
```

carat_group	avg_price	median_price	n	price_ratio	price_gap
1	4,406	4,780	23	1	835
1	5,242	4,864	1,558	1	0

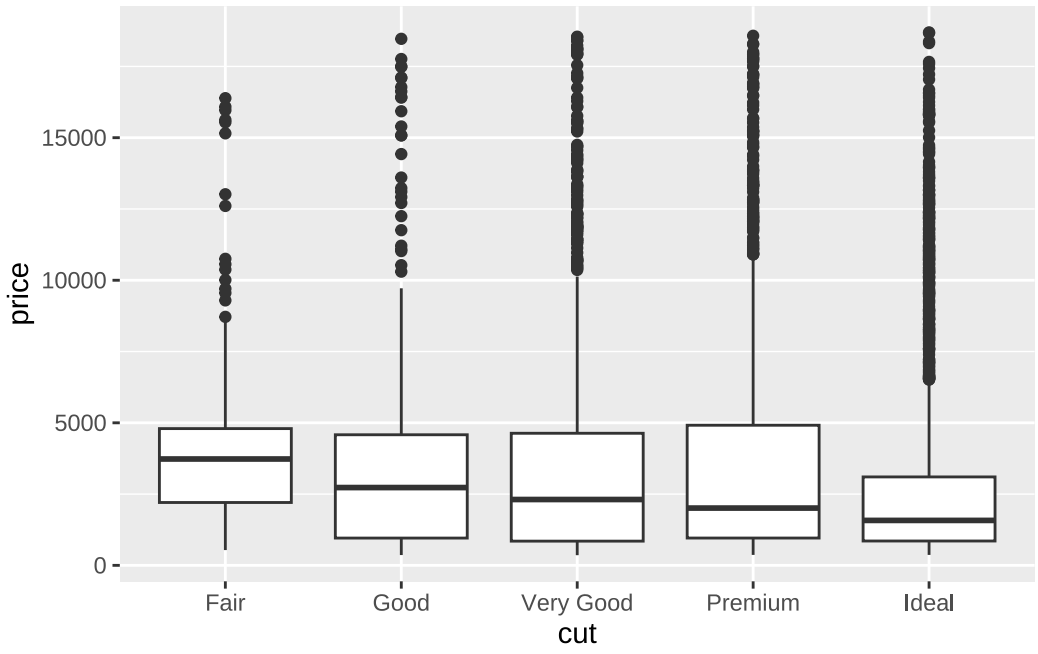
6.2.3 控制 color 变量后的 cut 与 price 间关系分析

```
diamonds  %>%
filter(color == "D")  %>%
ggplot(aes(x = cut, y = price))  +
geom_boxplot()
```

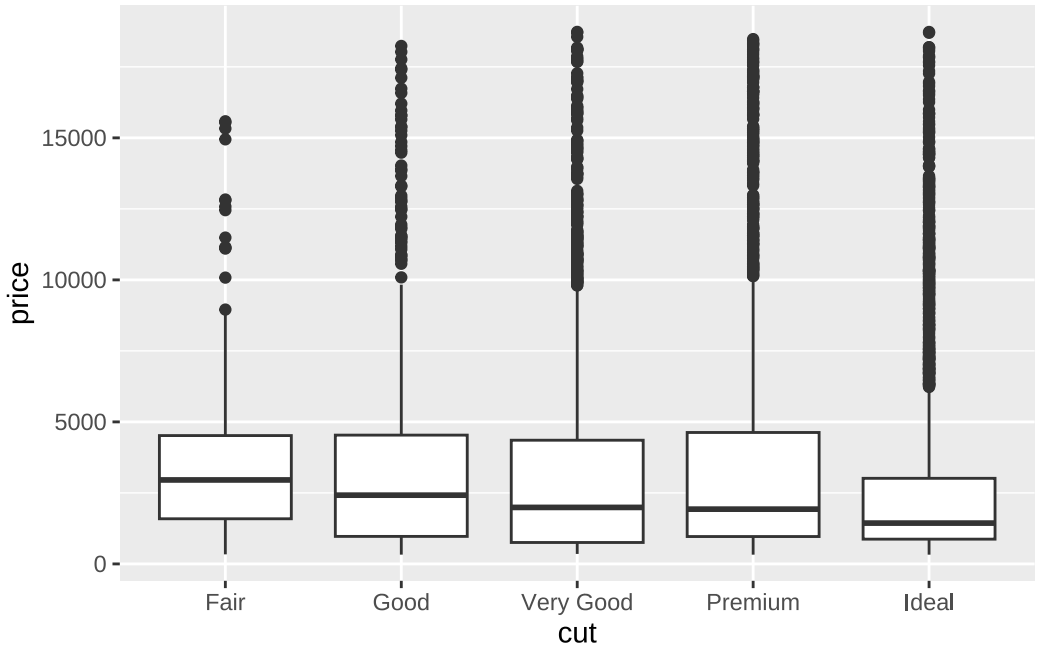



```
diamonds %>%
group_split(by = color)  %>%
map(~ggplot(.x, aes(x = cut, y=price)) +
geom_boxplot())
```

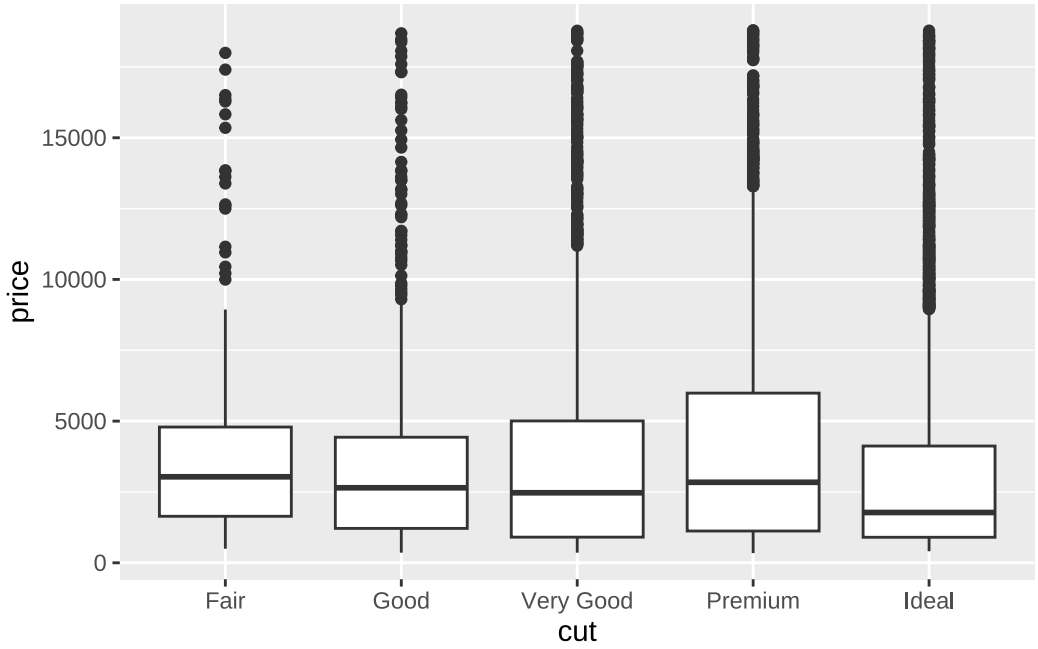
```
[[1]]
```


price
15000
10000
5000
0
Fair
Good
Very Good
Premium
Ideal
cut

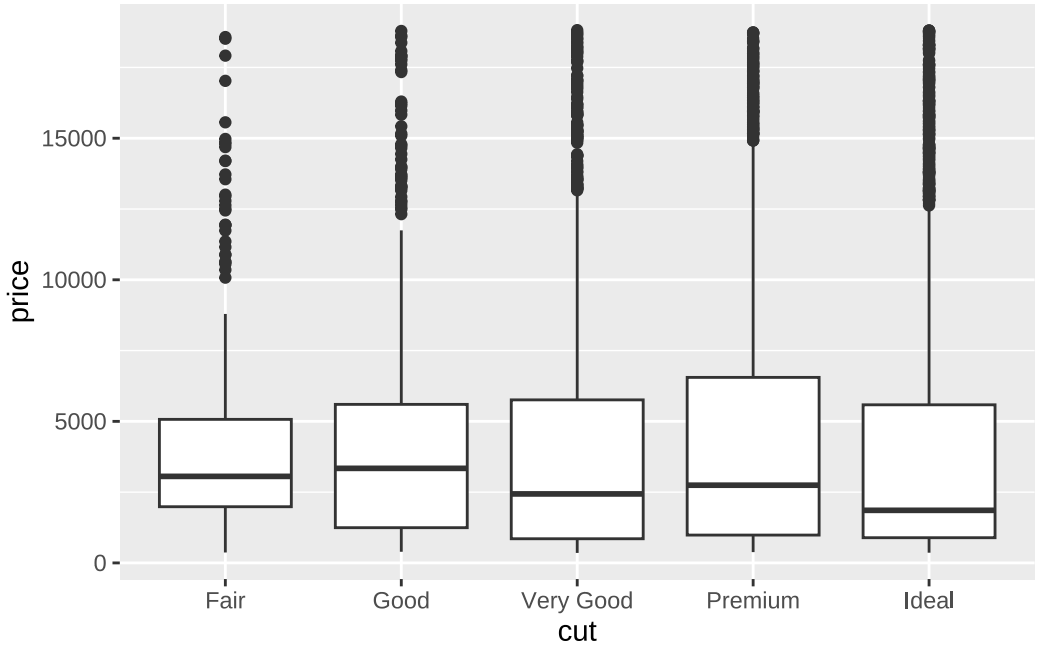
```
[[2]]
```


price
15000
10000
5000
0
Fair
Good
Very Good
Premium
Ideal
cut

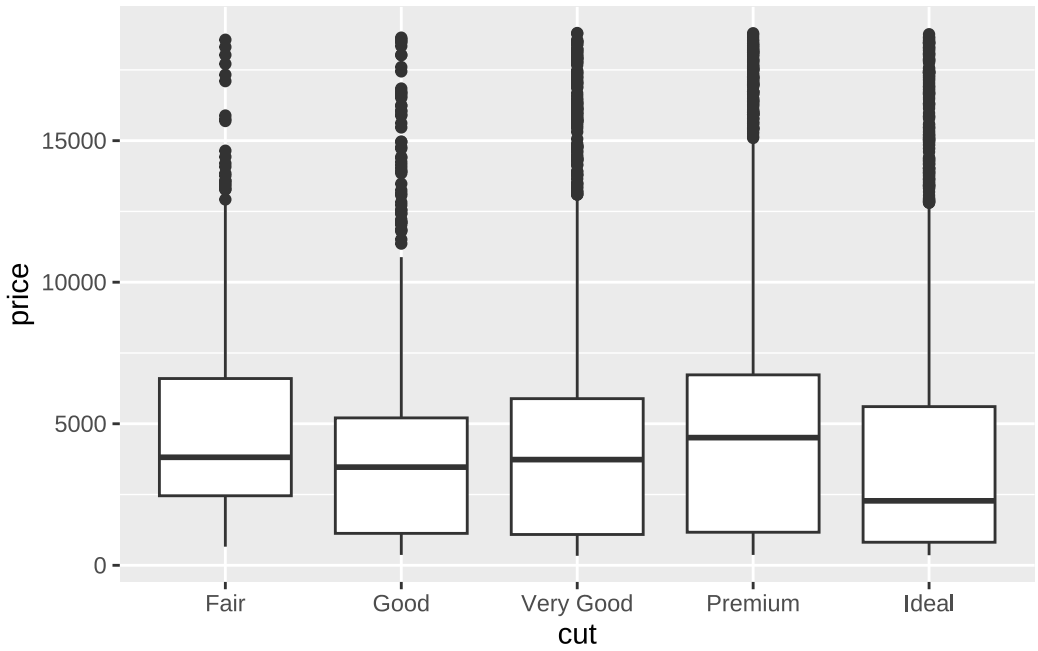
```
[[3]]
```


15000
10000
5000
0
price
Fair
Good
Very Good
Premium
Ideal
cut

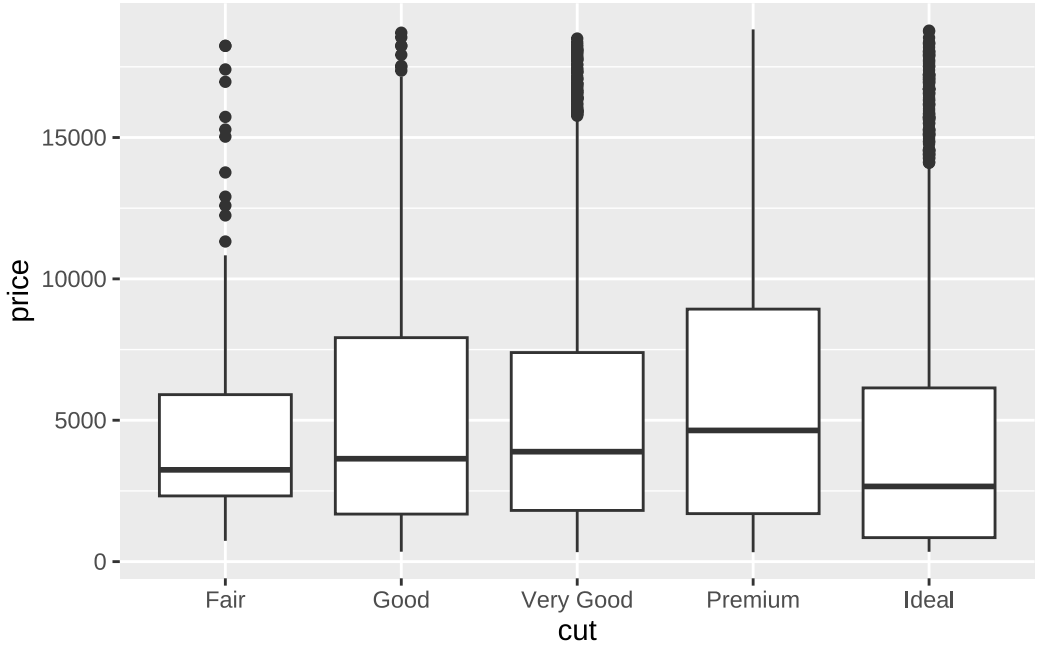
```
[[4]]
```


15000
10000
5000
0
price
Fair
Good
Very Good
Premium
Ideal
cut

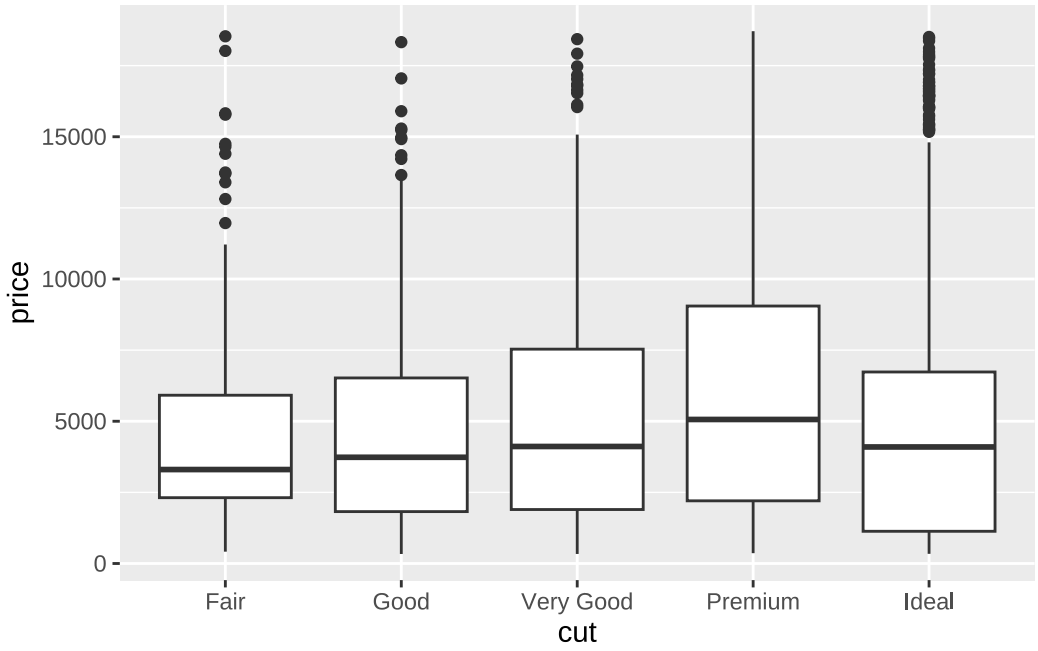
```
[[5]]
```


15000
10000
5000
0
price
Fair
Good
Very Good
Premium
Ideal
cut

```
[[6]]
```

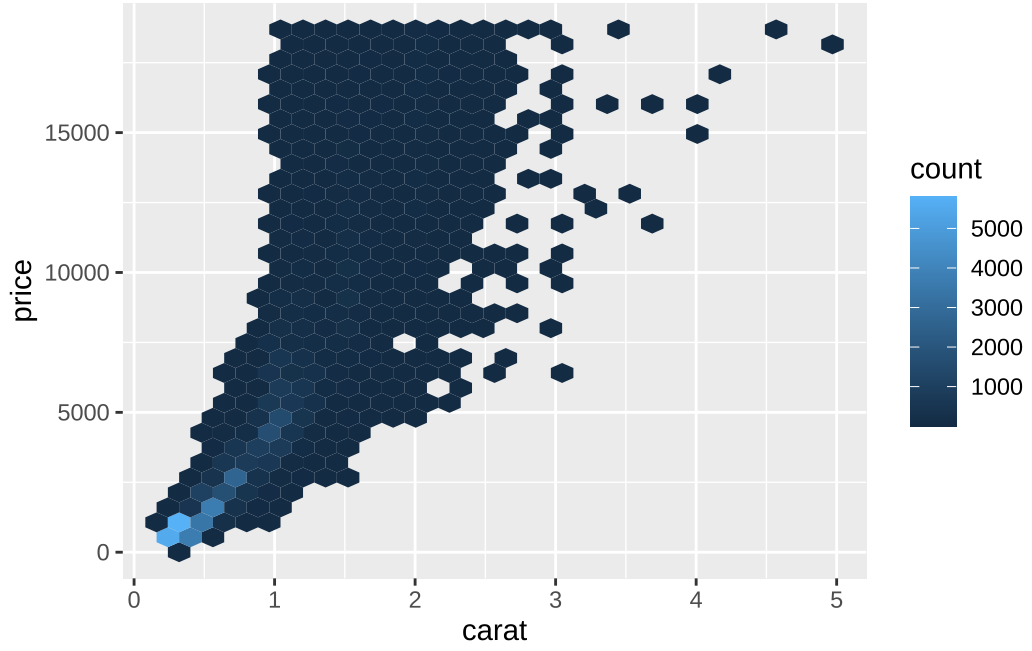

price
15000
10000
5000
0
Fair
Good
Very Good
Premium
Ideal
cut

```
[[7]]
```


price
15000
10000
5000
0
Fair
Good
Very Good
Premium
Ideal
cut

6.3 度量 carat 与 price 之间的关系

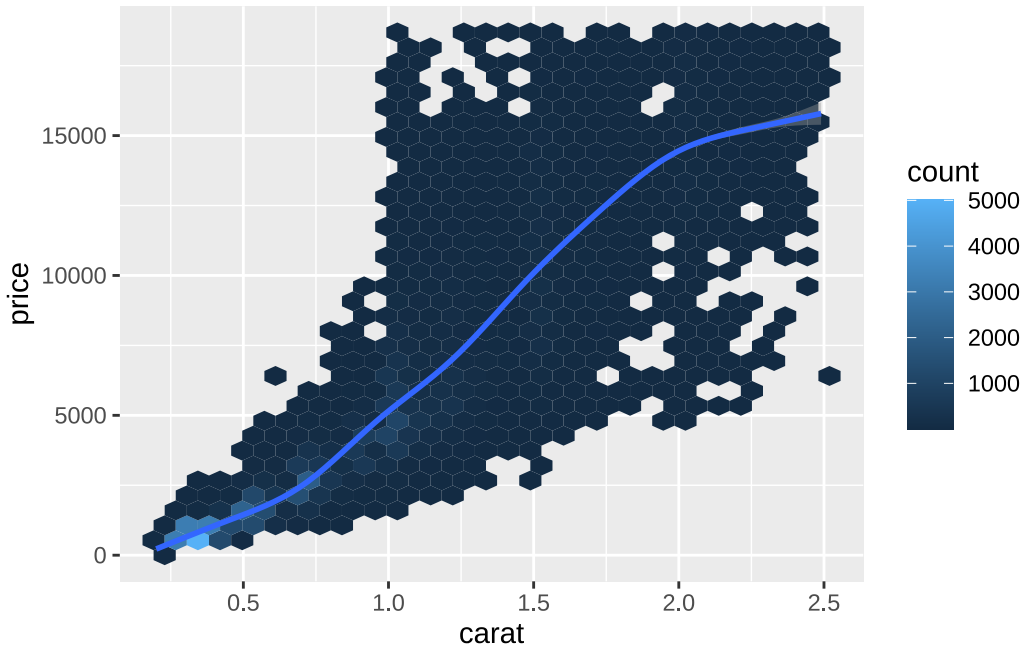
```
diamonds  %>%
ggplot(aes(x = carat, y = price)) +
geom_hex()
```



- 去噪

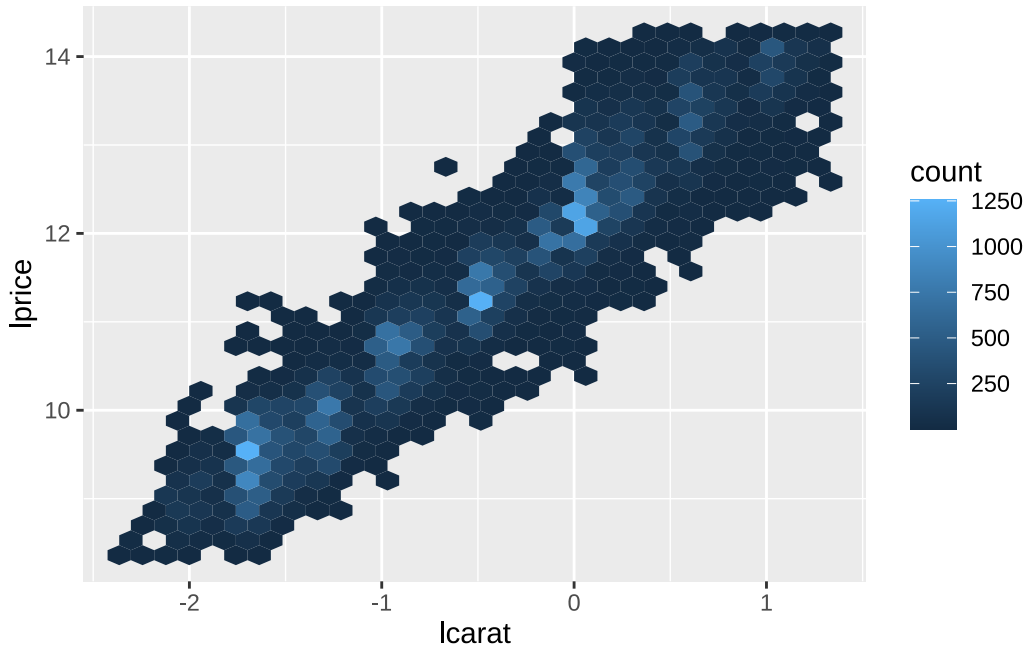
```
diamonds  %>%
filter(carat<2.5) %>%
ggplot(aes(x = carat, y = price)) +
geom_hex() +
geom_smooth()
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```




```
diamondsNew  <- diamonds  %>%
filter(carat < 2.5)  %>%
mutate(lcarat = log2(carat), lprice = log2(price))
```

```
diamondsNew %>%
ggplot(aes(x = lcarat, y = lprice)) +
geom_hex()
```

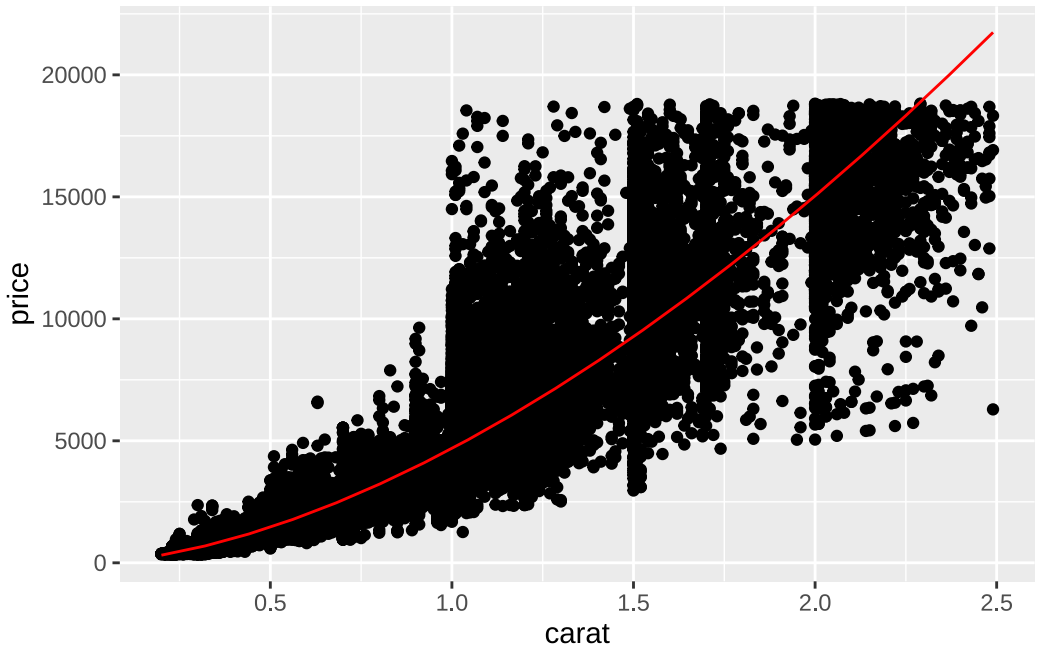


- 建立线性回归模型

```
modellm <- lm(lprice~lcarat, data = diamondsNew)
```

```
library(modelr)
grid <- diamondsNew  %>%
data_grid(carat = seq_range(carat, 20))  %>%
mutate(lcarat = log2(carat))  %>%
add_predictions(modellm, "lprice")  %>%
mutate(pricePred = 2^lprice)
```

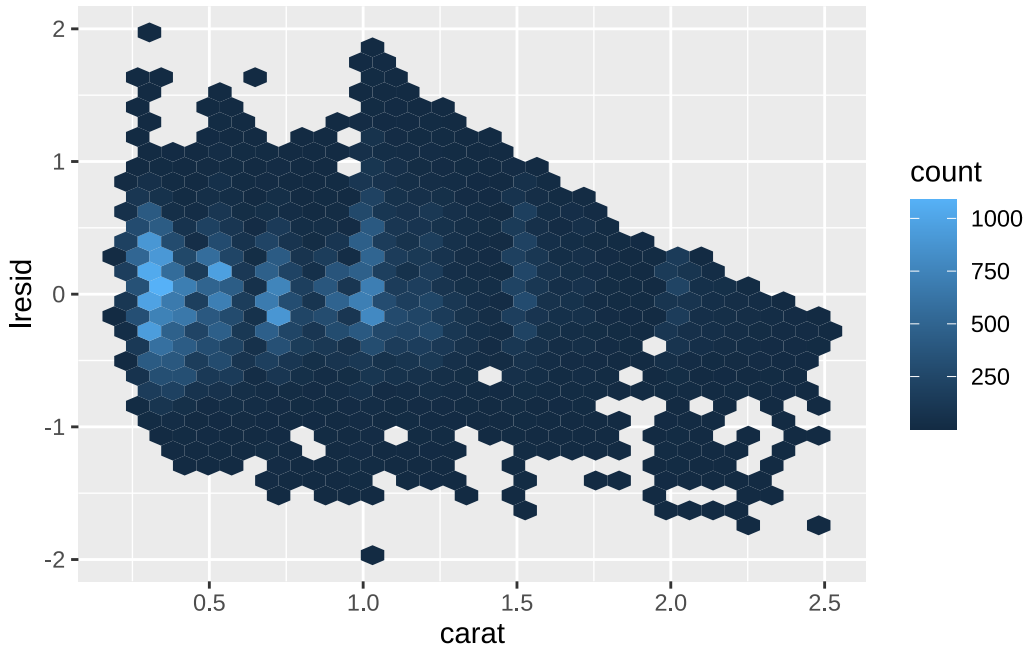
```
ggplot(diamondsNew, aes(x = carat, y = price)) +
geom_point() +
geom_line(data = grid, aes(x = carat, y = pricePred), col = "red")
```



残差分析

```
diamondsNew <- diamondsNew  %>%
add_residuals(modellm, "lresid")

ggplot(diamondsNew, aes(x= carat, y = lresid)) +
geom_hex()
```



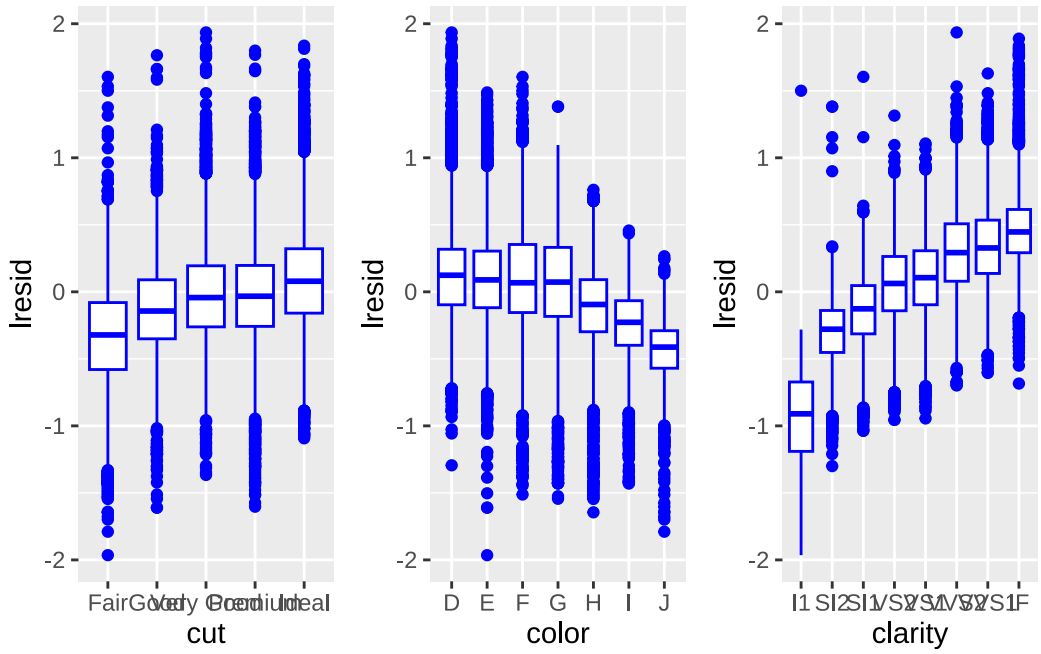
- cut color clarity 与残差的关系图

```
p1 <- ggplot(data = diamondsNew, aes(cut, lresid)) +
geom_boxplot(color = "blue")
```

```
p2 <- ggplot(data = diamondsNew, aes(color, lresid)) +
geom_boxplot(color = "blue")
```

```
p3 <- ggplot(data = diamondsNew, aes(clarity, lresid)) +
geom_boxplot(color = "blue")
```

```
grid.arrange(p1, p2, p3, ncol = 3)
```




```
mlm <- lm(lprice ~ lcarat + cut + color + clarity, data = diamondsNew)
diamondsNew %>%
data_grid(cut, .model = mlm) %>%
add_predictions(mlm, "prevalue")
```

```
# A tibble: 5 x 5
  cut       lcarat color clarity prevalue
  <ord>      <dbl> <chr> <chr>      <dbl>
1 Fair      -0.515 G     VS2         11.2
2 Good      -0.515 G     VS2         11.3
3 Very Good -0.515 G     VS2         11.4
4 Premium   -0.515 G     VS2         11.4
5 Ideal     -0.515 G     VS2         11.4
```

```
diamondsNew  %>%
group_by(by = clarity)  %>%
summarise(nNum = n())
```

```
# A tibble: 8 x 2
```

```
  by     nNum
  <ord> <int>
1 I1      704
2 SI2    9105
3 SI1   13054
4 VS2   12254
5 VS1    8169
6 VVS2   5066
7 VVS1   3655
8 IF     1790
```

```
diamondsNew  %>%
group_by(by = color)  %>%
summarise(nNum = n())
```

```
# A tibble: 7 x 2
  by     nNum
  <ord> <int>
1 D      6771
2 E      9792
3 F      9537
4 G     11277
5 H      8264
6 I      5387
7 J      2769
```

- 连续值用的中位数
- 离散值用的是众数?
- 考试题

7 案例 2: 航班数据集分析

```
library(magrittr)
library(tidyverse)
```

```
Warning: 程辑包'ggplot2'是用R版本4.3.3 来建造的
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.3     v readr     2.1.4
v forcats   1.0.0     v stringr   1.5.0
v ggplot2   3.5.2     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.0
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x tidyr::extract()   masks magrittr::extract()
x dplyr::filter()    masks stats::filter()
x dplyr::lag()       masks stats::lag()
x purrr::set_names() masks magrittr::set_names()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become e
```

```
library(modelr)
```

```
library(nycflights13)
head(flights)
```

```
# A tibble: 6 x 19
   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
```

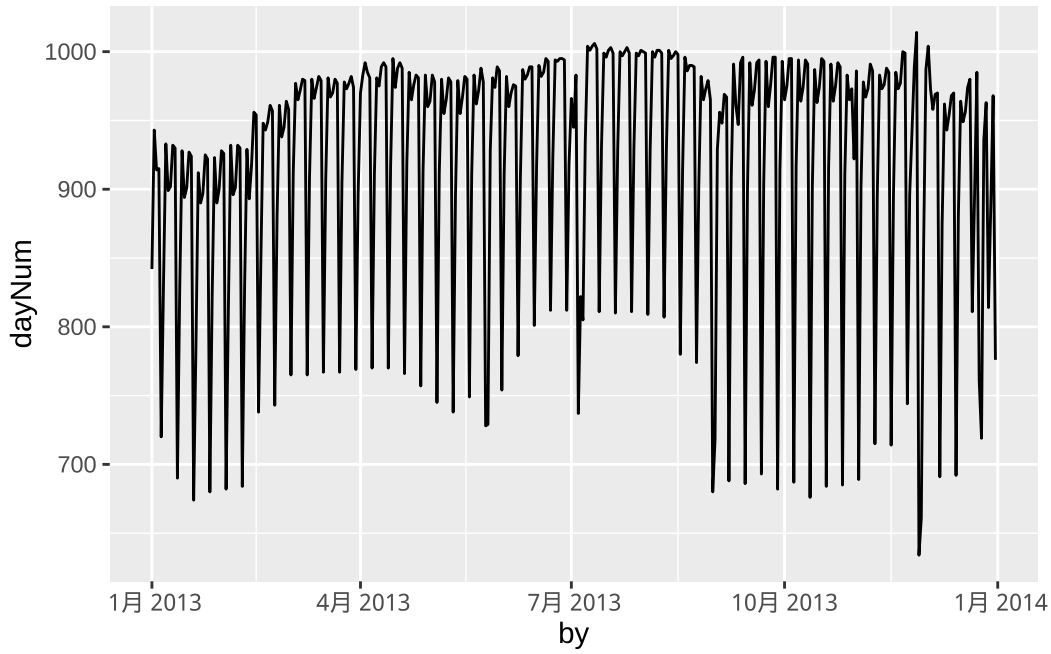
```
  <int> <int> <int>    <int>          <int>     <dbl>    <int>          <int>
1  2013     1     1      517            515         2      830            819
2  2013     1     1      533            529         4      850            830
3  2013     1     1      542            540         2      923            850
4  2013     1     1      544            545        -1     1004           1022
5  2013     1     1      554            600        -6      812            837
6  2013     1     1      554            558        -4      740            728
# i 11 more variables: arr_delay <dbl>, carrier <chr>, flight <int>,
#   tailnum <chr>, origin <chr>, dest <chr>, air_time <dbl>, distance <dbl>,
#   hour <dbl>, minute <dbl>, time_hour <dttm>
```

```
library(lubridate)
flightsNew <- flights %>%
mutate(date = make_date(year, month, day))
```

```
daily <- flightsNew  %>%
group_by(by = date)  %>%
summarise(dayNum = n())
head(daily)
```

```
# A tibble: 6 x 2
  by         dayNum
  <date>      <int>
1 2013-01-01    842
2 2013-01-02    943
3 2013-01-03    914
4 2013-01-04    915
5 2013-01-05    720
6 2013-01-06    832
```

```
ggplot(daily, aes(by, dayNum)) +
geom_line()
```




```
dailyNew <- daily  %>%
mutate(week = wday(by,label=TRUE))
```

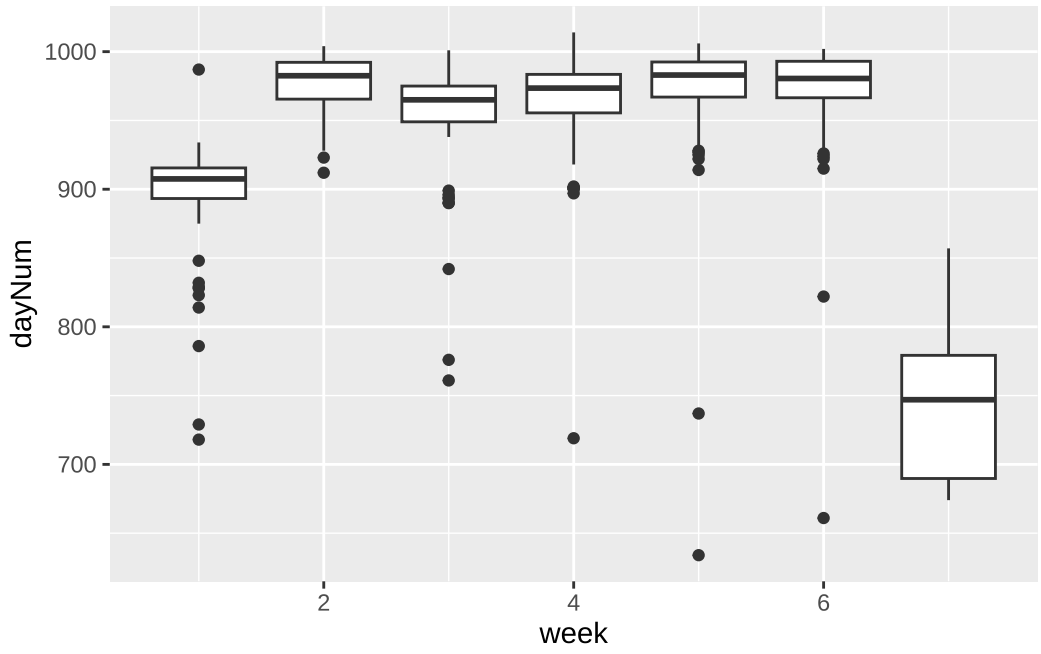
```
dailyNewTest <- daily  %>%
mutate(week = wday(by))
```

- 为什么会出现一个箱子?（石学成）

```
class(dailyNewTest$week)
```

```
[1] "numeric"
```

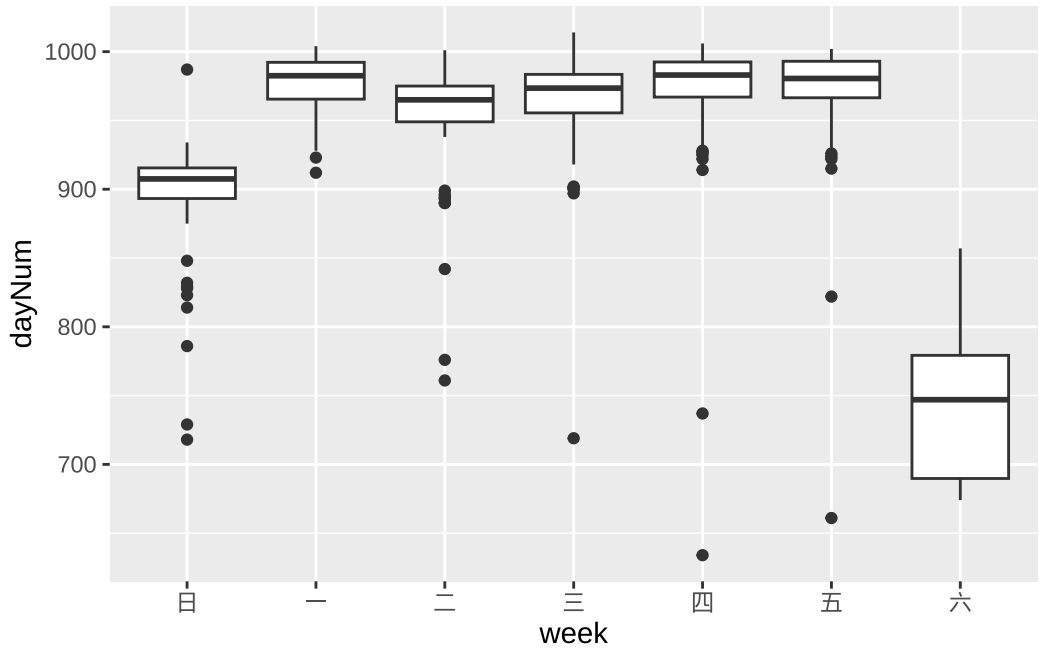
```
ggplot(dailyNewTest, aes(week, dayNum)) +
geom_boxplot(aes(group=week))
```




```
class(dailyNew$week)
```

```
[1] "ordered" "factor"
```

```
p1 <- ggplot(dailyNew, aes(week, dayNum)) + geom_boxplot()
p1
```




```
mod = lm(dayNum~week, data = dailyNew)
mod
```

```
Call:
lm(formula = dayNum ~ week, data = dailyNew)

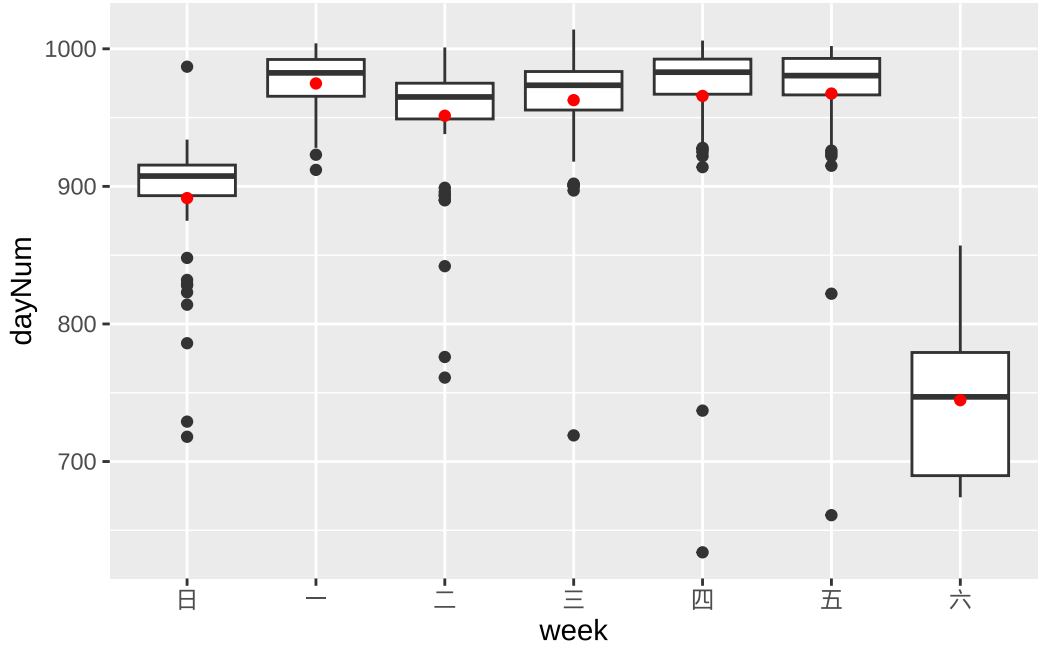
Coefficients:
(Intercept)        week.L        week.Q        week.C        week^4        week^5
    922.595       -83.322      -155.111       -62.834       -80.126        -4.967
     week^6
    -16.934
```

```
grid <- dailyNew %>%
  data_grid(week) %>%
  add_predictions(mod, "numPred")
grid
```

```
# A tibble: 7 x 2
```

```
  week  numPred
  <ord>   <dbl>
1 日       891.
2 一       975.
3 二       951.
4 三       963.
5 四       966.
6 五       967.
7 六       745.
```

```
p1 +
geom_point(data = grid, aes(week, numPred),col="red")
```

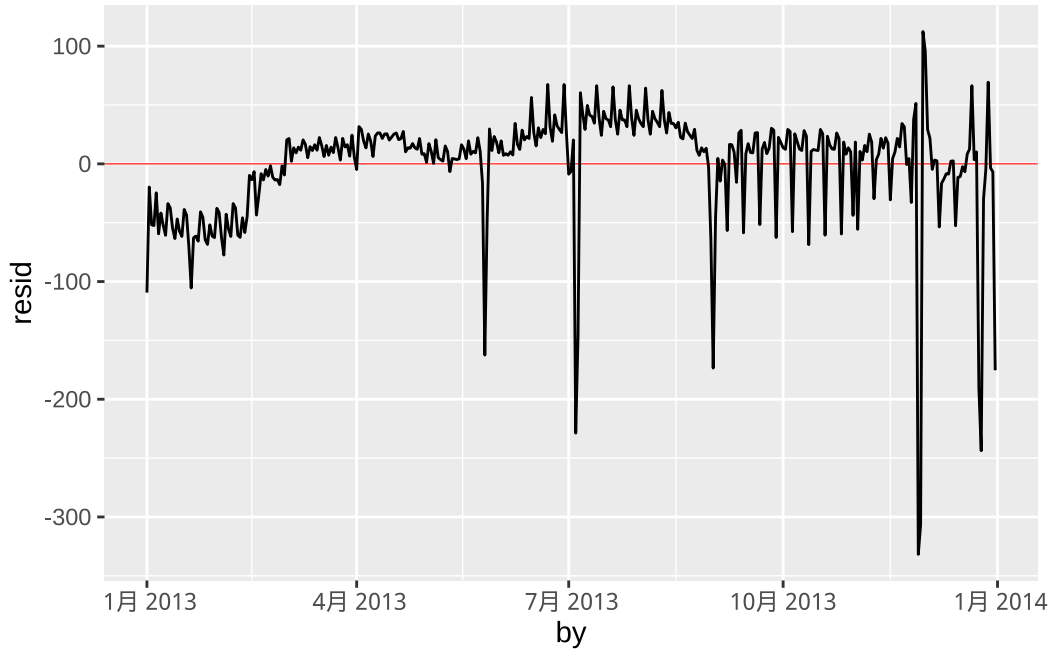


- 残差图

```
dailyRes <- dailyNew %>%
  add_residuals(mod)

ggplot(dailyRes, aes(by, resid)) +
```

```
  geom_ref_line(h=0, col = "red", size = .2)+
  geom_line()
```



找到残差比较大的点

```
dailyRes  %>%
  arrange(-abs(resid))
```

```
# A tibble: 365 x 4
   by         dayNum week  resid
   <date>      <int> <ord> <dbl>
 1 2013-11-28    634 四    -332.
 2 2013-11-29    661 五    -306.
 3 2013-12-25    719 三    -244.
 4 2013-07-04    737 四    -229.
 5 2013-12-24    761 二    -190.
 6 2013-12-31    776 二    -175.
 7 2013-09-01    718 日    -173.
 8 2013-05-26    729 日    -162.
```

```
 9 2013-07-05    822 五    -145.
10 2013-11-30    857 六     112.
# i 355 more rows
```

```
head(dailyRes)
```

```
# A tibble: 6 x 4
  by         dayNum week   resid
  <date>      <int> <ord>  <dbl>
1 2013-01-01    842 二    -109.
2 2013-01-02    943 三     -19.7
3 2013-01-03    914 四     -51.7
4 2013-01-04    915 五     -52.5
5 2013-01-05    720 六     -24.6
6 2013-01-06    832 日     -59.5
```

```
#filter(abs(resid)>200)
```

8 参数估计

8.1 矩估计

8.2 极大似然估计

8.3 区间估计

8.3.1

从自动机床加工的同类零件中抽取 16 件，测得长度分别为 (单位:cm):

12.15, 12.12, 12.01, 12.08, 12.09, 12.16, 12.06, 12.13, 12.07, 12.11, 12.08, 12.01, 12.03, 12.01, 12.03, 12.06

假设零件长度服从正态分布，求总体长度方差与标准差的 95% 置信区间.

```
a = c(12.15, 12.12, 12.01, 12.08, 12.09, 12.16, 12.06, 12.13, 12.07, 12.11, 12.08, 12.01,12.03,
n = length(a)
alpha = 0.05
leftPonit = (n-1)*var(a) / (qchisq(1-alpha/2, n-1))
rightPonit = (n-1)*var(a) / (qchisq(alpha/2, n-1))
leftPonit
```

```
[1] 0.001331471
```

```
rightPonit
```

```
[1] 0.005844649
```

8.3.2 双正态总体均值差的区间估计 (方差已知)

两台机床生产同一个型号的滚珠，从甲机床生产的滚珠中抽取 8 个，从乙机床生产的滚珠中抽取 9 个，测得这些滚珠的直径（mm）如下: 甲机床:

$$15.0, 14.8, 15.2, 15.4, 14.9, 15.1, 15.2, 14.8$$

乙机床:

$$15.2, 15.0, 14.8, 15.1, 15.6, 14.8, 15.1, 14.5, 15.0$$

若两台机床生产的滚珠直径的标准差分别是 0.18 与 0.24 求这两台机床生产的滚珠直径均值差置信度为 0.90 的置信区间.

```
x = c(15.0, 14.8, 15.2, 15.4, 14.9, 15.1, 15.2, 14.8)
y = c(15.2, 15.0, 14.8, 15.1, 15.6, 14.8, 15.1, 14.5, 15.0)
xbar = mean(x)
ybar = mean(y)
n1 = length(x)
n2 = length(y)
alpha = 0.1
mupoint = qnorm(1 - alpha/2)
leftPoint = (xbar - ybar) - sqrt((0.18^2)/n1 + (0.24^2)/n2) * mupoint
rightPoint = (xbar - ybar) + sqrt((0.18^2)/n1 + (0.24^2)/n2) * mupoint
leftPoint
```

```
[1] -0.1292567
```

```
rightPoint
```

```
[1] 0.2070344
```

8.3.3 双正态总体均值差的区间估计 (方差相等且未知)

机床厂某日从两台机床加工的零件中，分别抽取若干个样品，测得零件的尺寸分别如下（单位：cm）：
A 台：

$$6.2, 5.7, 6.5, 6.0, 6.3, 5.8, 5.7, 6.0, 6.0, 5.8, 6.0$$

B 台:

$$5.6, 5.9, 5.6, 5.7, 5.8, 6.0, 5.5, 5.7, 5.5$$

假设两台机器加工的零件尺寸均服从正态分布,
且方差相等, 取置信度为 0.95, 试求两台机器加工的零件平均尺寸之差的区间估计.

```
x = c(6.2,  5.7,  6.5,  6.0, 6.3,  5.8, 5.7, 6.0,  6.0,  5.8,  6.0 )
y = c(5.6,  5.9,  5.6,  5.7,  5.8,  6.0,  5.5, 5.7,  5.5)
n1 = length(x)
n2 = length(y)
xbar = mean(x)
ybar = mean(y)
sw2 = ((n1-1) * var(x) + (n2 - 1) * var(y))/(n1+n2-2)
sw = sqrt(sw2)
alpha = 0.05
tpoint = qt(1-alpha/2, n1+n2-2)
leftPoint = (xbar - ybar) - tpoint * sw * sqrt(1/n1+1/n2)
rightPot = (xbar - ybar) + tpoint * sw * sqrt(1/n1+1/n2)
leftPoint
```

```
[1] 0.09120869
```

```
rightPoint
```

```
[1] 0.2070344
```

- 如果不告诉两正态总体方差相等，应该如何做?

```
snx = var(x)
sny = var(y)
Fvalue = snx / sny
Fvalue
```

```
[1] 2.133333
```

```
Fleftpoint = qf(alpha/2, n1-1, n2-1)
Frightpoint = qf(1-alpha/2, n1-1, n2-1)
Fleftpoint
```

```
[1] 0.2594107
```

```
Frightpoint
```

```
[1] 4.295127
```

- Fvalue 未落入拒绝域，接受原假设，可认为两正态总体方差相等

9 假设检验

9.0.1 第一类错误

- 拒真概率：当原假设为真时，检验统计量落入拒绝域的概率

$$P(X \in W|H_0 = TRUE)$$

- α

9.0.2 第二类错误

- 取伪概率：当备择假设为真时，检验统计量未落入拒绝域的概率

$$P(X \notin W|H_1 = TRUE)$$

- β
- 假验的效能：正确接受备择假设的概率

$$(1 - \beta) \times 100\%$$

9.0.3 p 值

9.0.4 样本量

- 效能分析：决定统计学检验的效能和计算揭示一定大小效应所需要的最小样本量
  - α: 第一类错误概率
  - β: 第二类错误概率

  - d: 效应的大小，即所研究的效应相对样本的标准差 σ 的大小
  - n: 样本大小

- 上面四个参数，只要固定三个，就可以确定剩余参数的值

9.1 题目

9.1.1 灯泡寿命是否为理论值

某型灯泡寿命 X 服从正态分布, 从一批灯泡中任意取出 10 只, 测得其寿命分别为 (单位:h)

$$1490, 1440, 1680, 1610, 1500, 1750, 1550, 1420, 1800, 1580$$

能否认为这批灯泡平均寿命为 1600h（$\alpha = 0.05$）

```
a = c(1490, 1440, 1680, 1610,1500, 1750, 1550, 1420, 1800, 1580)
xbar = mean(a)
n = length(a)
tvalue = (xbar - 1600) / (sd(a)/sqrt(n))
tvalue
```

```
[1] -0.442742
```

```
alpha = 0.05
tpoint = qt(1-alpha/2, n-1)
tpoint
```

```
[1] 2.262157
```

- tvalue 未落入拒绝域，接受原假设

9.1.2 改进工艺后方差是否为理论值

某炼钢厂铁水含碳质量分数 X 在正常情况下服从正态分布. 现对操作工艺进行了改革又测量了 5 炉铁水, 含碳质量分数分别为:

$$4.421, 4.052, 4.357, 4.287, 4.683$$

是否可以认为由新工艺炼出的铁水含碳质量分数的方差仍为 0.108^2

```
a = c(4.421, 4.052, 4.357, 4.287, 4.68)
n = length(a)
chiValue = ((n-1) * var(a)) / (0.108^2)
chiValue
```

```
[1] 17.68872
```

```
alpha = 0.05
chiLeftPoint = qchisq(alpha/2, n-1)
chiRightPoint = qchisq(1-alpha/2, n-1)
chiLeftPoint
```

```
[1] 0.4844186
```

```
chiRightPoint
```

```
[1] 11.14329
```

- chiValue 落入拒绝域，拒绝原假设

9.1.3 药物药效分析

某种物种在处理前与处理后取样分析其含脂率如下: 处理前:

$$0.19, 0.18, 0.21, 0.30, 0.660.42, 0.08, 0.12, 0.30, 0.27$$

处理后:

$$0.15, 0.13, 0.00, 0.07, 0.24, 0.19, 0.04, 0.08, 0.20, 0.12$$

假定处理前后含脂率都服从正态分布, 且相互独立, 方差相等. 问处理前后含脂率的均值有无显著差异（$\alpha = 0.05$）

- 建立原假设与备择假设

$$H_0 : \mu_1 = \mu_2, H_1 : \mu_1 \neq \mu_2$$

- 检验统计量

$$T = \frac{\bar{X} - \bar{Y}}{s_w^* \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

- 拒绝域

$$D = \left\{ w \middle| |Z| > \mu_{\alpha/2} \right\}$$

```
a = c(0.19, 0.18, 0.21, 0.30, 0.66, 0.42, 0.08, 0.12, 0.30 , 0.27)
b = c(0.15, 0.13, 0.00, 0.07, 0.24, 0.19, 0.04, 0.08, 0.20, 0.12)
n1 = length(a)
n2 = length(b)
xbar = mean(a)
ybar = mean(b)
sw = sqrt(((n1-1)*var(a) + (n2-1)*var(b))/(n1+n2-2))
# varx = sum((a - mean(a))^2)/(n1-1)
# vary = sum((b - mean(b))^2)/(n2-1)
# sw = sqrt(((n1-1)*varx + (n2-1)*vary)/(n1+n2-2))
tvalue = (xbar-ybar)/(sw * sqrt(1/n1+1/n2))
tvalue
```

```
[1] 2.595701
```

```
alpha = 0.05
tAlpha = qt(1-alpha/2, n1+n2-2)
tAlpha
```

```
[1] 2.100922
```

- tvalue 落入拒绝域，拒绝原假设，均值有显著差异

9.1.4 断裂强度方差

为了考察温度对某物体断裂强度的影响, 在 70 度与 80 度下分别重复作了 8 次试验, 得断裂强度的数据如下 (单位：Mpa):

70 度:

$$20.5, 18.8, 19.8, 20.9, 21.5, 19.5, 21.0, 21.2$$

80 度:

$$17.7, 20.3, 20.0, 18.8, 19.0, 20.1, 20.2, 19.1$$

假设断裂强度服从正态分布, 试问两种情形下断裂强度方差是否相同（$\alpha = 0.05$）

```
a = c(20.5, 18.8, 19.8, 20.9, 21.5, 19.5, 21.0, 21.2)
b = c(17.7, 20.3, 20.0, 18.8, 19.0, 20.1, 20.2, 19.1)
varX = var(a)
varY = var(b)
Fvalue = varX/varY
Fvalue
```

```
[1] 1.068966
```

```
n1 = length(a)
n2 = length(b)
alpha = 0.05
Fleft = qf(alpha/2,n1-1, n2-1)
Fright = qf(1-alpha/2,n1-1, n2-1)
Fleft
```

```
[1] 0.2002038
```

```
Fright
```

```
[1] 4.994909
```

- Fvaleu 未落入拒绝域，接受原假设

10 课程总结

参考资料 (Wickham and Grolemund 2023)

参考资料

Wickham, Çetinkaya-Rundel, Hadley, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. R for Data Science: Import, Tidy, Transform, Visualize,; Model Data.